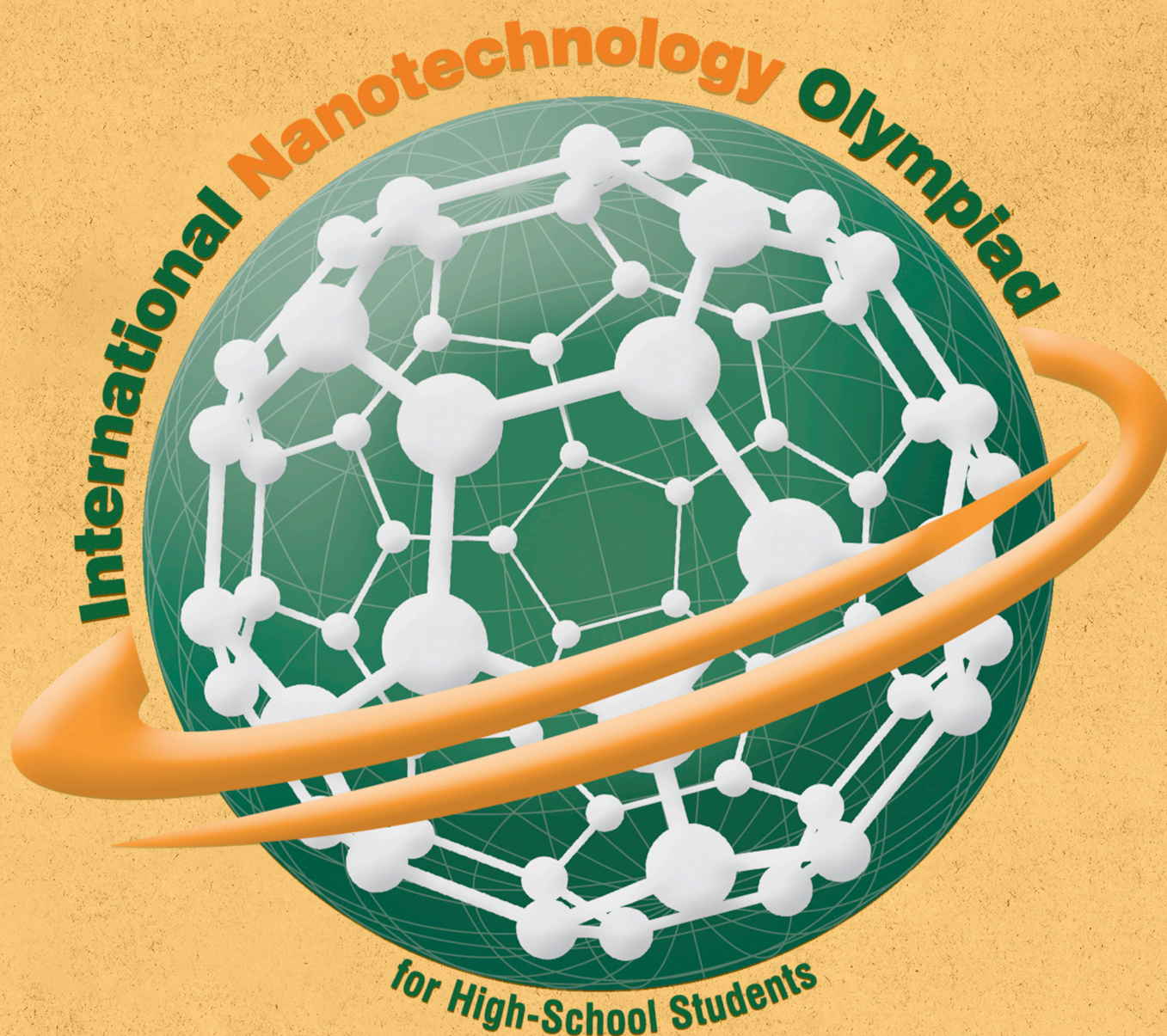


Role of of Nanotechnology in Sustainable Environmental Management



1st international Nanotechnology Olympiad for students
2025 August

Role of Nanotechnology in Sustainable Environmental Management

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Introduction

Environmental Crises of the 21st Century

With the Industrial Revolution in the 18th century, large cities grew and many people moved from small towns to these new industrial areas. This change brought new lifestyles and increased pressure on the environment. People began producing goods on a large scale, which led to faster use of natural resources. One major problem was the use of fossil fuels like coal, oil, and gas. Factories, power plants, cars, and machines burned these fuels and released large amounts of pollution into the air—such as carbon dioxide, sulfur dioxide, and nitrogen oxides.

At the same time, advances in medicine helped reduce infectious diseases, which caused the world population to grow quickly. To feed more people, farming increased, but it also caused problems. Farmers used chemical fertilizers and pesticides, and water was used in harmful ways. These practices polluted the soil and water and released greenhouse gases.

Another serious problem was deforestation—cutting down forests for farming and building. Forests help protect the environment, so removing them made things worse.

Today, environmental crises affect all parts of nature—air, water, and soil. These problems are not limited to one country or region; they are global. Bitter events and incidents in the field of the environment in recent years have made people more worried about the health of our planet.

The major environmental problems today include:

- **Climate change:** Caused by greenhouse gases from burning fossil fuels. This leads to higher global temperatures, melting ice, rising sea levels, and stronger storms and droughts. It also changes where animals and plants can live and grow.
- **Water pollution:** Caused by chemicals, heavy metals, microplastics, and sewage entering rivers, lakes, and groundwater. It harms fish and makes water unsafe for people.
- **Air pollution:** Caused by harmful gases and particles from cars, factories, and agriculture. It leads to breathing problems, heart disease, and damages crops and wildlife.
- **Soil pollution:** Caused by chemicals from farms and factories. Toxic materials enter the soil and reduce its quality, making it harder to grow food.
- **Light and noise pollution:** Too much light and loud sounds in cities affect both humans and animals, causing stress and sleep problems.
- **Deforestation:** Cutting down trees for farming or building destroys animal habitats and increases carbon in the air.

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- **Loss of biodiversity:** Many species of plants and animals are disappearing because of pollution, climate change, and hunting.
- **Plastic and microplastic waste:** Excess use of plastic pollutes the land and sea, harming animals and even entering the food we eat.

These problems are serious. They affect our health, safety, food supply, and economy. They are mostly caused by human activities—factories, farming, overpopulation, and poor waste management.

Before finding solutions, we must fully understand what these problems are, how they happen, and what damage they cause.

The Importance of Studying Pollution in Water, Air, and Soil

Studying and controlling pollution in water, air, and soil is a key step in overcoming environmental crises. These three resources—water, air, and soil—are essential for life. Their cleanliness affects the health of people, animals, plants, and the entire ecosystem.

Unlike some short-term solutions, focusing on these three areas creates long-lasting improvements. Here are the main reasons why this focus matters:

Air Pollution

Air pollution is one of the world's biggest threats. According to the World Health Organization (WHO), around 7 million people die each year from breathing polluted air. Tiny particles (called PM2.5) and harmful gases like nitrogen dioxide and carbon monoxide cause heart disease, lung cancer, and breathing problems. Air pollution also leads to acid rain, ozone layer damage, and other serious environmental issues.

Water Pollution

Water pollution is also a global crisis. More than 2 billion people around the world do not have access to clean drinking water (UNICEF and WHO, 2023). Pollutants such as heavy metals, nitrates, pesticides, and disease-causing microbes contaminate water. These cause illnesses, poisoning, and other serious health problems. In areas like the Middle East, polluted water limits farming and safe drinking water supply.

Soil Pollution

Although it gets less attention, soil pollution is just as dangerous. It degrades soil fertility, contaminates groundwater, and enters the food chain. According to a 2022 UN report, about 22 % of the world's farmland is degraded due to pollution and erosion. Pesticides, heavy metals, and microplastics kill helpful soil microbes and reduce food safety.

Pollution doesn't just harm the environment—it also threatens human health, food supplies, economic stability, and security:

- **Human Health:** Air and water pollution cause respiratory, digestive, and skin diseases. Climate change increases infectious disease spread and extreme weather like floods and droughts.
- **Food Security:** Soil and water pollution reduce crop yields and damage the food chain.
- **Economy:** A clean environment supports industries and businesses. Environmental damage reduces productivity, wastes resources, and raises recovery costs.

Protecting forests, reducing greenhouse gases, and conserving wildlife are all great—but they don't always solve pollution. For example, cutting CO₂ emissions won't help if people are still breathing dirty air daily. Cleaning soil doesn't matter if groundwater remains toxic. That's why governments and international organizations invest heavily in cleaning water, air, and soil.

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Nanotechnology—working with materials at the scale of 1–100 nanometers—is one of the most promising sciences of the 21st century. It offers clever, cost-effective solutions to environmental problems. The special features of nanomaterials—high surface area, reactivity, tiny size, and tunable design—make them invaluable tools for:

1. Cleaning Water

- Nanoparticles like zero-valent iron, manganese oxide, and carbon nanotubes can remove heavy metals, harmful organic chemicals, bacteria, and even medication residues.
- Nanostructured filters, such as graphene membranes or polymer filters infused with nanoparticles, can trap particles, microbes, metal ions, and complex molecules.
- Photocatalytic materials like TiO₂ use sunlight to break down dyes, pesticides, and drugs into harmless substances. They often use less energy and produce less waste than traditional treatments.

2. Purifying Air

- Nano-catalysts made from platinum, nickel, palladium, or cerium oxide can transform toxic gases in car exhausts or factory emissions into harmless CO₂ and water.
- Nano-adsorbents like zeolites and graphene structures are used in air filters, factories, labs—and even masks.
- Nano biosensors detect air pollution instantly, giving real-time alerts for dangerous gases.

3. Restoring Soil

- Soil is essential for growing crops and feeding the world. But it suffers from industrial, oil, and agricultural pollution.
- Magnetic nanoparticles and activated carbon can capture difficult pollutants like polycyclic aromatic hydrocarbons (PAHs) without harming soil structure.
- Nano-enhanced phytoremediation uses plants treated with nanoparticles to suck up heavy metals and organics more effectively.
- Nano-biochar, made from carbon-based nanomaterials, helps improve soil structure, retain water, and keep pollutants from spreading.

Broader Nanotechnology Roles includes:

- Waste management & recycling: Nano-enzymes and catalytic particles help break down plastics, medicines, and hazardous waste.
- Clean energy: Nanomaterials boost solar cell efficiency, improve lithium-ion batteries, support fuel cells, and enhance clean fuel production by splitting water into hydrogen using catalysts.

What makes nanotechnology a powerful tool for environmental protection is not just its ability to remove pollution. It also helps monitor the environment, prevent pollution, and even repair ecosystems. Nanosensors, made from tiny particles of gold, silver, or metal oxides, can detect very small amounts of pollutants in real time. These sensors help experts make fast and smart decisions. They can be used to detect pesticides in soil, harmful drugs in water, greenhouse gases in the air. Some nanosensors can also connect to smart systems and large databases to give more detailed analysis and reports.

Nanotechnology offers a new and better path compared to old, expensive, and slow methods. It has the potential to become the backbone of sustainable environmental solutions in our century.

Because of growing pollution, a rising global population, and limited resources, we urgently need safe, affordable, and effective nanomaterials. Nanotechnology is not just a way to react to problems; it helps build smart and protective ecosystems. Without doubt, students and future

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researchers who learn and apply nanotechnology will play a big role in protecting the health of our planet.

Overview of the Booklet and Educational Goals

This booklet is designed to help students prepare for the International Olympiad on Science and Nanotechnology. It teaches important ideas about nanotechnology and how it helps reduce environmental pollution. The content is written in a scientific, modern, and easy-to-understand style.

The main goals of the booklet are to:

- Introduce useful and commonly used nanomaterials,
- Explain how these materials are made,
- Show how they help remove pollution from water, soil, and air.

This booklet shows that nanotechnology is not only a research tool, but also a real-world solution to environmental crises.

Structure of the Booklet

Section 1: (chapter1 and 2) Understanding Pollution

- Introduces different types of pollution and the problems each one causes.
- Explains why studying and solving pollution is so important.

Section 2: Nanomaterials and Mechanisms Used in the Removal of Environmental Pollutants

chapter 3: Basics of Nanotechnology

- Explains the key features of nanomaterials:
 - High surface area,
 - Tiny size,
 - High reactivity,
 - Ability to change surface chemistry.
- Describes why nanomaterials are better than traditional methods.
- Shows which countries are leading in environmental nanotechnology.

Chapter 4: Important Types of Environmental Nanomaterials

- Metal nanoparticles like TiO_2 , ZnO , Fe_3O_4
- Carbon-based nanomaterials like graphene and carbon nanotubes
- Nanocomposites
- Nanomembranes

Chapter 5: How Nanomaterials Remove Pollution

- Explains the main processes like:
 - Adsorption
 - Photocatalysis
 - Advanced oxidation
 - Ion exchange
 - Membrane filtration
- Discusses how nanomaterials can stabilize and deactivate pollutants, especially in soil.

Section 3: How Nanomaterials Are Made

Chapter 6:

- Top-down methods (like Ball Milling, Laser Ablation)
- Bottom-up methods (like Sol-Gel, Co-precipitation)
- Green synthesis using plant extracts (eco-friendly)

Chapter 7:

- How scientists study and check nanomaterials using tools like:
 - SEM and TEM microscopes,
 - Spectroscopy tools like FTIR, UV-Vis, and XRD,
 - Surface analysis tools like BET and XPS,
 - Zeta potential and particle size analysis.

These methods help prove the materials are working well and are safe for the environment.

Section 4: Practical Uses of Nanotechnology for Pollution Control

Chapter 8: Nanotech in Water Purification

- Covers nanomembranes, photocatalysts, nano-adsorbents, nanofiltration, nanosensors, and heavy metal removal from polluted water.

Chapter 9: Nanotech for Soil Cleanup

- Includes using nanoparticles to clean contaminated soil, Removing toxic metals, Improving soil health with nanomaterials.

Chapter 10: Nanotech in Air Pollution Control

- Shows how nanotech helps remove harmful gases and particles using:
 - Gas-phase photocatalysts,
 - Nano-filters,
 - VOC capture,
 - Greenhouse gas reduction.

Section 5: Safety, Economy, and the Future

Chapter 11 and 12: safety and sustainable development

- Discusses toxicity and safety of nanomaterials,
- Talks about economic aspects and cost-benefit analysis,
- Explains how some countries have succeeded in using nanotech for environmental protection.

Chapter 13: The Future of Nanotech and the Role of Young Scientists

The final chapter looks at:

- Scientific and technical challenges,
- Limits of current technologies,
- Future research needs,
- How students and young scientists can help lead the way.

This chapter aims to inspire the next generation to take part in solving the world's environmental problems through nanotechnology.

This booklet is not just a study guide for an Olympiad. It's a learning journey to help students develop a scientific, solution-based, and responsible view of the environment. It tries to build a connection between what students learn in theory and what is happening in the real world.

Section One: Basic concepts and environmental challenges

Chapter 1 – Environmental Pollutants and their effects on human health and environment

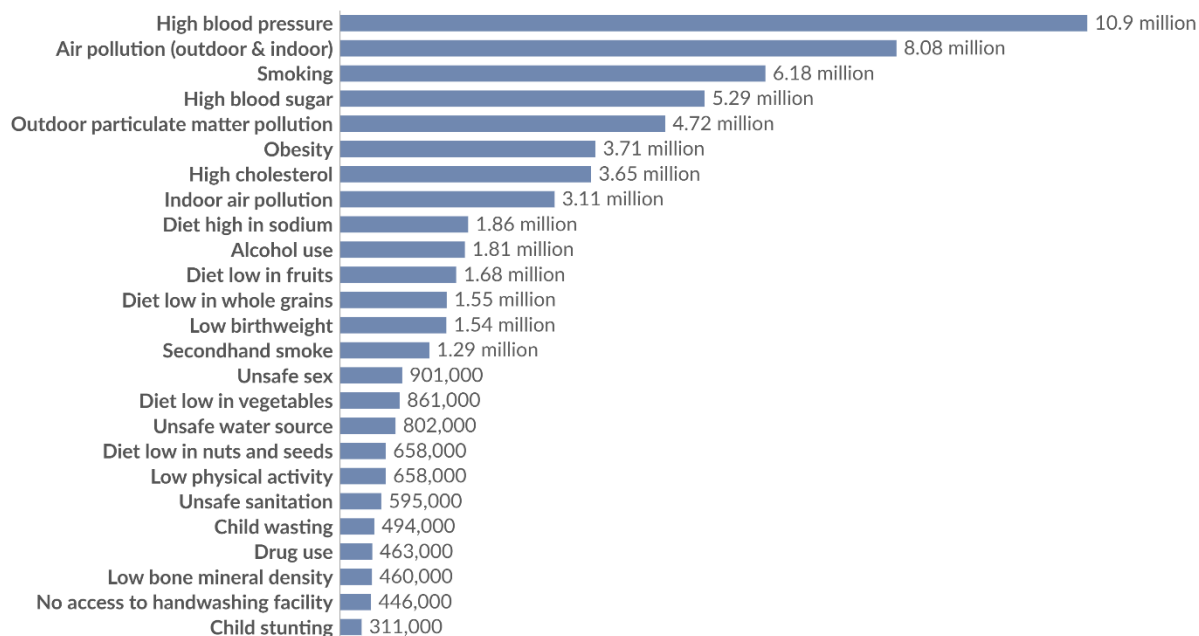
1-1 Definition of Pollutants and Their Classification

Pollution means unwanted waste from natural processes or human activities that are released into the air, soil, water, or oceans without considering the costs or consequences. These pollutants are a serious threat to human health and the Earth, and they put the stability of modern societies at risk. Pollution includes the following types:

- Air pollution by fine particles (PM_{2.5}), ozone, sulfur oxides, and nitrogen oxides
- Freshwater pollution
- Ocean pollution by mercury, nitrogen, phosphorus, plastics, and oil waste
- Soil pollution by lead, mercury, pesticides, industrial chemicals, electronic waste, and radioactive materials

According to the latest international reports and the World Health Organization (WHO), air pollution caused approximately 7-8 million premature deaths worldwide in 2019, with estimates holding steady through 2024. (Figure 1) This makes air pollution the second leading cause of non-communicable disease-related deaths globally, with roughly 1 in 8 deaths linked to air pollution. Every year, over 700,000 children under the age of 5 die due to air pollution, primarily from respiratory issues like pneumonia. Most of these deaths occur in low- and middle-income countries, particularly in WHO's South-East Asia and Western Pacific Regions. The economic cost of air pollution has also risen, with health damages estimated at \$8.1 trillion annually, equivalent to 5–6% of global Gross Domestic Product (GDP). These costs stem from lower work productivity, higher healthcare expenses, premature deaths, and reduced life expectancy. Over 89% of air pollution-related deaths and a significant portion of the economic burden occur in low- and middle-income countries, where poor healthcare systems, reliance on polluting fuels, and weak environmental regulations exacerbate the crisis. (Figure 2)

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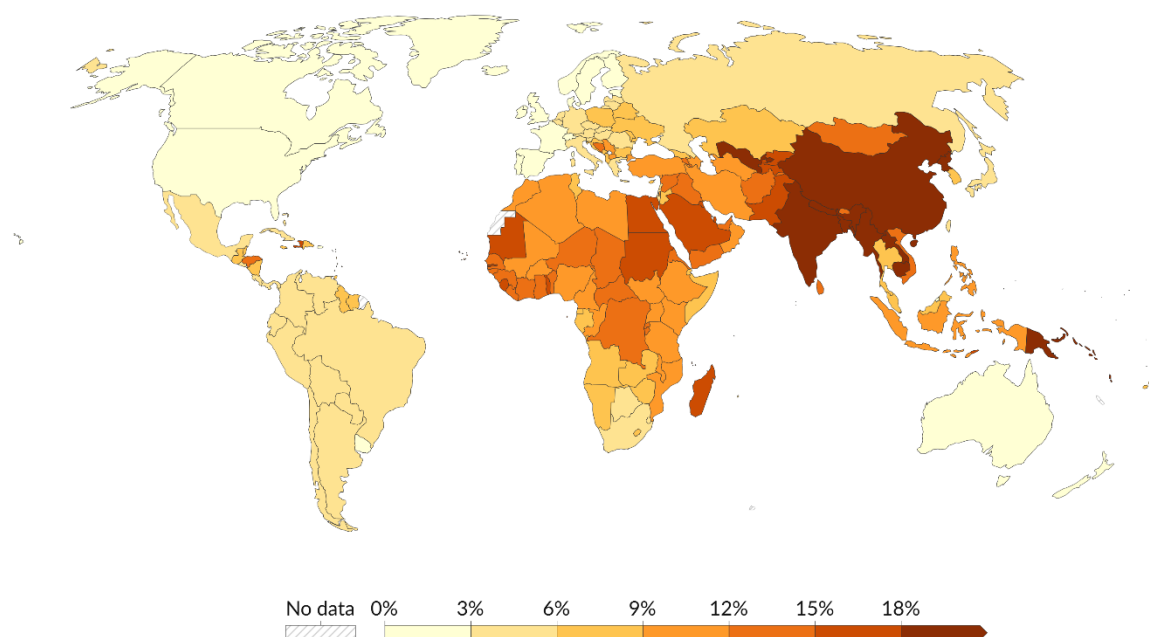


Data source: IHME, Global Burden of Disease (2024)

OurWorldinData.org/causes-of-death | CC BY

Note: Risk factors¹ are not mutually exclusive. The sum of deaths attributed to each risk factor can exceed the total number of deaths.

Figure 1. Deaths by risk factor, World, 2024



Data source: IHME, Global Burden of Disease (2024)

OurWorldinData.org/air-pollution | CC BY

Figure 2 share of deaths attributed to air pollution 2024

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While deaths from traditional pollution (like indoor air pollution from solid fuels, unsafe drinking water, and poor sanitation) have decreased, modern pollution (like outdoor air pollution, lead poisoning, and chemical pollution) is on the rise and needs serious action to prevent and reduce it.

There are several ways to classify pollutants. This chapter covers two common classifications.

First Classification: Traditional vs. Modern Pollutants

1. **Traditional Pollutants:** These are mainly linked to poor living conditions and underdevelopment. Common types include: indoor air pollution from burning solid fuels, polluted drinking water sources, poor sewage and sanitation systems
2. **Modern Pollutants:** These come mainly from industrial activities and urban development. They are divided into the following groups:

Outdoor Air Pollutants

- Fine particles smaller than 2.5 microns (PM_{2.5})
- Ground-level ozone (O₃)
- Nitrogen and sulfur oxides

Chemical Pollutants

- Heavy metals like lead and mercury
- Persistent organic compounds
- Chemical pesticides

Occupational Pollutants

- Exposure of workers to:
 - Cancer-causing substances
 - Industrial fine particles

Second Classification: Soil, Water, and Air Pollutants

This second type of classification, which will be used in later chapters of this booklet, divides pollutants into soil, water, and air categories.

1-2 Soil Pollutants

Soil pollution by toxic elements is a global concern for human health and food safety. Toxic elements in soil can come from natural sources or human activities. There are two main sources of toxic elements:

Natural Sources:

Volcanic emissions

Dust transport across continents

Weathering of metal-rich rocks

Human-made Sources:

Mining and metal smelting

Use of chemicals that contain metals

Industrial use of sewage sludge

Burning of fossil fuels

Military activities

Electronic waste (production, use, and disposal)

High amounts of toxic elements like Cr, Zn, Ni, Co, Cd, Cu, Mn, and Pb are also found in industrial wastewater. Use of phosphate fertilizers increases the amount of toxic elements in soil, especially cadmium (Cd). Toxic elements can also enter soil through biological materials such as compost, urban sewage sludge, and animal manure. These biological sources contain many toxic elements that pollute the soil and later reach water and groundwater, including Cd, Zn, Cr, Ni, and Pb. Using these materials repeatedly causes a build-up of toxic elements in the soil. Some elements like Zn, Pb, Cu, and As build up in soil due to the use of chemical pesticides and fungicides, which are used to prevent or treat plant diseases. This results in toxic elements entering food crops. The amount of toxic elements in soils varies widely depending on the strength of human sources and the level of industrial development in each region.

Generally, Soil pollution matters because it causes:

Loss of biodiversity

Reduced plant cover and climate-related issues

Lower soil fertility, which reduces crop production and causes economic problems

High cleanup costs for polluted soil

If soil pollutants enter the human food chain, they may cause severe health effects in humans. These effects are briefly described below by pollutant type:

Lead (Pb):

Lead is a metal that is easily absorbed by the body. It binds to sulfhydryl groups in proteins, interferes with calcium metabolism, and causes toxicity in the blood, bones, nervous system, digestive system, and kidneys. It's important to know that children absorb more than 50% of the lead they get from food, while adults absorb only about 15%. This makes lead especially dangerous for children. Most of the lead that enters the body is stored in the bones and teeth, where it causes weak bones and prevents proper healing after fractures. A smaller amount of lead stays in the blood, where it can cause anemia (low red blood cells) and kidney damage. The most serious damage caused by lead is to the nervous system. It can lead to movement disorders, learning difficulties, and mental health problems. In severe cases, it can cause blindness, insanity, seizures, or coma. A very important point is that in adults, lead-related problems can often be reversed, but in children, brain damage from lead is usually permanent. Lead damages the white matter of the brain, especially in children. If the peripheral nerves (outside the brain and spinal cord) are damaged, it can lead to muscle weakness, especially in the fingers and forearm muscles, which can cause wrist drop (the hand hangs limp and cannot be lifted).

Mercury (Hg):

Like lead, mercury has a strong attraction to sulfhydryl groups in proteins and causes damage to the nervous system, digestive system, and kidneys. Humans have used mercury for thousands of years — in cave paintings, hat-making, and even in treating syphilis and as a diuretic. Mercury waste from gold mining and inorganic mercury released from the Earth's crust can be changed into methylmercury by bacteria. Methylmercury enters the food chain, and its main source is contaminated fish. About 90% of the mercury we eat is absorbed into the body. It can cause nausea, stomach pain, and bloody diarrhea. Mercury is eliminated through the kidneys, which can damage them. Mercury is fat-loving, so it builds up in the brain. It can cross the placenta and reach the fetus. A developing brain is very sensitive to mercury, and it can cause severe problems in movement, thinking, senses, and behavior. Pregnant women should avoid eating fish that may be contaminated with mercury to protect their unborn children.

Arsenic (As):

Arsenic also binds to sulfhydryl groups in proteins, causing damage to the digestive system, nervous system, skin, and heart. Arsenic occurs naturally in soil and water, and it is also used in wood preservatives, pesticides, and other agricultural chemicals. Symptoms of arsenic poisoning include severe stomach pain, diarrhea, heart and lung problems, and nerve damage. Arsenic causes skin problems, especially on the palms and soles. It is also known to cause various types of cancer, though the exact mechanism is still unclear.

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Cadmium (Cd):

Cadmium causes harm through unknown mechanisms by producing reactive oxygen species (ROS) that are toxic to the kidneys, lungs, and bones. Cadmium enters the environment mainly through industrial waste, especially from nickel-cadmium batteries. It can cause emphysema (lung disease) and kidney poisoning. Cadmium exposure is also linked to a bone disease called Itai-Itai, which causes severe pain and bone weakness.

Industrial and Agricultural Exposure:

Exposure to chemical substances in industrial and agricultural settings can cause serious health problems. Organic compounds like polycyclic aromatic hydrocarbons (PAHs) (produced by burning fossil fuels or during industrial processes) and persistent pesticides such as DDT and PCBs are associated with:

Hormonal disorders (endocrine disruption)

Peripheral nerve damage (neuropathy)

Increased cancer risk

These chemicals can interfere with enzymes and hormone systems, leading to long-term health effects. Additionally, chemicals like Bisphenol A (BPA), which is used in plastics and epoxy resins, are known endocrine disruptors. BPA is linked to metabolic diseases such as diabetes and high blood pressure. Dangerous particles like asbestos and silica, found in mining and construction industries, can cause lung diseases such as pneumoconiosis and lung cancer. These risks not only affect the workers, but also their families and communities, who may be indirectly exposed to these harmful substances.

1-3 Water Pollutants

Although there is no single standard for classifying water pollutants, scientific sources and some global organizations have provided different systems of classification. One common method divides water pollutants into three main groups: Suspended solids, Waterborne pathogens and Chemical pollutants. However, the International Environmental Protection Agency categorizes these pollutants based on their chemical nature and by-products into the following main groups:

- (a) Organic pollutants
- (b) Inorganic pollutants
- (c) Disinfection by-products
- (d) Microorganisms
- (e) Radioactive isotopes

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Even though these categories overlap, this lesson covers all of them to keep the discussion complete.

1-3-1 Organic Pollutants:

Organic pollutants often come from synthetic chemicals, agricultural fertilizers, and industrial products such as greases, paints, and petroleum products. One important group is phenolic compounds, found in medicines, dyes, and antioxidants. These chemicals can change the taste of fish and drinking water and contaminate it. Another group of organic pollutants includes polycyclic aromatic hydrocarbons (PAHs), which are dangerous and hard to remove.

1-3-2 Inorganic Pollutants:

Inorganic pollutants include heavy metals, metal salts, and other compounds that do not contain carbon. A common example is fluoride. While it's added to water to prevent tooth decay, too much fluoride can cause stains on teeth and other side effects. Heavy metals are another dangerous type of inorganic pollutant. They can build up in body tissues and cause serious health problems. For example, lead and copper can enter water from the corrosion of water pipes.

1-3-3 Microorganisms (Pathogens):

Microorganisms are biological pollutants—commonly known as germs or microbes. These include:

Viruses

Bacteria

Parasites

Protozoa (single-celled organisms)

Even when chlorine is used to disinfect water, biofilms—thin layers of microorganisms—can still form in old or worn-out water pipes. These biofilms often contain specific bacteria like proteobacteria, actinobacteria and firmicutes. Besides biofilms, other microorganisms can also enter the water and cause digestive diseases or weaken the immune system. Some common bacteria found in drinking water include Shigella, Salmonella and E. coli. Since many pathogens are hard to grow in labs, scientists often link some disease outbreaks to unknown microbes. Diseases caused by waterborne pathogens include intestinal infections, lung diseases, flu, urinary tract infections and fever and bloody diarrhea. Each microbe has a Minimum Infective Dose (MID) — the smallest number of organisms needed to cause illness. For example, the MID of E. coli bacteria and protozoa has been reported to be 100 and 10, respectively.

1-3-4 Disinfection By-products:

Chlorine and other strong chemicals are often used to kill microbes in water. However, some organisms like Cryptosporidium are very resistant to these chemicals. Worse still, some disinfectants react with materials in water, forming new toxic substances. In these cases, the

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disinfectant may remove old pollution but also create new, dangerous by-products. This kind of secondary pollution can happen when disinfectants react with non-target materials or break down chemicals into harmful parts.

1-3-5 Radioactive Pollutants:

Radioactive pollutants are elements with unstable nuclei, which release high-energy ionizing radiation. Even tiny amounts of radioactive substances like radium or uranium in water can increase the risk of cancer.

1-3-6 Other types of water pollutants:

Other types of pollutants include:

- **Suspended Solids (TSS):** Total Suspended Solids (TSS) in water include broken-down plant or animal matter, sand and soil, dust and debris. Most TSS particles are smaller than 60 micrometers. These particles make water cloudy and can harm aquatic life. TSS particles can also carry microorganisms, helping spread disease. Chemically, they may be organic, inorganic, or colloidal (tiny particles that don't settle easily).
- **Nutrient Pollutants:** Nutrient pollutants are substances that feed harmful microbes or disturb natural ecosystems. One big problem is that negatively charged nutrients like nitrates and phosphates are not well absorbed by soil, so they reach surface waters easily. These compounds disrupt aquatic ecosystems by encouraging excessive algae growth, which reduces oxygen levels in water and kills aquatic animals. Other health risks from nitrates include childhood cancer, miscarriage, stomach cancer. Nitrates can also react with food compounds like amines and amides, forming N-nitroso compounds, which are dangerous. The safe limit for nitrates in drinking water is 6 mg per liter, set by health organizations. Although phosphorus and nitrogen aren't toxic to humans, they help algae grow, which reduces oxygen in water and kills marine life.
- **Hospital Waste:** Hospital waste is also known as "new-emerging pollutants". These include chemicals and drugs from hospitals that pollute surface waters. Common treatment methods for hospital waste in water:

Filtration

Oxidation

Crystallization

Floatation

Sedimentation and precipitation

Chemical synthesis

Reverse osmosis

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Ion exchange

Adsorption

Electrochemical methods

The best tools to evaluate the danger of hospital waste are:

Environmental Risk Assessment (ERA)

Quantitative Structure–Activity Relationship (QSAR)

1-4 Air Pollutants

Although some natural events like volcanoes and wildfires can release pollutants into the environment, most air pollution comes from human activities. Dangerous chemicals can sometimes get into the air by accident, but many air pollutants come from factories and other human activities. These pollutants can harm human health and the environment. An air pollutant is any substance that can hurt people, animals, plants, or materials. For people, air pollutants can cause more deaths, serious diseases, or health risks. (Figure3)

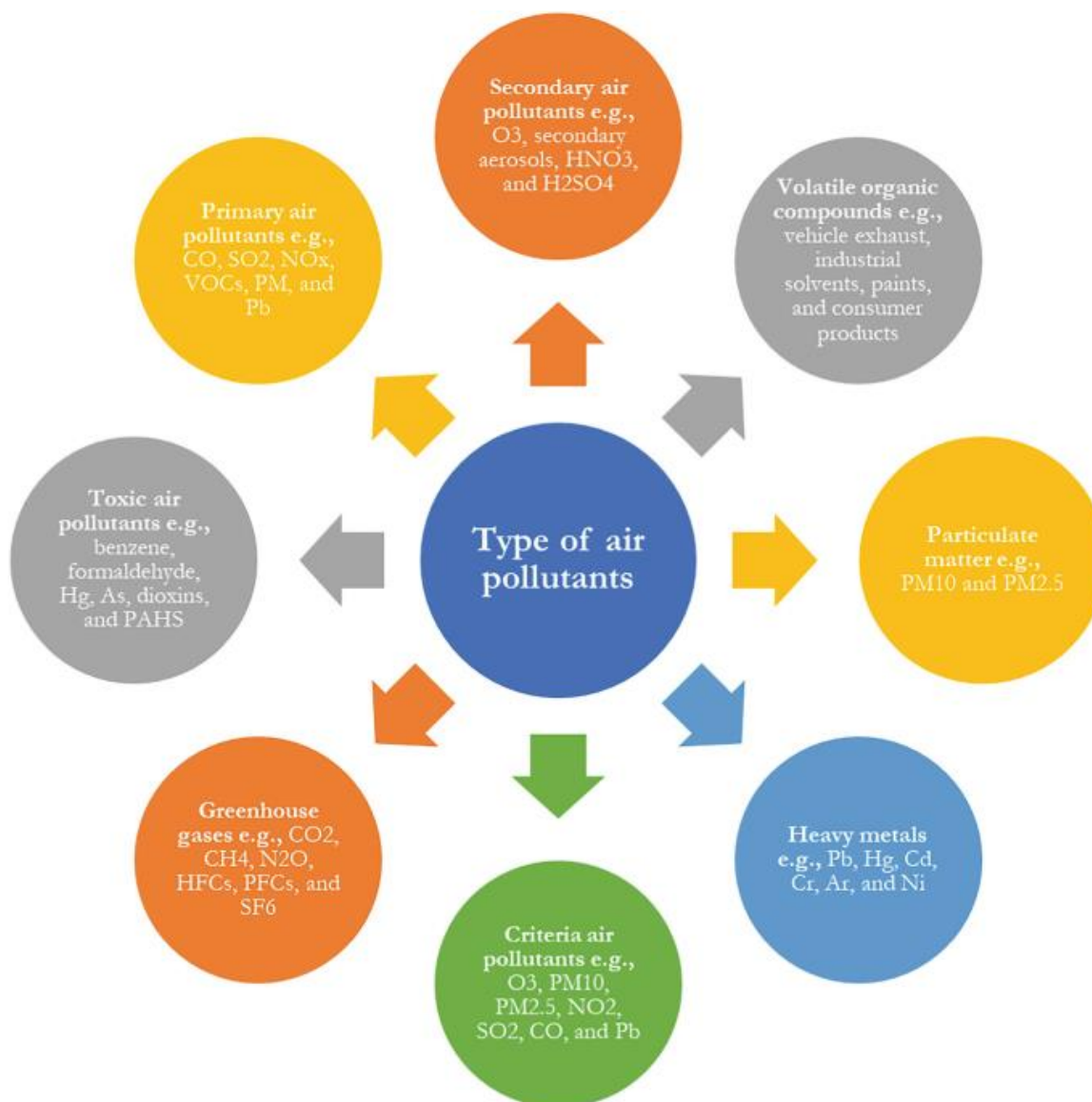


Figure 3 types of air pollutants

The main change in the air's composition comes from burning fossil fuels, which are used for energy and transportation. There are many different air pollutants. They are different in their chemical makeup, how they react, how long they stay in the environment, how far they can travel, and how they affect people and animals. However, we can group them into four main types:

1. Gaseous Pollutants (like SO₂, NO_x, CO, ozone, and volatile organic compounds)
2. Persistent Organic Pollutants (like dioxins)
3. Heavy Metals (like lead, mercury)
4. Particulate Matter (PM)

1-4-1 Gaseous Pollutants

Gaseous pollutants are a big part of air pollution and mostly come from burning fossil fuels. Nitrogen oxides (NO_x) are released as NO, which quickly changes into NO₂ in the air. The main sources are cars and factories. Ozone in the lower atmosphere forms when NO₂ and volatile organic compounds (VOCs) react in sunlight. While human SO₂ comes from the combustion of sulfur-containing fossil fuels (mainly coal and heavy oils) and the smelting of sulfur-containing ores, volcanoes and oceans are considered the main natural sources, accounting for only about 2% of total emissions. Finally, an important category of compounds for which combustion of fuels, are the main sources of emission, are called volatile organic compounds (VOCs). This category includes chemical species of an organic nature such as benzene. Although most gaseous pollutants are inhaled and mainly affect the respiratory system, they can also cause blood problems (CO, benzene) and cancer.

1-4-2 Persistent Organic Pollutants

Persistent organic pollutants are a group of toxic chemicals that remain in the environment for long periods and their effects are intensified by moving up the food chain (biomagnification). They include pesticides as well as dioxins, furans, and PCBs. In general, the general term "dioxins" is used to describe polychlorinated dibenzo-dioxins (PCDDs) and polychlorinated dibenzo-furans (PCDFs), while polychlorinated biphenyls (PCBs) are known as "dioxin-like compounds" and can act similarly to dioxins in terms of toxicity. Dioxins are formed during incomplete combustion and whenever chlorine-containing materials (such as plastics) are burned. Dioxins released into the atmosphere tend to settle in soil and water, but due to their insolubility in water, they do not contaminate groundwater sources. Most of the dioxins present in plants come from air and dust or pesticides and enter the food chain, where, due to their stable binding ability to lipids, they accumulate in the body.

1-4-3 Heavy Metals

As mentioned earlier, heavy metals include base metallic elements such as lead, mercury, cadmium, silver, nickel, vanadium, chromium, and manganese. They are natural components of the Earth's crust; they cannot be decomposed or destroyed and can be transported through the air and enter human water and food sources. In addition, they enter the environment through various sources such as combustion, wastewater discharge, and manufacturing facilities. To some extent, they enter the human body and, as trace elements, are necessary for maintaining normal metabolic reactions. However, at higher concentrations (although relatively low), they can become toxic. Most heavy metals are dangerous because they tend to bioaccumulate in the human body. Bioaccumulation means an increase in the concentration of a chemical substance in a living organism over time, compared to its concentration in the environment. Compounds accumulate in living organisms when their uptake and storage are faster than their breakdown (metabolism) or excretion.

1-4-4 Particulate Matter (PM)

Particulate matter (PM) is a general term used to describe a type of air pollutant that includes complex and diverse mixtures of particles suspended in breathable air. These particles vary in size and composition and are produced by various natural and human activities. The main sources of particle pollution include factories, power plants, incinerators, motor vehicles, construction activities, fires, and natural wind-blown dust. The size of these particles varies ($PM_{2.5}$ and PM_{10} for aerodynamic diameters smaller than 2.5 micrometers and 10 micrometers, respectively) and various categories have been defined: ultrafine particles (smaller than 0.1 micrometers), fine particles (smaller than 1 micrometer), and coarse particles (larger than 1 micrometer). The size of the particles determines in which part of the respiratory system they will settle: PM_{10} particles mainly settle in the upper respiratory tract, while fine and ultrafine particles can reach the alveoli of the lungs. So far, no single component has been identified that can explain most of the effects of PM. Among the parameters that play an important role in causing health effects are the size and surface area of the particles, their number, and their composition. The composition of PM is diverse, as they can absorb and transfer a wide range of pollutants. However, their main components include metals, organic compounds, materials of biological origin, ions, reactive gases, and the carbon core of the particles. There is strong evidence that ultrafine and fine particles, compared to coarse particles, are more dangerous in terms of mortality and cardiovascular and respiratory effects. In addition, the metal content, the presence of polycyclic aromatic hydrocarbons (PAHs), and other organic compounds such as endotoxins mainly play a role in PM toxicity.

Next, we will generally discuss the health effects of pollutants in this category:

Ozone is One of the most widespread air pollutants. Ozone damages the epithelial cells of the airways and alveolar cells by creating free radicals. Low levels of ozone are usually tolerated by healthy individuals, but in people with asthma and emphysema, especially if combined with particulate matter, it is dangerous. It should be noted that children are especially sensitive to ozone.

Sulfur dioxide, particulate matter, and acid aerosols cause inflammatory reactions in the airways. It should be noted that larger particles are trapped in the nose or in the cilia and are less dangerous.

Carbon monoxide: The tendency of this gas to hemoglobin is 200 times that of oxygen, and the main problem is the formation of carboxyhemoglobin, which is very stable and gradually reduces the level of oxygen in the blood, leading to hypoxia, and due to hypoxia, the central nervous system becomes paralyzed.

Smoke from burning organic materials: In closed spaces, smoke from burning organic materials makes a person prone to lung infections.

Radon: Radon is also a radioactive gas that, in closed environments especially homes and uranium mine, causes lung cancer. It should be noted that smoking greatly increases the carcinogenic effect of radon.

Bioaerosols: Bioaerosols include a wide range of microbial and biological agents that can cause infectious and allergic diseases. These agents are more common in families with low socioeconomic status, mainly due to poor housing conditions, inadequate ventilation, and greater exposure to pollution sources.

Bioaerosols include:

1. Pathogenic microbial agents such as:
 - Bacteria that cause Legionnaires' disease (*Legionella pneumophila*)
 - Viruses that cause viral pneumonia and colds
2. Allergens from animals and the environment such as:
 - Droppings of domestic animals (such as birds or rodents)
 - Dust mites, which are common causes of allergic rhinitis and asthma
 - Molds and fungi, which can cause eye irritation, airway inflammation, and worsen asthma.

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Chapter 2: Challenges of Pollutants

2-1 Technical problems in removing pollutants with conventional methods

2-1-1 Current solutions to soil pollution

- Soil Dressing:

Soil dressing is a very effective method to prevent the contamination of agricultural products by soil pollutants. This method involves covering contaminated agricultural soil with uncontaminated soil. There are various methods to improve contaminated soils through soil dressing. However, if the thickness of the soil dressing is not enough, the roots of agricultural products reach the contaminated soil and the effect of the solution decreases. Experiments conducted so far show that the required thickness of soil dressing depends on the type of soil, method, and environment. It should also be noted that if the land is cultivated too deeply (mixing with contaminated soil) or the concentration of pollutants in the irrigation water is high, the effect of soil dressing decreases over time. Although soil dressing is very effective in reducing the absorption of cadmium by rice, several problems have been mentioned in this method. The biggest problem is the cost; the unit cost of soil dressing is very high. Second, supplying soil dressing materials is difficult; it is no longer easy to obtain large amounts of high-quality mountain soil containing an appropriate amount of clay. In addition, soil dressing reduces the fertility of paddy fields and agricultural lands, because the mountain soil used for soil dressing is often infertile and lacks humus. Therefore, it is necessary to restore soil fertility by using large amounts of organic fertilizers for many years. Also, soil dressing raises the surface of paddy fields and agricultural lands and requires the construction of embankments and improvement of irrigation and drainage facilities.

- Water management and control of pollutant absorption:

When the paddy field is flooded and the soil is in reducing conditions, hydrogen sulfide is produced, which is in equilibrium with sulfate ions. Hydrogen sulfide, depending on pH, is dissociated into sulfide ions and precipitates with pollutant ions as sulfide salt.

But when the land is drained and the soil is in oxidative conditions, the sulfide salt is ionized to sulfate salt, which is soluble in water. This means that the solubility of pollutants changes depending on the redox potential (Eh) of the soil, and the Eh of the paddy field soil can be controlled through water management. As a result, controlling the absorption of pollutants by agricultural products is possible through water management. In addition, increasing the pH increases the cation exchange capacity (CEC) of the soil, which has a variable charge and increases the adsorption of heavy metal ions to clay. Therefore, the absorption of heavy metals by rice can be suppressed by applying materials that increase the soil pH, such as calcium carbonate, calcium silicate, autoclaved lightweight concrete (ALC), or fused silicate phosphate. However, if the soil acts as a strong buffer, the application of materials may not be sufficient to increase the soil pH. For such soil, controlling the absorption of cadmium by rice through the application of materials may be limited and sometimes close to zero.

- Remediation technology based on chemical washing method:

Soil washing is a remediation technology that involves mixing contaminated soil with a detergent in liquid form, penetration and removal of the pollutant, and then processing the soil in a remediation system. (Figure4) This chemical method can be used to effectively remove the pollutant and recover the soil in a relatively short period. Although studies on soil remediation

based on washing have been conducted by several private companies, many of these studies have focused on sites of old factories, etc., and include the transfer of contaminated soil to processing facilities for remediation, where the volume of contaminated soil and its heavy metal content are reduced by separating the clay fraction, which has a higher concentration of heavy metals. However, the application of these methods in paddy fields and agricultural lands is difficult.



Figure 4 Contaminated Soils Wash Plant & Washing Equipment

- Remediation technology based on phytoextraction:

It has been found for some time that some plants can effectively absorb cadmium present in the soil. For example, tall goldenrod (*Solidago altissima* L) from the Asteraceae family and pennycress (*Thlaspi caerulescens* L) from the Brassicaceae family are considered capable of absorbing large amounts of heavy metals. Recently, sorghum (*Sorghum bicolor* L) from the Poaceae family and kenaf (*Hibiscus cannabinus* L) from the Malvaceae family have been identified as examples of heavy metal-absorbing plants. By cultivating these plants in agricultural lands contaminated with heavy metals, the heavy metals are absorbed and removed from the soil. Today, this technology is considered a low-cost and environmentally friendly soil remediation technology.

While phytoextraction is a low-cost and environmentally friendly method, it is time-consuming. Since the remediation of soil contaminated with heavy metals in one stage by phytoextraction is difficult, several harvests may be necessary.

2-1-2 Current solutions to water pollution

Treating polluted water is one of the most important environmental and health challenges in today's world. With population growth, industrial expansion, and excessive use of resources, water pollution by various chemical, microbial, and heavy metal contaminants has increased. To address this problem, several methods for water treatment have been developed, each with its own advantages and limitations.

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Common Water Treatment Methods

- **Physical Treatment:** Includes processes such as filtration, sedimentation, and flotation, which are used to remove suspended particles, silt, and some larger pollutants.
- **Chemical Treatment:** Involves the use of chemicals like chlorine, ozone, or coagulants to remove dissolved pollutants, microorganisms, and heavy metals.
- **Biological Treatment:** Utilizes bacteria and microorganisms to break down organic matter and some biological pollutants.
- **Advanced Treatment:** Includes technologies such as reverse osmosis, ultrafiltration, nanofiltration, advanced oxidation, and ultraviolet radiation to remove resistant and emerging pollutants.

Challenges and problems of each water treatment methods are summarized in table1.

Table 1 Challenges and problems of water treatment methods

Method Type	Main Challenges and Problems
Physical	Filter clogging, need for frequent cleaning, inability to remove dissolved and microbial pollutants
Chemical	Production of toxic sludge, high cost, residual chemicals in water, need for precise dosing
Biological	Sensitivity to temperature and pH changes, low efficiency in removing heavy metals and emerging pollutants
Advanced	High energy consumption, expensive, need for advanced equipment, membrane fouling and degradation

Common Problems:

- **Incomplete Removal of Emerging Pollutants:** Many new pollutants such as pharmaceuticals, hormones, and persistent organic compounds are not completely removed by traditional methods and require more advanced technologies.
- **Generation of Secondary Waste:** Some methods, such as chemical coagulation and reverse osmosis, produce sludge or concentrated pollutant solutions that are challenging to dispose of.
- **Operational and Maintenance Costs:** Many modern technologies require high initial investment and maintenance costs, which limit their large-scale application.

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- **Energy Consumption:** Some methods, like membrane systems and advanced oxidation, consume a lot of energy and are not always environmentally or economically optimal.
- **Equipment Durability and Longevity:** Clogging, corrosion, and wear of treatment equipment, especially in membrane and filtration systems, reduce efficiency and increase repair costs.

In summary, each water treatment method has its own strengths and weaknesses, and choosing the right method depends on the type of pollutant, environmental conditions, financial resources, and the final goal. To achieve safe and sustainable water, combining several technologies and developing innovative and cost-effective solutions is essential. Additionally, attention to managing secondary waste and optimizing energy consumption are fundamental requirements for the future of water treatment.

2-1-3 Current solutions to air pollution

Various materials have been developed for the removal of air pollutants, including porous adsorbents, thermal catalysts, and photocatalysts. The use of adsorbents can be a cost-effective and easy method for absorbing air pollutants, but after complete saturation of the adsorbents with air pollutants, frequent replacement may be required. In addition, air pollutants can be re-released during the regeneration process. On the other hand, air pollutants can be removed with thermal catalysts under dark conditions, but this method is usually very costly because it requires high thermal energy.

Over the past few decades, TiO_2 has attracted considerable attention and, due to its outstanding ability in photocatalytic decomposition with UV stimulation, chemical stability, and cost-effectiveness, has become a promising material for the removal of air pollutants. Today, this material is used for indoor and outdoor air purification. For example, in Belgium, efforts have been made to use TiO_2 as a top layer for concrete sidewalks. This photocatalytic concrete can effectively reduce the concentration of air pollutants.

However, the stability of TiO_2 catalysts in long-term performance remains a serious challenge. It is widely known that the surface of TiO_2 can easily be deactivated by the accumulation of partially oxidized air pollutants at the catalytic active sites. Prolonged irradiation with UV light in the absence of pollutants can help regenerate the catalytic surface, but this method is not suitable for outdoor applications where pollutants are always present. More importantly, when the surface of TiO_2 is exposed to oily pollutants present in outdoor environments, these pollutants are difficult to wash off with water due to the insufficient hydrophilicity of the TiO_2 surface (The TiO_2 surface is inherently hydrophilic in the absence of oily pollutants.). As a result, photocatalytic reactions are inhibited due to the surface being covered by oily pollutants.

2-1-4 Emerging pollutants (Eps) and new challenges

Emerging pollutants are natural or synthetic chemical substances that are usually not included in environmental monitoring programs, but can have adverse effects on ecosystems and human health. Examples include pharmaceuticals, pesticides, disinfectants, nanoparticles, and microplastics. In general, their sources can be point sources (such as industrial and municipal wastewater) or non-point sources (such as agricultural runoff and atmospheric deposition). Main challenges in managing Eps include:

a) Monitoring and analysis challenges

The diversity and high number of EPs: more than 700 substances have been identified in Europe, which are divided into 20 different categories.

1. Limitations of analytical methods:

- Current methods are often only suitable for specific categories of EPs.
- For some EPs (such as hormones), the detection limit of existing methods is higher than dangerous concentrations.
- For some substances such as microplastics and nanoparticles, standard methods are still under development.

2. Lack of coordination in monitoring

- Existing data from different countries are not comparable due to differences in sampling and analysis methods.

b) Challenges of behavior and fate of EPs in the environment

- Complexity of EPs behavior
- The transfer and absorption of EPs in soil and water depend on various factors such as pH, organic matter, and the presence of sludge or fertilizer.
- Some EPs (such as nanoparticles) have unusual behaviors and current models are not able to predict them.
- Some EPs in the environment are converted to metabolites or by-products that may be more toxic than the original substance.

c) Modeling and risk assessment challenges

1. Lack of predictive models

- Current models are often designed for pesticides and are not suitable for new EPs (such as pharmaceuticals).

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- There is a need to develop multi-compartment models to simulate the transfer of EPs between water, soil, and sediments.
2. Combined risk assessment
- The combined effects of EPs (mixtures of several substances) are difficult to assess.

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Section Two: Nanomaterials and Mechanisms Used in the Removal of Environmental Pollutants

Chapter 3: Introduction to Nanotechnology and Its Role in the Environment

3-1 Basics of Nanotechnology

Nanotechnology refers to the study and engineering of materials at the nanoscale (1–100 nanometers), where manipulating matter at the atomic and molecular levels leads to new properties. Nanomaterials can be natural or synthetic and are often produced through chemical, biological, or green synthesis methods. These ultra-small particles, because of their very high surface-to-volume ratio, show different physical and chemical properties compared to larger-sized materials. The dimensional types of nanoparticles can be classified into zero-dimensional (quantum dots), one-dimensional (nanorods or nanotubes), two-dimensional (nanosheets), and three-dimensional structures. With rapid advancement over recent decades, nanoparticles have attracted global attention in environmental research and have opened up new functional opportunities compared to regular materials.

3-2 Unique Properties of Nanomaterials

Nanomaterials have several key properties that make them special for environmental cleanup. These include:

- **Extremely high surface area:** Because of their nanoscale size, these particles have a very large surface area relative to their volume. This leads to high capacity for absorption or chemical reaction with pollutants.
- **High catalytic activity and reactivity:** Many nanoparticles, such as titanium dioxide, have strong photocatalytic properties and can break down organic pollutants when exposed to UV light. They also have oxidation-reduction abilities and strong chemical bonding, which help speed up the breakdown of contaminants.
- **Surface modification (functionalization):** The surface of nanomaterials can be modified by attaching chemical groups or magnetic coatings to improve their ability to target specific pollutants. For example, coating iron nanoparticles (Fe_3O_4) with polymers or chemical agents prevents clumping and makes them easier to separate.

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- Magnetic properties (for some materials): Some metallic nanoparticles like Fe_3O_4 are strongly magnetic, allowing them to be pulled out of a solution using a magnetic field after they capture pollutants. This makes it easier to recover and reuse them.
- Special nanostructures: The way nanoparticles are structured (such as nanotubes, sheets, or spheres) affects how they work and increases their efficiency. These features give nanomaterials greater absorption and reaction capacity compared to larger particles.

3-3 Why is Nanotechnology Suitable for Removing Pollutants?

Because of these unique features, nanotechnology is a powerful tool for cleaning the environment. The main reasons nanotechnology works well for pollutant removal include:

- Better absorption and catalytic efficiency: Due to their large surface area, nanoparticles can absorb many pollutants or help break them down through reactions like photocatalytic oxidation.
- Surface tunability: The surface of nanoparticles can be modified with chemical groups or magnetic coatings to improve their ability to interact with different types of pollutants, from microbes to heavy metals.
- Works in different environments: Nanomaterials can be used in water, soil, and air. For example, nanosensors or biosensors can quickly detect the presence of pollutants. Nanofiltration membranes with nanometer-sized pores are useful for separating tiny particles.
- Easy recovery: In many methods, after absorbing pollutants, nanoparticles can be removed without creating dangerous waste—often using magnetic fields or changing the environment. This makes the purification process cleaner and more cost-effective.
- Sustainability and less use of chemicals: Nanotechnology can often remove pollutants using fewer chemicals. For example, nanofiltration and photocatalysis need fewer additives than traditional methods. This makes the technology a sustainable option for water purification.

3-4 Global Applications and Examples

Figure 5 illustrates how nanomaterials are used in environmental cleanup. Different nanomaterials like carbon nanotubes, graphene, and metal nanoparticles are used to target pollutants in air, soil, and water. For example, titanium dioxide nanoparticles, as photocatalysts under UV light, can break down dyes and toxic organic compounds into simpler substances. Also, iron nanoparticles (Fe^0 or Fe_3O_4), due to their large surface area and redox ability, can remove heavy metals like chromium, lead, and cadmium from water. These nanoparticles can then be separated using a magnet. Carbon nanotubes and graphene oxide also show excellent performance in absorbing organic and ionic pollutants like fluoride ions, because of their porosity and large surface area.

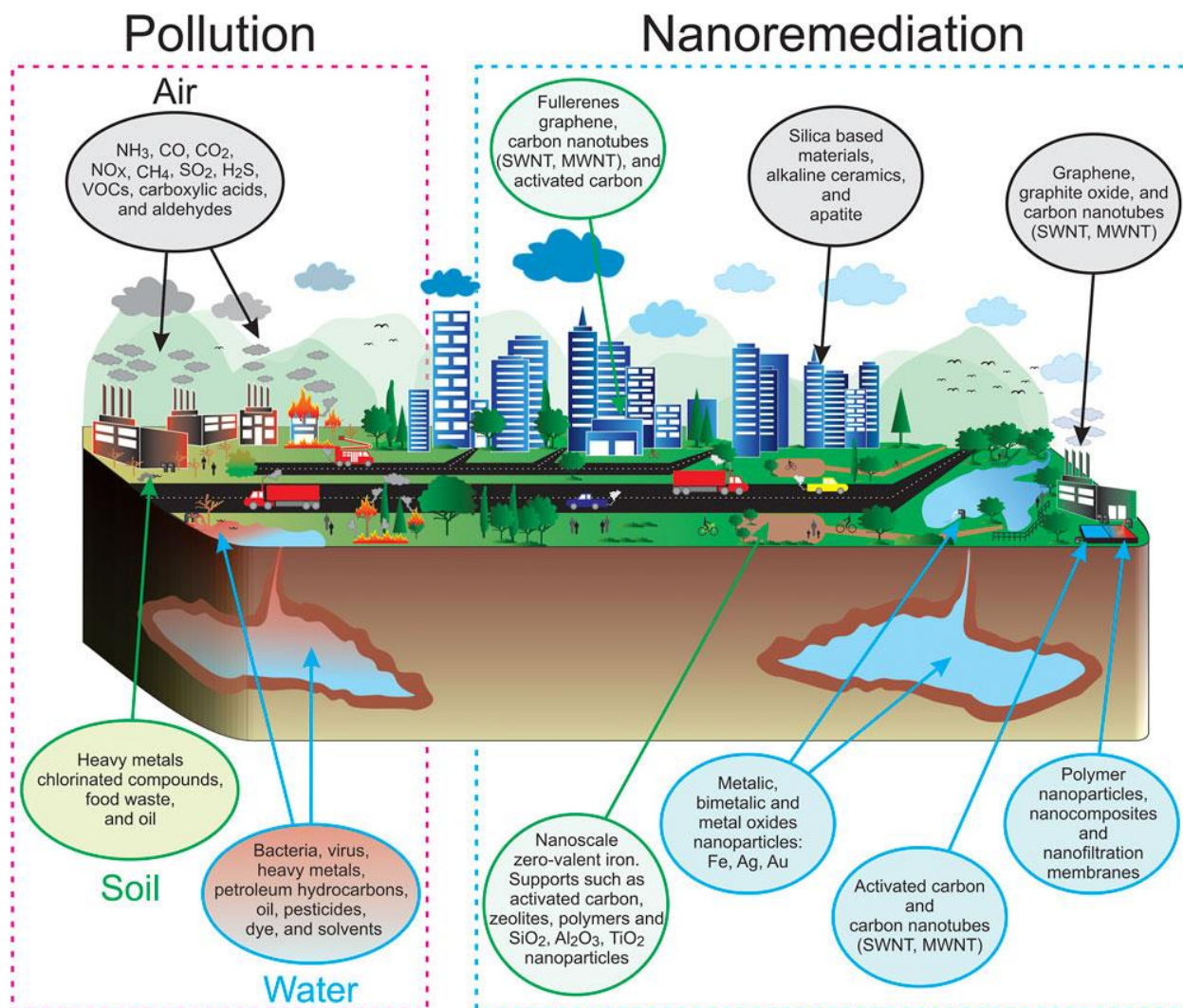


Figure 5 schematic of nanomaterials application in environmental cleanup

In addition, nanofiber membranes and nanosensors help separate and monitor pollutants in real-time. There are many global examples of nanotechnology being used in environmental work. In the United States and Europe, nanocomposites made of titanium dioxide and carbon nanotubes are used to clean industrial wastewater. In China, large-scale projects using graphene-based compounds and iron nanoparticles are being tested for cleaning rivers and industrial sewage. In India and South Africa, purification systems using titanium dioxide and carbon nanoparticles are being studied to solve water shortages. Brazil and Malaysia are investing in using nanotechnology in agriculture to improve fertilizers and soil sensors.

3-5 Position of Countries Worldwide

At the international level, countries like China and the United States are leading research in nanotechnology, especially for environmental applications. For example, a report in 2019 showed that China and the U.S. had the highest number of scientific articles on environmental

nanotechnology in top journals. Also, in investment and patents, China has become a leader over the past 20 years, with around 40% of the world's nanotechnology patents by 2024. Other countries also have national programs: the U.S. has the National Nanotechnology Initiative (NNI), which has invested over \$30 billion in research and development; Japan has a "Nanotechnology Framework"; and the European Union, in its Horizon Europe program, prioritizes renewable energy and water filtration using nanotechnology. Developing countries are also joining in. For example, India's "Nano Mission" focuses on practical research in water and health, while Brazil and Malaysia are promoting innovation in nano-agriculture. Iran has made notable progress in the practical application and commercialization of nanotechnology for environmental protection. Supported by government initiatives and knowledge-based companies, the country has developed technologies such as nanofiltration systems for wastewater treatment, nano-catalysts for air purification, and Iranian-made nanobubble systems for cleaning polluted water. Overall, research and industrial work in environmental nanotechnology are expanding, with international cooperation and government support continuing to drive the development of this strategic technology

Recent studies show that using nanomaterials (like metal oxides, carbon nanotubes, magnetic nanoparticles, and nanocomposites) in air and water purification improves efficiency and allows for effective pollutant removal. The findings suggest that nanotechnology offers a promising and cost-effective way to solve major environmental pollution problems. The notes above are based on up-to-date scientific sources and provide a comprehensive overview of the basics and current status of environmental nanotechnology.

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Chapter 4: Nanomaterials used in pollutant removal

4-1 Metallic Nanoparticles and Metal Oxides

Metallic nanoparticles and metal oxide nanoparticles (such as TiO_2 , ZnO , Fe_3O_4 , etc.) have shown remarkable effectiveness in pollutant removal due to their high surface-to-volume ratio and high reactivity. These nanoparticles can accelerate pollutant adsorption and degradation by providing a large active surface area and reactive functional groups. For instance, metallic and metal oxide nanoparticles with strong performance in pollutant breakdown and adsorption have been widely used in environmental remediation. Moreover, many studies have shown that adsorption and photocatalytic methods are more cost-effective and efficient than traditional chemical or physical methods.

4-1-1 Key Advantages and Properties

- **High Specific Surface Area and Reactivity:** Reducing particle size leads to an increased surface area and the emergence of unique quantum effects. As a result, the adsorption capacity and catalytic activity of nanoparticles increase significantly. Metallic nanoparticles have been effectively used to remove metal ions from wastewater.
- **Magnetic Properties (for Fe_3O_4):** Magnetic iron oxide nanoparticles not only possess catalytic activity but can also be easily separated from the environment using a magnetic field. This property enables easy recovery of the nanoparticles.
- **Dual Functionality:** These materials can act both as surface adsorbents for removing pollutants and as active catalysts, especially under light exposure. For example, TiO_2 and ZnO are two semiconductor photocatalysts that generate strong free radicals under UV light.
- **Ease of Use:** Materials like TiO_2 and Fe_3O_4 are stable, relatively inexpensive, and readily available. Simple synthetic methods such as co-precipitation and precipitation are commonly used to produce nanostructures of these compounds.

4-1-2 Practical Examples of Using Metallic and Metal Oxide Nanoparticles

- **Titanium Dioxide (TiO_2):** This well-known nanocatalyst can degrade organic pollutants in the presence of UV light by generating hydroxyl radicals. For instance, TiO_2 nanoparticles doped with neodymium and nitrogen were able to remove about 92% of the organic dye “Acid Blue AB-29” after 80 minutes of visible light exposure, while pure TiO_2 only achieved around 31% removal.
- **Zinc Oxide (ZnO):** ZnO nanoparticles also play an effective role in pollutant removal, especially as photocatalysts or adsorbents for metal ions. Research has shown that ZnO can adsorb heavy metals such as lead (Pb^{2+}) and cadmium (Cd^{2+}) from aqueous solutions. Additionally, ZnO has structures with active functional groups that aid in ion adsorption.

4-2 Nanocarbon Materials (Graphene, Carbon Nanotubes, Activated Carbon)

Polluted water containing toxic substances (such as heavy metals, dyes, pharmaceuticals, and persistent organic compounds) poses a serious threat to human health and the environment. Therefore, chemical adsorption using nanocarbon materials (including graphene, carbon nanotubes, and activated carbon) has attracted significant attention as an efficient method for water purification. These carbon-based materials have graphitic structures, extremely high surface areas, and unique chemical properties that enable effective pollutant removal.

This report focuses on the physicochemical aspects of nanocarbon performance, examining their structure, surface properties, adsorption mechanisms, and the impact of environmental parameters on their efficiency.

4-2-1 Comparison of the Performance of Graphene, Carbon Nanotubes, and Activated Carbon

- **Surface Area and Porosity:** Activated carbon, due to its network of fine pores, can reach a very high specific surface area (up to several thousand square meters per gram), offering numerous adsorption sites. Graphene also offers a large surface area (more than 1000 m²/g), making it a desirable option for pollutant adsorption. Carbon nanotubes (CNTs) have a moderate surface area, but through surface modification using acid treatment or by opening their tubular structure, their effective surface area can be significantly increased.
- **Adsorption Performance:** Carbon nanotubes (CNTs) have shown excellent performance in heavy metal adsorption due to their highly active surface and functionalized groups. Graphene and graphene oxide also perform well in adsorbing organic pollutants such as dyes, pharmaceuticals, and persistent organics. Activated carbon, because of its high porosity and graphitic structure, has a wide adsorption capacity for various contaminants including heavy metals, dyes, and organics.
- **Electrical Conductivity:** In processes where electron transfer is important, such as advanced oxidation processes, graphene and CNTs outperform activated carbon due to their high electrical conductivity. Activated carbon typically has lower conductivity.
- **Cost and Practical Application:** Activated carbon is relatively inexpensive, widely available, and commonly used in industry, although its regeneration and reuse can be energy-intensive and costly. Graphene and CNTs are more expensive and raise environmental concerns such as stability and sedimentation. Thus, efforts are being made to produce nanocarbon composites with polymers or nanoparticles to reduce costs, enhance stability, and improve functional performance.

4-2-2 Practical Examples of Nanocarbon Applications

- **Graphene/Graphene Oxide (GO):** These two-dimensional materials provide a broad surface area suitable for pollutant adsorption. For instance, a composite made of

GO/ZnTiO₃/TiO₂ exhibited an adsorption capacity of 78 mg of methylene blue per gram and showed only a 15% drop in performance after five cycles of use. Other studies indicate that graphene-based nanomaterials can remove over 95% of dye pollutants like methylene blue.

- Carbon Nanotubes (CNTs): Multi-walled carbon nanotubes (MWCNTs) have been used as adsorbents for organic substances. A review study reported that standard MWCNTs could adsorb up to 100 mg of methylene blue per gram, while MWCNTs functionalized with –SH groups reached an adsorption capacity of 166.7 mg/g, showing a significant improvement after modification.
- Activated Carbon: Due to its extremely high surface area and porosity, activated carbon is widely used for pollutant adsorption. Studies show that for removing dyes and organic pollutants, activated carbon provides a low-cost and effective adsorption method. Additionally, magnetic activated carbon (e.g., combined with Fe₃O₄) has been employed to allow easier separation of the adsorbent after use.

4-3 Nanocomposites

In recent years, nanocoatings and nanocomposites have emerged as promising new adsorbents. These materials have special physicochemical properties such as high surface area, adjustable porosity, and active functional groups, which give them a strong capacity to remove pollutants. Many nanocomposites can also be easily separated and reused, reducing the total cost of the process.

Nanocomposites are materials made by combining different types of nanomaterials. By using the synergistic effects between the components, their pollutant removal performance improves. In other words, combining the best properties of each component into one structure increases their adsorption or degradation abilities.

4-3-1 Structure and Composition of Nanocomposites

Nanocomposites used in water treatment often combine metal or metal oxide nanoparticles (like TiO₂, ZnO, Fe₃O₄, MgO, etc.) with polymeric or carbon-based carriers. For example, chitosan-based nanocomposites usually contain metal oxide nanoparticles, carbon nanotubes, or graphene. In these composites, the antibacterial properties and functional groups of chitosan combine with the surface area and magnetic properties of the nanoparticles.

Graphene and its derivatives (like graphene oxide) are also used as supports and enhancers due to their high surface area and excellent electrical conductivity. Adding graphene oxide to matrices like chitosan or alginate can significantly increase the adsorption capacity of the resulting composite.

Two-dimensional materials (like graphene) and tube-like structures (such as carbon nanotubes) are also great choices for nanocomposites because of their large active surface areas. Polymer additives like polyaniline, alginate, polyacrylamide, and ion-exchange resins improve surface functional groups and mechanical stability, enhancing overall performance.

4-3-2 Surface Properties and Morphology

A key reason for the high efficiency of nanocomposites in pollutant removal is their large specific surface area and adjustable porosity. A higher surface area provides more active sites for adsorption. Especially in carbon-metal nanocomposites, increasing surface area can significantly improve adsorption performance. For example, combining graphene oxide with TiO_2 increases surface area and improves the adsorption of organic pollutants.

The density and distribution of surface functional groups like $-\text{OH}$, $-\text{COOH}$, and $-\text{NH}_2$ also play an important role. These groups help with ion suspension and complex formation with pollutants. The shape and morphology of nanocomposites (such as cubic, spherical, tubular, or sheet-like) also affect adsorption and are often analyzed using scanning electron microscopy (SEM) or X-ray diffraction (XRD).

In general, carbon-based materials like graphene, activated carbon, and carbon nanotubes are not only useful for pollutant adsorption but are also applied in electrochemical processes due to their high conductivity.

Also, nanoparticles like silica oxide, titanium dioxide, and other metal oxides often have porous structures that support both monolayer (Langmuir) and multilayer (Freundlich) adsorption mechanisms. The combination of high surface area and optimal porosity significantly increases their adsorption potential.

4-3-3 Practical Examples of Nanocomposites

- **GO- TiO_2 Nanocomposite:** This structure combines graphene oxide (GO) with titanium dioxide (TiO_2). In this composite, GO acts as a support and increases the surface contact and dispersion of TiO_2 nanoparticles, improving the overall performance. Studies have shown that this composite can efficiently remove organic dyes like *crystal violet* under visible light, with much better performance than pure TiO_2 . It has also been effective in removing metal ions like lead and cadmium after just two hours of light exposure. (Figure 6)

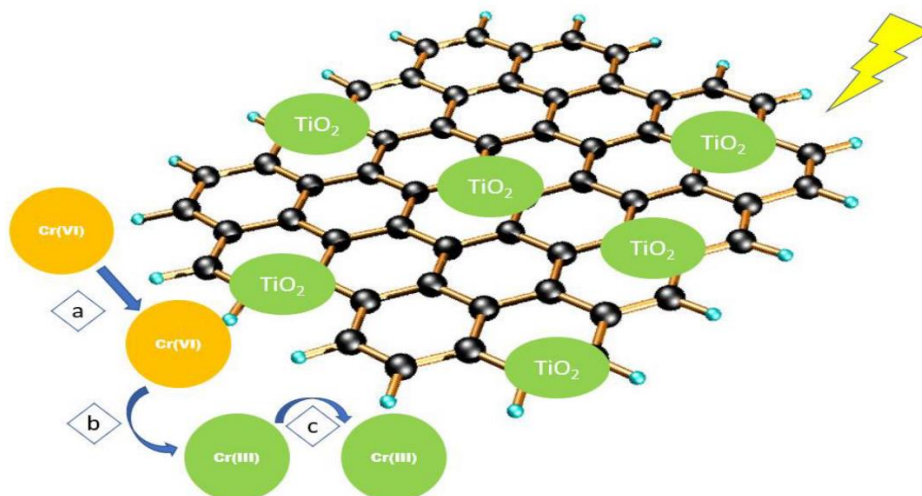


Figure 7 GO/TiO₂-Related Nanocomposites as Photocatalysts for Pollutant Removal in Wastewater Treatment

- **Fe₂O₃–TiO₂ on Reduced Graphene Oxide (rGO):** Another example is a nanocomposite combining iron oxide (Fe₂O₃) and titanium dioxide on rGO. This structure has been used to remove arsenic ions from water, showing high adsorption capacity for both arsenic (III) and arsenic(V). The improved performance is due to the increased number of active sites at the contact points between the components, allowing more effective pollutant adsorption.

4-4 Nanomembranes and Adsorbent Materials

Many of the nanomaterials introduced earlier can also be used as adsorbent materials or as parts of membranes to help remove organic and inorganic pollutants from wastewater.

4-4-1 Adsorption Process

Adsorption is one of the key methods for water purification. In this process, pollutants gather physically or chemically on the surface of a material called an adsorbent. The adsorption of pollutants usually happens in three main steps:

1. Transport of the pollutant from the liquid (wastewater) to the surface of the adsorbent
2. Attachment or binding of the pollutant to the surface
3. Penetration of the pollutant into the structure of the adsorbent

Nanoparticles are excellent adsorbents because of their high surface area and the ability to be functionalized (modified with special chemical groups). Also, their nanoporous structures make it easier for pollutants to enter and bind inside the particles.

Another advantage of nano-adsorbents is that they can often be recovered and reused. For example, self-assembled iron oxide nanoparticles can remove dyes and heavy metal ions from wastewater. If these nano-adsorbents have magnetic properties, they can be easily separated from water using a magnet. After separation, the adsorbents can be cleaned and regenerated using a process called catalytic combustion, typically at around 300°C. In this process, heat causes the catalyst to oxidize (burn off) the adsorbed pollutants, so the material can be reused again.

4-4-2 Nanofiltration

Besides adsorption, filtration is also a major method for water treatment. In filtration, a membrane or filter is used to separate solid particles from liquids. Nanofiltration is a special separation method that uses pressure to push water through a membrane with very small pores (in the nanometer range). It is widely used today for purifying water, desalination (removing salt from water) and removing small organic molecules and ions.

Key features of nanofiltration:

- Effective separation of pollutants based on size and electric charge
- High permeability to water
- Lower energy consumption compared to methods like reverse osmosis (which requires higher pressure)

For example, nanofiltration typically uses 7 to 30 atmospheres of pressure, while reverse osmosis needs 20 to 100 atmospheres. That's why nanofiltration is considered a cost-effective and efficient method in the water purification industry.

4-4-3 Examples of Nanomembranes and Adsorbents in Use

- Ceramic nanomembranes: These membranes have nanoscale pores and are very effective in removing pollutants. In some studies, ceramic membranes combined with nano-catalysts removed over 98% of organic dyes. They were also highly effective in removing pharmaceutical compounds and nano-pollutants. In one case, these membranes removed over 84% of antihistamine drugs from water.
- Hybrid membranes (GO/TiO₂/Ag): Research has shown that adding nanoparticles to membrane structures can boost separation efficiency. For example, a membrane made of graphene oxide (GO), titanium dioxide (TiO₂), and silver nanoparticles (Ag) achieved 99.6% efficiency in separating an oil–water emulsion under ultraviolet light. As the emulsion passed through the membrane, the pollutants were also broken down.
- Magnetic adsorbents: Magnetic iron oxide nanoparticles (Fe₃O₄) have been used as adsorbents for metal ions. Studies have shown that these nanoparticles can effectively

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remove heavy metals like cadmium, chromium, copper, nickel, lead, and zinc from water. For example, they were able to remove about 99% of lead ions from water at a pH of 4.5. The main benefit of these adsorbents is that they can be quickly separated from water using a magnet.

In the adsorption process, pollutants are captured physically or chemically on another material. This method can be used to clean both organic and inorganic contaminants. Nanomaterials have more accessible surface area than bulk materials, can be functionalized, and often have nano-scale pores that help the adsorption process.

Common materials used for adsorption include:

- Carbon nanostructures
- Bio-based adsorbents
- Metal oxide adsorbents
- Zero-valent iron nanoparticles (nZVI)
- Hydrogels

In nanofiltration, a membrane with nano-sized pores prevents the passage of pollutants. By using membranes with different surface charges, we can selectively separate particles. These membranes are made from polymers or inorganic ceramics, each with its own advantages and disadvantages. Nanomaterials like carbon nanostructures and metal oxides are also used to make these membranes.

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Chapter 5: Mechanisms of pollutant removal by nanomaterials

5-1 Adsorption and Chemical absorption

The treatment of dye-containing wastewater, due to its high pollution potential, is one of the major concerns of environmental health experts. Among the various methods proposed for dye removal, adsorption has attracted much attention from researchers due to its unique advantages. The most important factor in the adsorption process is selecting an appropriate adsorbent that can remove the highest amount of pollutants in the shortest possible time. Adsorption is a separation process in which certain components of a fluid phase are transferred onto the surface of a solid adsorbent. Typically, fine adsorbent particles are held in a fixed bed, and the fluid continuously flows through the bed until the solid becomes nearly saturated and can no longer perform the intended separation. Adsorption can be defined as the tendency of fluid phase molecules to adhere to a solid surface. In this process, porous solids are used, which offer a very large surface area due to the presence of internal pores—meaning that not only the outer surface, but also internal surfaces contribute to the total area. As the surface area of the adsorbent increases, the adsorption capacity also increases. Therefore, highly porous particles are used in adsorption systems. In an adsorption process, the material being removed is the adsorbate, and the suspended substances are known as the adsorptive.

Chemical absorption or Chemisorption is a type of surface adsorption in which strong chemical bonds—either covalent or ionic—are formed between the adsorbate and the surface of the adsorbent. Unlike physical adsorption, which is based on weak van der Waals forces, Chemical absorption involves high adsorption energy (typically between 40 to 800 kJ/mol) and usually occurs as a monolayer and is irreversible. Chemical absorption is highly selective and strongly depends on the chemical structure and composition of the surface. It typically occurs more effectively at higher temperatures, as forming chemical bonds requires activation energy.

Examples of this process include:

- adsorption of hydrogen on the surface of nickel during hydrogenation reactions,
- adsorption of CO gas on platinum surfaces in catalysts,
- and adsorption of heavy metal ions like lead and cadmium via surface functional groups such as –OH or –COOH on nanomaterials.

This type of adsorption is widely used in:

- environmental processes such as water purification,
- removal of heavy metals and organic pollutants,
- industrial catalysts,
- gas sensors,
- and storage of gases like hydrogen and carbon dioxide.

5-2 Catalytic Degradation (Photocatalysts and Nano-Catalysts)

In catalytic degradation processes, nanomaterials act as active catalysts to convert pollutants into

harmless products. In particular, in photocatalysis, semiconductor nanomaterials such as titanium dioxide (TiO_2) or zinc oxide (ZnO) absorb ultraviolet or visible light and generate electron–hole pairs. The holes react with water or hydroxide ions to produce highly reactive hydroxyl radicals ($\bullet\text{OH}$), and the electrons reduce dissolved oxygen to superoxide radicals. These radical species are extremely reactive and break down organic pollutants into carbon dioxide and harmless inorganic compounds. Photocatalysis has environmental advantages, as it only requires sunlight and a semiconductor, and does not produce harmful chemicals during the process.

For example, TiO_2 nanoparticles are among the most important photocatalysts. Under light exposure, they produce charge carriers (electrons and holes) that initiate strong redox reactions, decomposing dyes and complex organic compounds into CO_2 and harmless ions.

5-3 Advanced Oxidation

In advanced oxidation processes (AOPs), nanomaterials assist strong oxidizing agents such as hydrogen peroxide (H_2O_2), ozone, or persulfate to generate highly reactive oxidizing radicals, such as hydroxyl radicals ($\bullet\text{OH}$) or sulfate radicals ($\text{SO}_4^{\bullet-}$), capable of breaking down and degrading pollutants. These processes possess extremely high oxidative power and can convert organic contaminants into water, carbon dioxide, and simple mineral salts.

A well-known example is the Fenton reaction, in which ferrous ions (Fe^{2+}) derived from nanoscale iron oxides convert hydrogen peroxide into hydroxyl radicals.

In general, AOPs involve systems based on hydrogen peroxide (e.g., Fenton and Fenton-like reactions), ozone, or UV/ H_2O_2 combinations. Nanocatalysts, such as magnetic iron oxide (Fe_3O_4) or iron oxide (Fe_2O_3), play a key role. For instance, magnetic iron nanoparticles (Fe_3O_4), in the presence of hydrogen peroxide, can produce hydroxyl radicals and effectively degrade pollutants such as phenols or organic dyes within a few hours.

5-4 Ion Exchange and Complexation

In ionic adsorption mechanisms, ionic pollutants (usually heavy metals) are removed through ion exchange with mobile ions within the nanomaterial structure, or by forming complexes with surface functional groups.

Simply put, nanostructures like zeolites or activated carbon contain mobile ions such as Na^+ , K^+ , or Ca^{2+} that can be exchanged with pollutants in solution. Additionally, their surfaces may carry functional groups like amines ($-\text{NH}_2$), carboxyls ($-\text{COOH}$), or thiols, which can bind metal ions and immobilize them.

For example, aluminosilicate frameworks with regular porous structures—especially nano-zeolites—have high surface area and uniform pores, making them highly capable of ion exchange. Studies have shown that modified nano-zeolite compounds can significantly increase the adsorption capacity for heavy metals like Pb^{2+} and Cd^{2+} , where the pollutant ions replace sodium ions in the zeolite structure and are thus removed from solution.

5-5 Microbial Removal Using Nanostructures

Combining nanomaterials with microbial or enzymatic methods is one of the effective strategies in pollutant removal. In these systems, nanomaterials can serve as carriers or immobilizing supports for microorganisms or enzymes. For example, enzymes can be immobilized on the surface of magnetic nanoparticles to enhance their thermal and structural stability, and enable easy separation after the reaction.

Additionally, biological materials can either synthesize or coat active nanoparticles. For instance, biosynthesized copper nanoparticles produced by copper-resistant bacteria have been shown to degrade azo dyes like *Reactive Black-5* and *Congo Red* with efficiencies of 80–95%.

In microbial degradation processes, magnetic nanomaterials such as Fe_3O_4 are used as supports for enzymes like laccase. In one example, a Fe_3O_4 /chitosan composite served as a support for enzyme immobilization via covalent bonding, enabling the degradation of phenolic pollutants like 2,4-dichlorophenol over multiple cycles with efficiencies above 75%.

In general, the use of nanomaterials in microbial treatment enhances enzyme loading, structural stability, and recoverability of enzymes or microorganisms.

5-6 Pollutant Transformation and Stabilization

Sometimes, nanomaterials transform pollutants into less harmful forms or stabilize them in the environment. For instance, nanoscale zero-valent iron (nZVI) acts as a strong reducing agent and can convert toxic compounds into inert forms. An example is the reduction of hexavalent chromium (Cr^{6+}) to trivalent chromium (Cr^{3+}), which is less soluble and precipitates as chromium hydroxide.

Heavy metal ions such as Pb^{2+} or Cd^{2+} can also be stabilized by forming surface complexes on nanoparticles like iron oxides or activated carbon. This stabilization reduces the bioavailability and mobility of the metals in the environment.

Another example includes hydroxyapatite nanoparticles, which can remove toxic metal ions by forming insoluble phosphate phases such as pyromorphite (in the case of lead), or similar compounds for cadmium. In this process, metal ions are either precipitated or exchanged with calcium ions (Ca^{2+}) in the hydroxyapatite structure.

Thus, nanomaterials not only help in removing pollutants, but also significantly reduce their toxicity and mobility in the environment.

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Section three: synthesis and characterization of nanomaterial for environmental application

Chapter 6: Methods for the Synthesis of Environmental Nanomaterials

Introduction

Nanomaterials are tiny particles with sizes measured in billionths of a meter. They have unique properties that make them very useful in solving environmental problems such as water purification, pollution control, and energy storage. To use these nanomaterials effectively, scientists must first create them through various synthesis methods. This chapter introduces the audience to the main methods for fabricating environmental nanomaterials and, by focusing on practical examples and real-world applications, helps to understand how these methods work beyond theory.

This chapter covers two broad categories of synthesis: Top-down methods, which breakdown larger materials into nanoparticles, and Bottom-up methods, which build nanoparticles from atoms or molecules. The chapter will also explore green synthesis—an eco-friendly approach using plants—and compare these methods in terms of efficiency, cost, and environmental impact.

6.1 Top-Down Methods

Top-down methods start with bulk materials and reduce them to nanoscale particles. Think of carving a sculpture from a large block of stone, but at a scale millions of times smaller!

6.1.1 Ball Milling

Ball milling or mechanical milling is a synthesis method where large pieces of material are ground into nanoparticles by tumbling them with hard balls inside a rotating container. This method is widely used because it is simple and can produce large amounts of nanomaterials. Iron oxide nanoparticles produced by ball milling are used in environmental cleanup to remove heavy metals like lead and arsenic from contaminated water. The high surface area of these nanoparticles allows them to adsorb toxic metals efficiently. For instance, researchers have shown that ball-milled iron oxide nanoparticles can reduce lead concentration in water by over 90%, making water safer for drinking and irrigation. For more details, check the relevant chapter in chapter 4.

Advantages:

- Cost-effective and scalable
- Does not require complex chemicals

Limitations:

- Possible contamination from the grinding media
- Less precise control over particle size and shape

6.1.2 Laser Ablation

Laser ablation uses intense laser pulses to vaporize material from a solid target into a surrounding liquid or gas, forming nanoparticles (See Figure 7). This method produces very pure nanoparticles without the need for chemical reagents. Silver nanoparticles made by laser ablation are used in water filters to kill bacteria and other harmful microorganisms. Because the process avoids chemical contaminants, the nanoparticles are ideal for applications in drinking water treatment and medical devices.

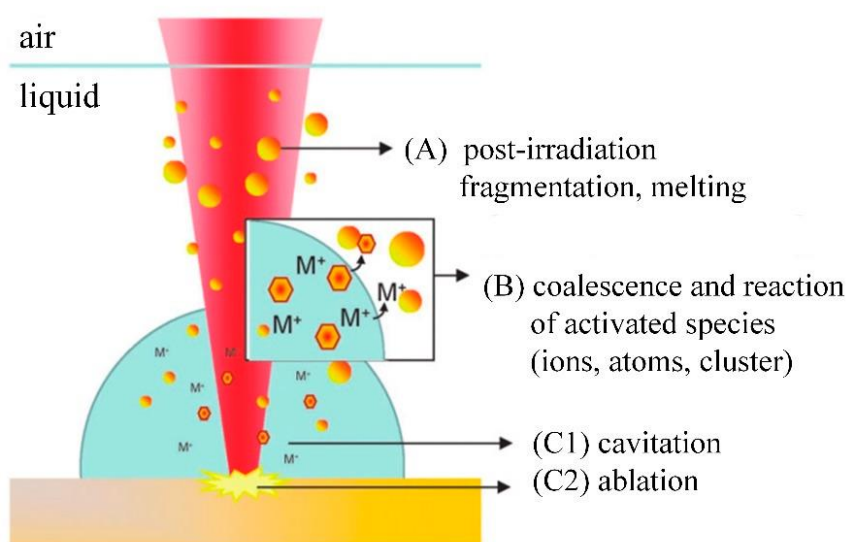


Figure 7 An overview of laser ablation assisted by ultrasonic waves and the processes occurring during it:

- (A) Post irradiation: Colloidal particles interact with the laser beam;
 - (B) The generated nanoparticles undergo polymerization and react with the colloidal particles;
 - (C1) An expanding cavitation bubble is created, inside which crystallized nanoparticles form through nucleation and coalescence;
 - (C2) The incoming laser pulse is absorbed by the bulk material, leading to plasma formation.
- (<https://doi.org/10.1039/c4cp03316g> , Copyright 2014, Royal Society of Chemistry)

Advantages:

- Produces high-purity nanoparticles
- Environmentally clean, no chemical waste

Limitations:

- Expensive equipment
- Limited production scale compared to ball milling

6.2 Bottom-Up Methods

Bottom-up approaches build nanomaterials by assembling atoms or molecules, similar to constructing a building brick by brick!

6.2.1 Sol-Gel, Co-precipitation, and Hydrothermal Methods

These methods were discussed in detail in the first booklet. Here, we briefly review them with a focus on practical applications.

- **Sol-Gel:** Involves transforming a liquid solution into a solid gel that contains nanoparticles. This method is used to create metal oxide nanomaterials like titanium dioxide (TiO_2), which are applied in sensors and water treatment. For example, TiO_2 nanoparticles synthesized by sol-gel are used to break down organic pollutants in wastewater through photocatalysis, a process activated by light.
- **Co-precipitation:** This method mixes solutions containing metal ions to form solid nanoparticles. It is commonly used to produce magnetic nanoparticles such as magnetite (Fe_3O_4), which can remove pollutants from water by magnetic separation. For instance, magnetite nanoparticles synthesized this way have been successfully used to extract arsenic from groundwater.
- **Hydrothermal:** Uses high temperature and pressure inside a sealed vessel to grow nanomaterials with controlled shapes and sizes. Hydrothermal synthesis is applied in making zinc oxide (ZnO) nanorods that degrade harmful dyes in industrial wastewater. These nanorods act as catalysts, speeding up the breakdown of toxic chemicals under mild conditions.

6.2.2 Microwave-Assisted Synthesis

Microwave-assisted synthesis uses microwave radiation (with frequency: 300 MHz to 300 GHz) to heat reactions quickly and uniformly, speeding up the formation of nanomaterials (See Figure 8). This method is energy-efficient and can convert industrial waste into useful nanomaterials. Magnetic biochar (a form of charcoal) nanoparticles produced by microwave-assisted synthesis from agricultural waste have been used to remove heavy metals from polluted water. This approach not only recycles waste but also produces effective nanomaterials for environmental cleanup.

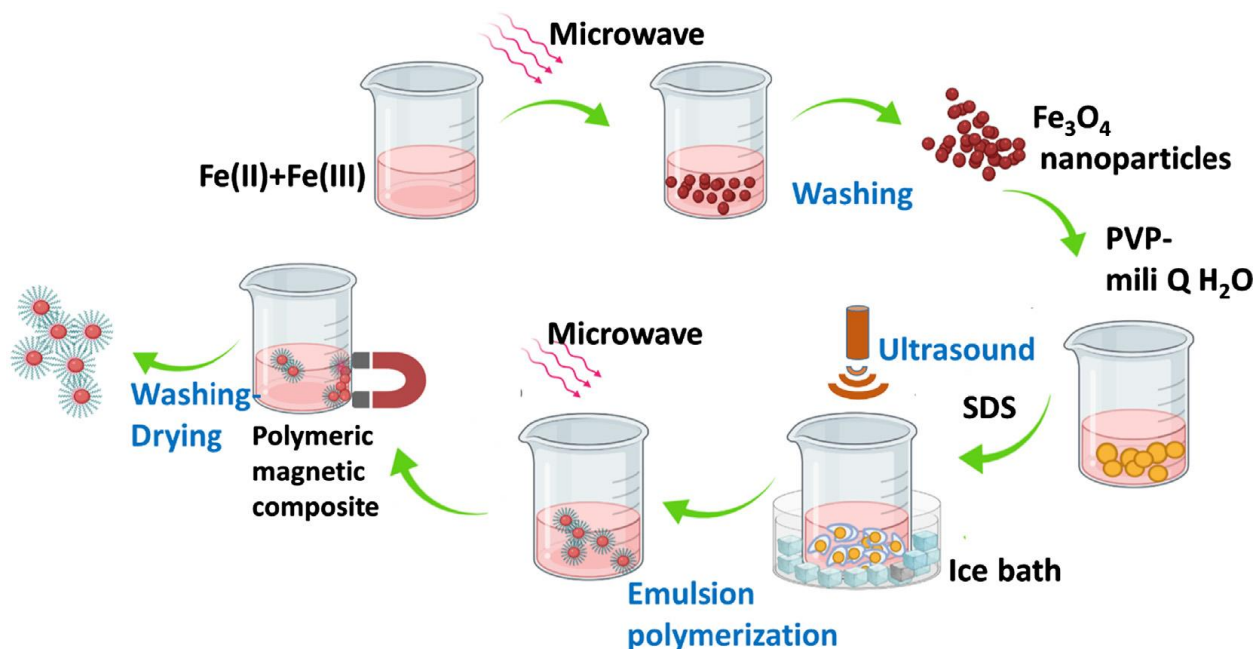


Figure 8 Diagram illustrating the microwave-assisted synthesis of magnetite nanoparticles and $\text{Fe}_3\text{O}_4/\text{poly}(\text{styrene-divinylbenzene-acrylic acid})$

<https://doi.org/10.1016/j.eurpolymj.2017.11.005> , Jaiswal et al. (2018)

6.3 Green Synthesis

Green synthesis is an environmentally-friendly method that uses natural substances from plants to produce nanomaterials. This method avoids toxic chemicals and harsh conditions, making it safer for people and the planet.

Plant extracts contain natural molecules like flavonoids, terpenoids, and phenols that act as both reducing agents (which convert metal ions into nanoparticles) and stabilizers (which prevent nanoparticles from clumping). The process usually involves drying plant material, extracting these molecules with hot water, and mixing the extract with a metal salt solution to form nanoparticles (See Figure 9). Silver nanoparticles synthesized using green tea extract have been applied in water disinfection due to their strong antibacterial properties. In another study, gold nanoparticles produced with neem leaf extract were used to develop sensors that detect environmental pollutants with high sensitivity.

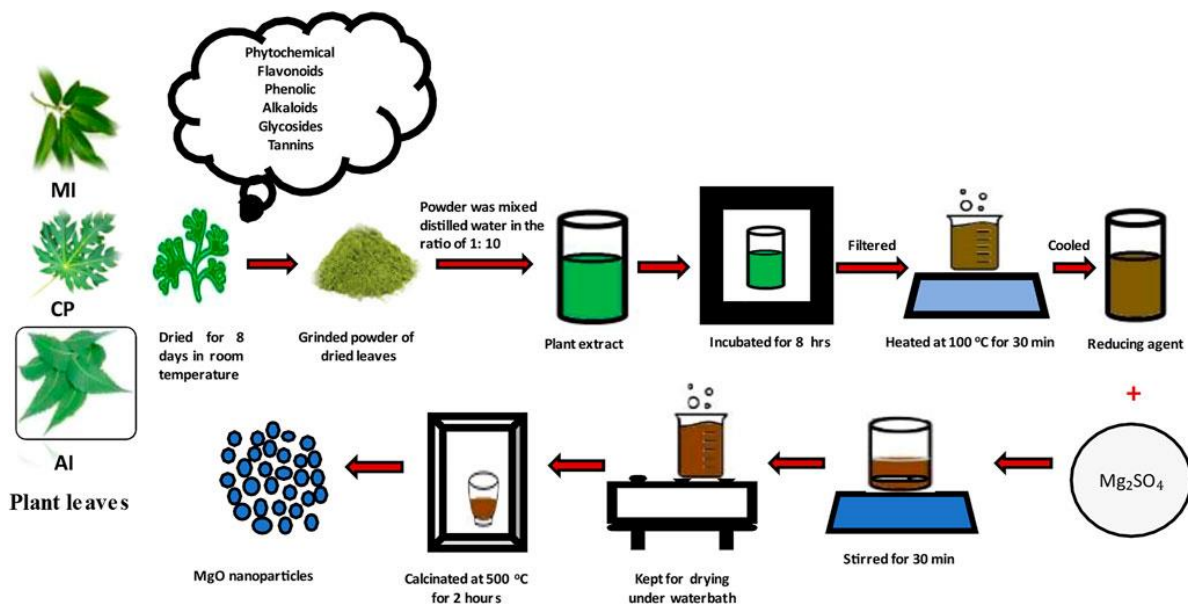


Figure 9 Green synthesis of MgO nanoparticles

(<https://doi.org/10.3389/fchem.2023.1143614> , Copyright 2023, Frontiers in Chemistry)

Advantages:

- Uses renewable, non-toxic materials
- Simple and low-cost
- Produces biocompatible nanoparticles suitable for medical and environmental uses

Limitations:

- Controlling particle size and shape can be challenging
- Scaling up production requires further research

6.4 Comparison of Synthesis Methods: Efficiency, Cost, and Environmental Compatibility

Top-down methods like ball milling are good for large-scale production but may use more energy and have less control over particle properties. Bottom-up methods offer precise control but can be more complex and costly. Green synthesis stands out as the most environmentally friendly, using natural materials to produce nanoparticles safely and sustainably (See Table 2).

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Table 2 Different Methods of Nanomaterials Synthesis with their properties

Method	Efficiency	Cost	Environmental Impact	Real-World Application Example
Ball Milling	High (bulk production)	Low	Moderate (energy use, wear)	Iron oxide nanoparticles for heavy metal removal
Laser Ablation	High purity	High	Low (no chemicals)	Silver nanoparticles for antibacterial water filters
Sol-Gel	Moderate to high	Moderate	Moderate (organic solvents)	TiO ₂ nanoparticles for wastewater treatment
Hydrothermal	High control	Moderate	Moderate (energy, pressure)	ZnO nanorods for dye degradation
Microwave-Assisted	High speed	Moderate	Low (energy efficient)	Magnetic biochar from waste for pollutant removal
Green Synthesis	Moderate	Low	Very low (eco-friendly)	Silver nanoparticles from plant extracts for disinfection

The synthesis of environmental nanomaterials involves a variety of methods, each with unique advantages and challenges. Top-down approaches such as ball milling and laser ablation are effective for producing nanoparticles from bulk materials, while bottom-up techniques like sol-gel, co-precipitation, and hydrothermal synthesis build nanoparticles from the ground up with precise control. Green synthesis using plant extracts offers a promising, sustainable alternative that aligns with environmental safety goals.

Understanding these methods through practical examples helps us appreciate how nanotechnology can be applied to real-world environmental problems, from cleaning polluted water to developing sensors for detecting contaminants. As you continue your journey in nanotechnology, knowing how nanomaterials are made will be key to innovating solutions for a cleaner, healthier planet.

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Chapter 7: Methods for Characterization and Analysis of Environmental Nanomaterials

Understanding the properties of nanomaterials is essential for their safe and effective use in environmental applications. Characterization is the process of examining nanomaterials to determine their size, shape, surface features, chemical composition, and stability. These characteristics influence how nanomaterials interact with pollutants, living organisms, and the environment. This chapter explains about the main methods used to characterize environmental nanomaterials that were produced in the previous chapter, with practical examples to show how these techniques are applied in real research and industry.

7-1 Microscopic Methods

Microscopy allows scientists to visualize nanomaterials at extremely high magnifications, revealing details that are invisible to the naked eye.

7-1-1 Scanning Electron Microscopy (SEM)

SEM uses a focused beam of electrons to scan the surface of a sample, producing detailed images of surface shape, texture, and size distribution. In environmental nanotechnology, SEM is often used to check if nanoparticles have the right size and surface structure for applications like pollutant adsorption. For example, SEM images of iron oxide nanoparticles used in water treatment reveal rough surfaces that increase the area available to trap heavy metals, helping researchers optimize synthesis methods for better performance.

7-1-2 Transmission Electron Microscopy (TEM)

TEM passes electrons through a very thin sample, providing images of the internal structure and precise size of nanoparticles. In studies of zinc oxide (ZnO) nanorods designed to degrade dyes in wastewater, TEM has shown that the length and diameter of the nanorods directly affect their ability to break down pollutants under sunlight. By adjusting synthesis conditions and confirming the nanorods' structure with TEM, researchers have improved the efficiency of pollutant removal.

7-1-3 Atomic Force Microscopy (AFM)

AFM uses a tiny probe that moves over a sample's surface to create a 3D map of its topography at the nanoscale. Unlike SEM and TEM, AFM can analyze samples in air or liquids, making it useful for studying nanomaterials in real environmental conditions. For instance, AFM has been used to examine the surface roughness of biochar nanoparticles used for removing heavy metals from water. The images showed that rougher surfaces improve adsorption capacity, guiding the design of more effective filters.

7-2 Spectroscopic Methods

Spectroscopy involves studying how nanomaterials interact with light or other electromagnetic radiation to reveal their chemical composition and structure.

7-2-1 Fourier Transform Infrared Spectroscopy (FTIR)

FTIR measures how nanomaterials absorb infrared light, identifying chemical bonds and functional groups on their surfaces. This is important because surface chemistry determines how nanoparticles interact with pollutants. For example, FTIR has confirmed the presence of plant-based molecules on silver nanoparticles made by green synthesis, which stabilize the nanoparticles and enhance their antibacterial properties in water purification.

7-2-2 X-ray Diffraction (XRD)

XRD identifies the crystal structure and purity of nanomaterials by measuring how X-rays scatter off atoms. The crystal structure influences material reactivity and effectiveness. In research on titanium dioxide (TiO_2) nanoparticles for pollutant degradation, XRD confirmed the presence of the anatase crystal phase, which is highly active in breaking down contaminants.

7-2-3 UV-Visible Spectroscopy (UV-Vis)

UV-Vis spectroscopy measures how nanomaterials absorb or scatter ultraviolet and visible light. It helps estimate nanoparticle size and concentration in solutions. For example, researchers have monitored the formation of gold nanoparticles synthesized with plant extracts by observing shifts in UV-Vis absorption peaks, allowing control over nanoparticle size for environmental sensing applications. Additionally, this method can be used to measure the concentration of contaminants in a solution, making it particularly useful for evaluating how effectively nanomaterials remove pollutants from aqueous environments by tracking changes in contaminant concentration before and after treatment.

7-3 Surface Analysis Methods

The surface of nanomaterials plays a critical role in their environmental behavior, so analyzing surface area and chemistry is essential.

7-3-1 Brunauer–Emmett–Teller (BET) Surface Area Analysis

BET measures the total surface area of nanomaterials by how much gas (usually nitrogen) they adsorb. A larger surface area usually means better pollutant adsorption. For example, BET analysis of graphene oxide nanosheets revealed a high surface area, making them highly efficient adsorbents for heavy metals in water treatment. Similarly, the surface area of TiO_2 nanoparticles, measured by BET, affects their photocatalytic activity in breaking down pollutants.

7-3-2 X-ray Photoelectron Spectroscopy (XPS)

XPS identifies elements and chemical states on nanoparticle surfaces by measuring electron energies emitted when X-rays hit the sample. This technique has been used to study the surface chemistry of nanoparticles after exposure to different environments. For instance, XPS analysis of

copper oxide nanoparticles revealed the presence of surface contaminants from the synthesis process, which can affect their environmental performance. XPS also helps confirm the successful functionalization of nanoparticles, such as the attachment of specific molecules to improve pollutant removal.

7-4 Zeta Potential and Dynamic Light Scattering (DLS) for Stability and Size

7-4-1 Zeta Potential

Zeta potential measures the electrical charge on nanoparticle surfaces in liquids. This charge affects stability—whether particles stay dispersed or clump together. For example, silver nanoparticles with zeta potential values greater than ± 30 mV are considered stable, as the electrostatic repulsion prevents aggregation. Stable dispersions are crucial for effective pollutant removal in water treatment.

7-4-2 Dynamic Light Scattering (DLS)

DLS estimates nanoparticle size distribution in suspension by analyzing how particles scatter light as they move randomly. DLS is widely used to monitor the size and aggregation of nanoparticles in environmental samples. For instance, DLS has been used to track the size and stability of magnetic nanoparticles in water, helping researchers optimize their use for pollutant removal.

7-5 Importance of Characterization in Environmental Applications

Characterization ensures that nanomaterials work safely and effectively in environmental settings. Knowing size and shape helps predict how nanoparticles move through water or soil and how organisms might absorb them. Understanding surface chemistry guides interactions with pollutants or living systems. Measuring stability ensures nanoparticles remain effective during storage and use. Confirming crystal structure links material properties to environmental functions like pollutant breakdown. Without thorough characterization, nanomaterials could be inefficient or even harmful when released into the environment.

Characterizing environmental nanomaterials involves a range of powerful techniques that reveal their size, shape, surface properties, chemical composition, and stability. Microscopic methods like SEM, TEM, and AFM provide detailed images, while spectroscopic methods such as FTIR, XRD, and UV-Vis help understand chemical structure and nanoparticle formation. Surface analysis tools like BET and XPS reveal surface area and chemical states, and zeta potential and DLS inform about stability and size in liquids. By combining these methods, scientists can design and use nanomaterials safely and effectively to address environmental challenges. Learning these techniques will help us see the careful work behind developing nanotechnology solutions that can improve our environment and everyday life.

Additional details can be found in the chapter on nanomaterial characterization methods, featured in the booklet available on the official Olympiad website.

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Section five: Applications of Nanotechnology in Removing Pollutants

Chapter 8: Detection and Removal of Water Pollutants

Nanotechnology has helped make big improvements in how we clean water. It has brought new ideas to solve the global water shortage. These improvements use materials made at a very small scale—called the nanoscale. At this size, we can control how small particles and molecules are separated from water.

When a material is made extremely small – on the order of just a few nanometers – its surface area soars compared to its volume. In other words, the surface-to-volume ratio becomes much larger. For example, only about 0.15% of the atoms in a 1 μm particle lie on the surface, but in a 6 nm particle nearly 20% of the atoms are surface atoms. This means nanoscale particles have vastly more “active sites” exposed where waterborne pollutants can attach. In turn, the material’s ability to adsorb or chemically react with contaminants rises dramatically. In fact, the large surface area of nanoparticles is key to their unusual chemical activity – chemical reactivity often increases at the nanoscale.(figure 10)

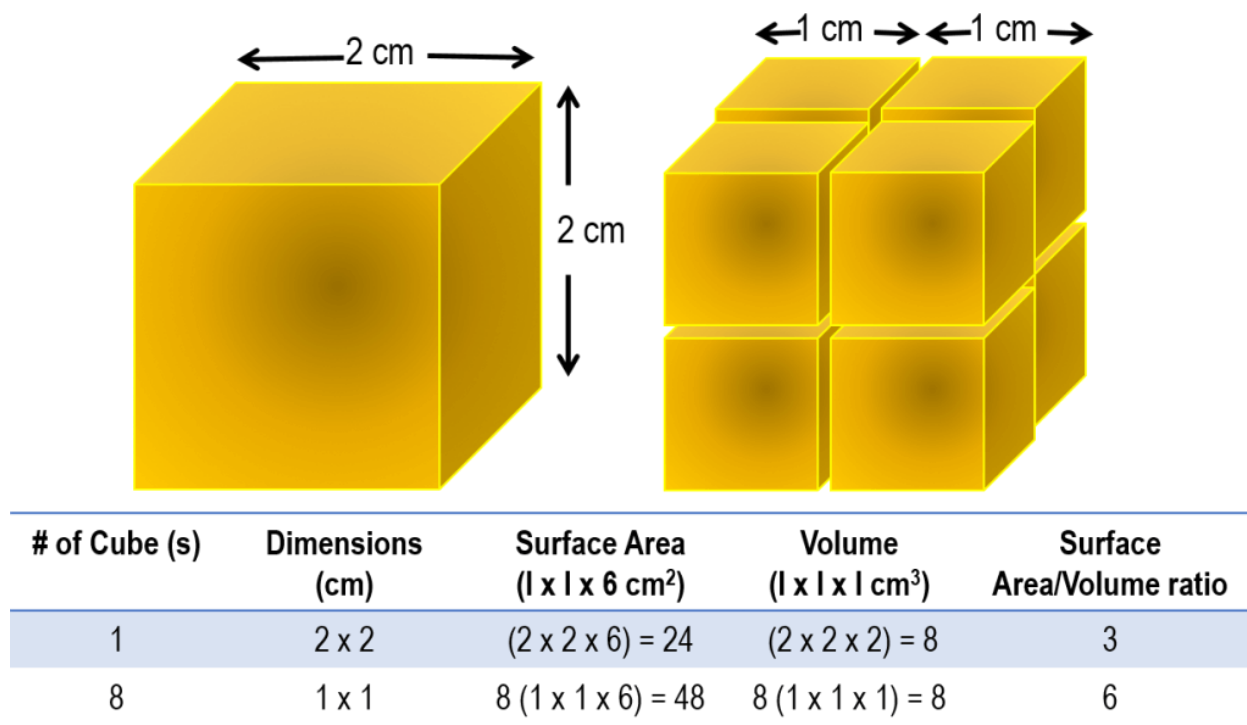


Figure 10- An illustration to depict the increase in surface area of a cube that accompanies a decrease in size.

In short, shrinking particles to the nanoscale reveals a much greater fraction of atoms to the environment, creating far more opportunity for molecules in wastewater to stick or react on the surface. Because nanoscale adsorbents present so much more surface to the water, they can capture contaminants both faster and in greater quantity. More surface area simply means more pollutant molecules can stick at once, so nanomaterials tend to have very high adsorption capacity.

Also, a small nanoparticle gives dissolved molecules a very short distance to travel before hitting the surface. This speeds up adsorption: in practice, nanoparticles often reach adsorption equilibrium in a much shorter time than larger particles, as pollutants find and occupy surface sites more quickly. Finally, the surfaces of nanoparticles often have many unsatisfied (high-energy) bonds, which strongly attract contaminants. Forces like Van der Waals attraction or electrostatic forces pull pollutants onto the high-energy surface more effectively.

Altogether, these effects – huge surface area, abundant active sites, and strong surface interactions – mean that nanoscale materials remove contaminants from water more efficiently and rapidly than their bulk counterparts.

8-1 Nanomembranes for Water Purification

In this part, we will learn about different types of nanomembranes based on what they are made of and the size of the holes in them. We will also learn how each type works and what it is used for.

8-1-1 Classification Based on material of nanomembranes

In a basic classification based on the material of nanomembranes, three advanced types of nanoscale membranes including nanostructured ceramics, organic-inorganic hybrids, and biomimetic systems are widely used in applications to remove pollutants.

Nanostructured Ceramic Membranes:

1. Zeolite Membranes: Zeolite membranes have an ordered crystal structure and are a great example of nanotechnology in materials. One type, called LTA zeolite, has more pores and lets water pass through more easily than other types. These membranes work using three main methods:

- Molecular sieving: This is based on the size of the ions (tiny charged particles). For example, sodium ions, which are about 0.36 nanometers wide, can pass through LTA zeolite pores that are 0.41 nanometers wide.
- Ion exchange: This happens by swapping aluminum and silicon atoms in the structure, which changes the pore size and surface charge.
- Competitive adsorption: Pollutants stick to the charged surfaces inside the membrane.

An important note: MFI-type zeolite crystals need to be lined up with the flow of water to work well. But LTA zeolite does not need to be aligned this way.

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2. Photocatalytic Membranes: These are polymer membranes coated with TiO_2 (titanium dioxide) nanoparticles, which are semiconductors. They clean pollutants in four steps:

1. UV light hits the membrane and creates electron-hole pairs (written as e^-/h^+).
2. Special chemical traps like oxygen and hydroxyl groups catch the charges and stop them from coming back together.
3. Reactive oxygen species (ROS) such as hydroxyl radicals are made.
4. These ROS attack the cell walls of germs and break down harmful organic compounds.

The advantage of using nanoparticles is that both the electron and the hole can reach the surface and take part in reactions. In larger materials, usually only one of them works effectively.

Organic–Inorganic Hybrid Membranes:

1. Mixed Matrix Membranes (MMMs): These membranes are made by mixing inorganic nanomaterials like zirconia and alumina into polymer materials like polysulfone. They offer two main advantages:

- They create special fast paths for water when the amount of filler material reaches a certain level (usually more than 15%).
- They help control pore morphology. For example, zirconia particles that are 0.9 micrometers wide create a more even distribution of holes by disturbing the interface between the materials.

2. Smart Nanofillers:

- Magnesium oxide (MgO) nanoparticles destroy the cell walls of both Gram-positive and Gram-negative bacteria.
- Zero-valent iron nanoparticles can clean up harmful chemicals like PCBs (polychlorinated biphenyls).
- Silver nanoparticles stop germs from growing and reduce biofouling (clogging caused by microbes).

Biomimetic Membranes (Membranes that Copy Nature):

1. Aquaporin-based Systems: Aquaporins are water channels found in living cells. They have two amazing features:

- Super selectivity: Each aquaporin can let 2 to 8 billion water molecules pass through every second.
- Osmotic flow: Water moves through them without needing high pressure. Scientists think that membranes with 75% aquaporin content can work 10 times better than regular reverse osmosis membranes.

Figure 11 represents an Aquaporins based membranes for water purification.

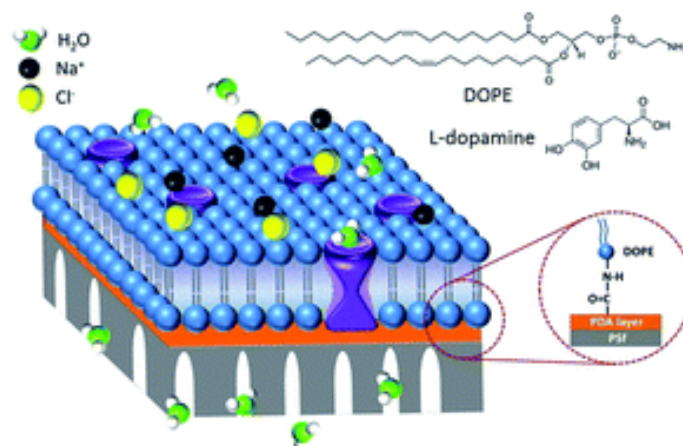


Figure 11. Fabrication of an aquaporin-based forward osmosis membrane through covalent bonding of a lipid bilayer to a microporous support

2. Carbon Nanotube Membranes: Carbon nanotubes stand straight up in these membranes and help water flow in two amazing ways:

- They reduce friction because of their smooth atomic surface, which lets water molecules slide through easily.
- They save 20% more energy because they need less pressure than normal membranes.

To make these membranes, we need special production methods like Chemical Vapor Deposition (CVD) to line up the nanotubes properly.

8-1-2 Classification Based on Chemical Uniformity:

Membranes can be grouped by how even their chemical composition is:

- Isotropic membranes: These have the same chemical makeup all the way through.
- Anisotropic membranes: These have different chemical or physical properties in different parts.

Types of Isotropic Membranes:

1. Microporous membranes with pore sizes from 0.1 to 5 micrometers. These are usually made by hitting thin plastic films with heavy metal ions. This creates small, controlled holes.
2. Dense nanostructured films: In these, particles move through by molecular diffusion, and movement depends on pressure gradient, concentration, or electric fields.
3. Charged membranes: These are made to repel ions with the same charge. For example: Cationic membranes have negatively charged pore walls, so they block anions (negatively charged ions). Anionic membranes do the opposite, blocking cations.

Types of Anisotropic Membranes:

Role of Nanotechnology in Sustainable Environmental Management

1. Phase separation membranes: although these membranes are uniform in chemical composition, but their pores are not evenly spread across the thickness of the membrane.
2. Composite membranes: These are different in both chemistry and structure. They are made using advanced methods like surface polymerization or plasma coating.

8-1-3 Membrane performance based on pore size

it is divided into five categories:

- Microfiltration membranes with pores between 0.1 and 5 micrometers. These can remove suspended particles, blood cells, and large bacteria.
- Ultrafiltration membranes with pores between 0.01 and 0.1 micrometers. These can filter large molecules like proteins and viruses, in addition to larger particles.
- Nanofiltration membranes with pores between 0.001 and 0.01 micrometers. These are between ultrafiltration and reverse osmosis and can remove specific ions like calcium and magnesium to reduce water hardness.
- Reverse osmosis membranes with very tight pores (0.0001 to 0.001 micrometers). These allow only very small particles to pass. In reverse osmosis, water molecules dissolve into the membrane material and diffuse through it under pressure gradient, while larger molecules and ions are rejected. These membranes can remove up to 99% of salts, minerals, and organic molecules.

8-1-4 Classification based on their material: organic or inorganic

1. Inorganic membranes: Ceramic membranes like TiO_2 and Al_2O_3 are strong, heat-resistant, and less likely to clog. A new type of ceramic membrane is the photocatalytic TiO_2 membrane, which can break down organic pollutants and kill microbes when UV light shines on it. Hybrid materials like $\text{TiO}_2\text{-SiO}_2$ can filter up to 94% better. Their super water-attracting (super hydrophilic) surface reduces how much pollution sticks to them.
2. Organic membranes: These are made from polymers like polysulfone and polypropylene. They are widely used but naturally repel water, which means they can clog easily with proteins or organic matter. Some new solutions are: changing the surface to make it more water-friendly, making it smoother, changing its charge, and using smart polymers that react to the environment.
3. Hybrid membranes: These combine polymeric materials with nanoparticles like Al_2O_3 or carbon nanotubes using methods such as phase inversion or in-situ polymerization. These special membranes can increase water flow by 40%, decrease clogging by 60% and make the membrane 3 times stronger.

In general, nanotechnology membranes have changed the world of water purification by working at the tiny molecular level. In the future, they will keep helping solve global water problems. LTA zeolite membranes with low-density structure are useful for turning salty water into fresh water. Photocatalytic membranes that clean themselves can greatly lower operating costs. Biomimetic

systems like aquaporins (water channels in cells) are great examples of how we can save energy in filtration.

8-1-5 Future improvements

- Increasing energy efficiency by up to 50% in reverse osmosis by changing membrane nanostructure.
- Making multi-purpose membranes that can filter, kill germs, and break down pollutants all at once.
- Designing smart surfaces that react to changes in the environment.
- Lowering production costs by 30% using self-assembling nanoparticles, which will help make these technologies available around the world.

Future research should focus on making it easier to produce membranes on a large scale, making them last longer, and balancing cost with performance. These advancements will not only help solve the water crisis but also show how nanotechnology, materials science, and environmental engineering work together to build a better future.

8-2 Photocatalytic Materials for Removing Organic Pollutants in Water Treatment

Photocatalytic processes are a special group of advanced oxidation technologies (AOPs) that play an important role in cleaning water. This method is very useful because it can completely break down toxic organic pollutants into harmless products like CO₂ and H₂O. (Figure12) It can also break down stable compounds, works well at room temperature, doesn't need oxygen to be added, and costs less money. Because of all these benefits, it is a better option than older methods like using ozone, activated carbon, or biological treatments. In this method, nanostructured semiconductors are used because they have a high surface area, more active sites, and better ability to speed up reactions.

The general basics of the photocatalytic process were explained in Chapter 5.

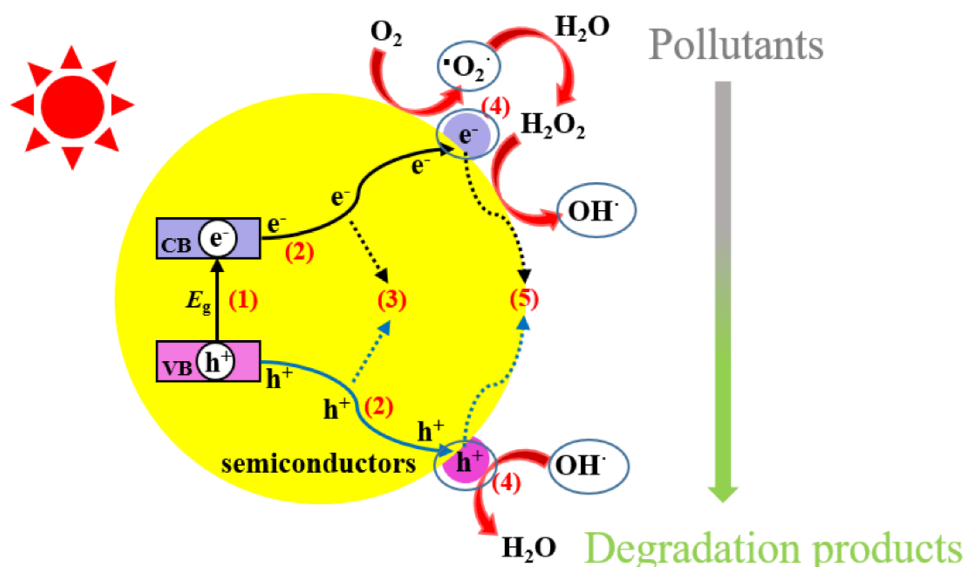


Figure 12 photocatalytic reactions

8-2-1 Types of Photocatalytic Materials

Semiconductor materials have electronic valence and conduction bands separated by a band gap (E_g), which is the minimum energy to excite an electron from bound states into conducting states. At the nanoscale, quantum confinement alters these bands: when a crystal's size approaches the exciton Bohr radius, its valence and conduction band edges shift to higher energy and split into discrete levels, effectively widening the band gap. In practice this means nanocrystals (quantum dots) show size-dependent optical and electrical properties – for example, smaller dots emit bluer light because their band gap is larger. Quantum dots of different sizes fluoresce in different colors under UV light, reflecting the changing band gap as the particle shrinks. These principles of band structure and quantum confinement underlie the unique conductivity and optical behavior of semiconducting nanomaterials.(figure 13)

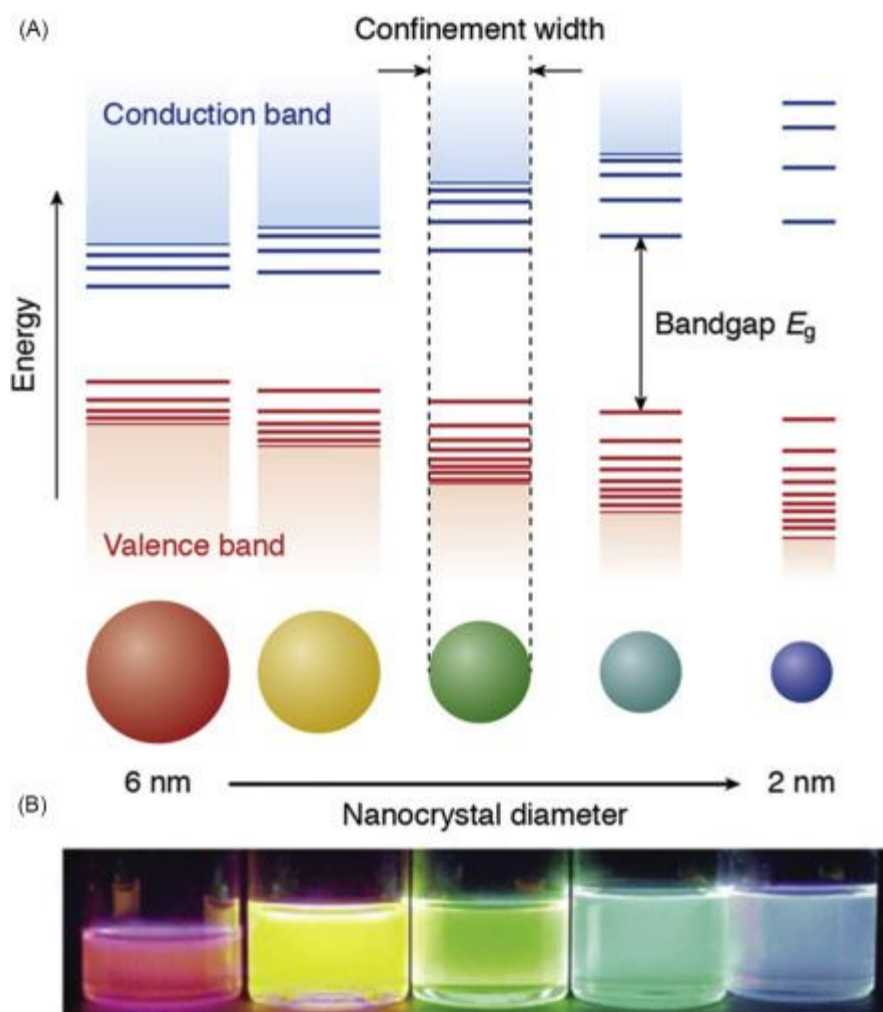


Figure 13-Schematic representation of the quantum confinement effects

In photocatalytic water treatment, semiconductor nanomaterials (e.g., TiO_2 or ZnO) use their band gap to absorb light and generate reactive charge carriers. When photons with energy equal to or greater than E_g hit the nanoparticle, electrons are excited from the valence band into the conduction band, leaving behind holes. These photogenerated electrons and holes migrate to the surface and produce radical species (for example, $e^- + \text{O}_2 \rightarrow \bullet\text{O}_2^-$ and $h^+ + \text{H}_2\text{O} \rightarrow \bullet\text{OH}$), which then oxidize and break down organic pollutants. Common photocatalysts are wide-band-gap metal oxides like TiO_2 (anatase ~ 3.2 eV) or ZnO , prized for their stability and low cost. At the nanoscale, these materials have enormously increased surface area and tunable band structures: more atoms at the surface and quantum-size effects improve light absorption and charge separation. In summary, by engineering a semiconductor's band gap and using nanostructures, photocatalysts can more effectively harvest light and accelerate the degradation of wastewater contaminants.

Common Semiconductors

- TiO_2 (Titanium Dioxide): It has an energy gap of 3.2 eV and works under UV light. It is chemically stable, non-toxic, and cheap. But it only uses about 4–5% of sunlight, which is a limitation.
- ZnO (Zinc Oxide): It has an energy gap of 3.37 eV and works well for removing dyes. One of its problems is that it dissolves in acidic conditions.
- CdS (Cadmium Sulfide): It has an energy gap of 2.4 eV and works under visible light. But it contains cadmium, which is toxic, and it is not stable under light.

Ways to Improve Performance

- Doping: This means adding small amounts of other elements. Adding metal ions like Fe^{3+} or Cr^{3+} (called cationic doping) reduces the energy gap so the material can use visible light. Adding nonmetals like N, S, or C (called anionic doping) creates new energy levels in the gap.
- Dye sensitization: This means attaching visible light-absorbing dyes to the surface of semiconductors like TiO_2 .
- Combining Semiconductors: Joining TiO_2 with semiconductors that have a smaller energy gap, like CdS , can help transfer electric charges better.

Factors That Affect How Well Pollutants Are Broken Down:

8-2-2 Environmental Factors

- Pollutant concentration: When there is too much pollutant, the surface of the catalyst becomes full, and light cannot pass through well. This makes the breakdown rate go down.
- Catalyst concentration: Adding more catalyst up to an optimal level (usually 0.5–2 g/L) increases the number of active sites. But if too much is added, it makes the water cloudy and blocks light.
- pH level: It changes the surface charge of the catalyst. For example, the surface charge of TiO_2 becomes zero at around pH 6.3. It also affects how pollutants behave and how energy bands shift.
- Temperature: The best range is 20–80°C. If it's too hot, electrons and holes recombine faster, which reduces performance.

8-2-3 Physical and Chemical Factors

- Particle size and shape: Tiny particles (less than 20 nanometers) with more surface area can absorb more light and pollutants.

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- Ions in the water: Metal ions like Fe^{3+} and Cu^{2+} compete with pollutants for space on the catalyst surface. Anions like Cl^- , SO_4^{2-} , and PO_4^{3-} stick to the surface and lower the activity of the reaction.
- Dissolved oxygen: It is very important because it catches electrons and stops recombination.
- Light intensity and exposure time: At low light levels (under 20 mW/cm^2), higher light means faster reactions. But at very high levels, recombination increases and the performance stops improving.

8-2-4 Photocatalytic Reactor Design

There are two main types of reactors: suspension reactors and fixed-film reactors.

- Suspension reactors: The catalyst particles float in the water. They have a big contact area with the water, but it's hard to remove the particles after the process.
- Fixed-film reactors: The catalyst is attached to solid surfaces like glass, metal, or ceramics. It's easy to recover the catalyst, but the contact area is smaller.

Light source: UV lamps (254–365 nm) or visible light can be used. Efficiency can be improved using reflectors or special windows made of quartz for UV light or Pyrex glass for visible light.

Operating conditions like flow rate, pollutant concentration, and layer thickness all affect the system's performance.

8-2-5 Real Applications in Water Treatment

Removing Organic Pollutants

Industrial dyes like methylene blue and rhodamine can be broken down more than 95% using TiO_2 under UV light. Pesticides and chlorinated hydrocarbons can also be changed into CO_2 and short-chain acids. Phenol-based compounds are broken down by hydroxyl radicals into quinones and carboxylic acids, which helps the environment.

Benefits of Nanotechnology

TiO_2 nanoparticles have a surface area of up to $250 \text{ m}^2/\text{g}$, which helps the reaction happen faster. Doped nanoparticles can work with visible light. For example, nitrogen-doped TiO_2 has an energy gap of 2.8 eV and works under sunlight. Nanocomposites (like combining TiO_2 with graphene or metal nanoparticles) help reduce recombination and improve the reaction.

Challenges in Industrial Use

- Recovering the catalyst: Suspension systems need membrane filtration (like ultrafiltration) to separate the catalyst after the reaction.
- Light source cost: Using sunlight is a low-cost solution, especially in sunny countries like Iran with 300 sunny days a year.
- Scaling up: We need to design larger reactors that are still efficient.

Nanostructured photocatalysts are ideal for cleaning water polluted with harmful organic compounds. They work well with sunlight, break down pollutants completely, and are chemically stable. To succeed with this technology, we need to focus on three things:

1. Choosing the right material: Make semiconductors with small energy gaps that can use visible light.
2. Controlling operating conditions: Adjust pH, catalyst concentration, and oxygen levels.
3. Reactor design: Improve light use and lower energy waste.

8-3 Nano adsorbents for Removing Heavy Metals from Water

Adsorption is one of the simplest, most effective, and cheapest methods for removing pollutants from water. Nano adsorbents use special features of nanomaterials like very high surface area (more than 500 square meters per gram), many active sites, and the ability to be chemically modified for special purposes. Because of this, they can remove heavy metals like lead, cadmium, arsenic, and chromium with very high efficiency. This technology has become very important in solving global problems like the lack of safe drinking water and pollution from industrial wastewater. The scientific basics of adsorption are explained in Chapter 5.

8-3-1 Types of Nano adsorbents and How They Work

- **Carbon-Based Nano adsorbents**

Carbon nanotubes (CNTs) are hollow tubes with high surface area. They have four different areas where pollutants can be adsorbed: inside the tubes, between the tubes, on the outer grooves, and on the outer surface. Adsorption happens faster on the outside. These nanotubes can remove 80% to 99% of heavy metals from water. The main ways they work include filling of holes, π - π interactions (for pollutants with ring structures), and ion exchange—especially after the surface is modified with oxygen-containing groups like carboxyl (-COOH) and hydroxyl (-OH).

The graphene family includes graphene, graphene oxide (GO), and reduced graphene oxide (rGO). GO has many polar (charged) groups like carboxyl and epoxide, which make it very good at adsorbing positive metal ions like Pb^{2+} . It can remove up to 95% of lead. On the other hand, rGO has fewer oxygen groups and is more water-repellent. It works better for nonpolar organic pollutants like methyl orange, removing around 95% of it.

- **Metal Oxide Nano adsorbents**

Magnetic iron oxide nanoparticles (Fe_3O_4) are naturally magnetic, so they can be removed from water easily using an external magnet. These are especially made to remove harmful forms of arsenic (As^{3+} and As^{5+}). They work mainly by electrostatic attraction and forming surface chemical complexes.

Advanced examples include composites like g- C_3N_4 with β -cyclodextrin. Computer studies (DFT theory) show that lead ions (Pb^{2+}) form strong chemical bonds with these materials. Methyl orange is also adsorbed using hydrogen bonds and π - π interactions.

- **Hybrid and Polymeric Nano adsorbents**

Carrageenan/silica nanocomposites combine natural polymers with mineral nanomaterials. They can adsorb high amounts of positively charged dyes like methylene blue (about 184 mg per gram). They mostly work through electrostatic attraction and hydrogen bonding.

Modified chitosan, a natural polymer, has many amino groups ($-\text{NH}_2$), which can bind with heavy metals. When chitosan is combined with nanomaterials like silica nanoparticles, it becomes stronger and can adsorb more pollutants.

8-3-2 Key Challenges and Technological Solutions

One major challenge is separating nano adsorbents from water after use because the particles are very small. Different solutions have been developed:

- Magnetic nanoparticles like Fe_3O_4 can be pulled out using a magnet.
- Carbon nanotubes and graphene sheets can be separated using a high-speed centrifuge (20,000 to 50,000 rounds per minute).
- Polymeric nano adsorbents can be attached to larger supports like fiberglass or polypropylene, making them easier to use in industrial columns.

Recovering and reusing nano adsorbents is important to reduce costs. The most common method is changing the pH of the solution:

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- For positively charged metals like Pb^{2+} and Cd^{2+} , acidic solutions (using diluted HCl or HNO_3) are used to remove the metals from the adsorbent.
- For negatively charged ions like CrO_4^{2-} , basic solutions (using NaOH) are used.

Studies show that after 5 cycles of use and regeneration, the adsorbent still works well, with less than 6% loss in its capacity.

One big challenge is the high cost of production, especially for single-walled carbon nanotubes, which are about 1,000 times more expensive than regular activated carbon. Promising solutions include:

- Developing low-cost production methods (like using pyrolysis of polypyrrole),
- Using agricultural waste (like rice husks or sugarcane ash) as raw materials,
- Improving recycling methods to make the adsorbents last longer.

8-3-3 Future Perspectives and Research Directions

Scientists are working on designing smart nano adsorbents that can react to changes in the environment like pH, temperature, or magnetic fields. These materials will be able to adsorb specific pollutants under ideal conditions and release them when environmental conditions change.

There is also interest in developing multi-functional nano adsorbents by combining different chemical groups. For example:

- Thiol groups for mercury (Hg),
- Amine groups for transition metals,
- Phosphonate groups for uranium (U).

Another important area is combining nano adsorbents with advanced water treatment technologies like nanostructured membranes, photocatalytic systems, and pressure-driven membrane processes (such as ultrafiltration). This will allow for multi-step treatment systems with very high efficiency.

Overall, nano adsorbents—with their high adsorption capacity, fast reaction time, and ability to be specially modified—are bringing major improvements to water treatment technologies. For this technology to succeed, we need:

- A deep understanding of how adsorption works at the molecular level (using computer tools like DFT),
- The design of smart materials that are mechanically and chemically stable,
- Practical solutions for separation, recovery, and affordable scaling.

Future research should focus on making nano adsorbents more environmentally friendly, lowering production costs using renewable resources and creating standard testing methods to check their safety and performance under real-world conditions.

8-4 The Use of Nanosensors in Detecting Water Pollutants:

Water pollution, especially from heavy metals like mercury, lead, cadmium, and also toxic organic compounds from industrial and farming activities, is now recognized as one of the most serious threats to human health and the environment. These pollutants can cause serious, irreversible damage to living systems even at very low concentrations, leading to chronic diseases such as cancer. In this situation, nanotechnology offers new and cost-effective solutions and plays a revolutionary role in water monitoring and purification. Nanosensors, as one of the most effective tools in this technology, use the special properties of materials at the nanoscale to quickly, accurately, and selectively detect a wide range of pollutants even at very low levels, such as parts per billion (ppb) or parts per million (ppm).

Figure 14 briefly shows the components of nanosensors used in environmental monitoring.

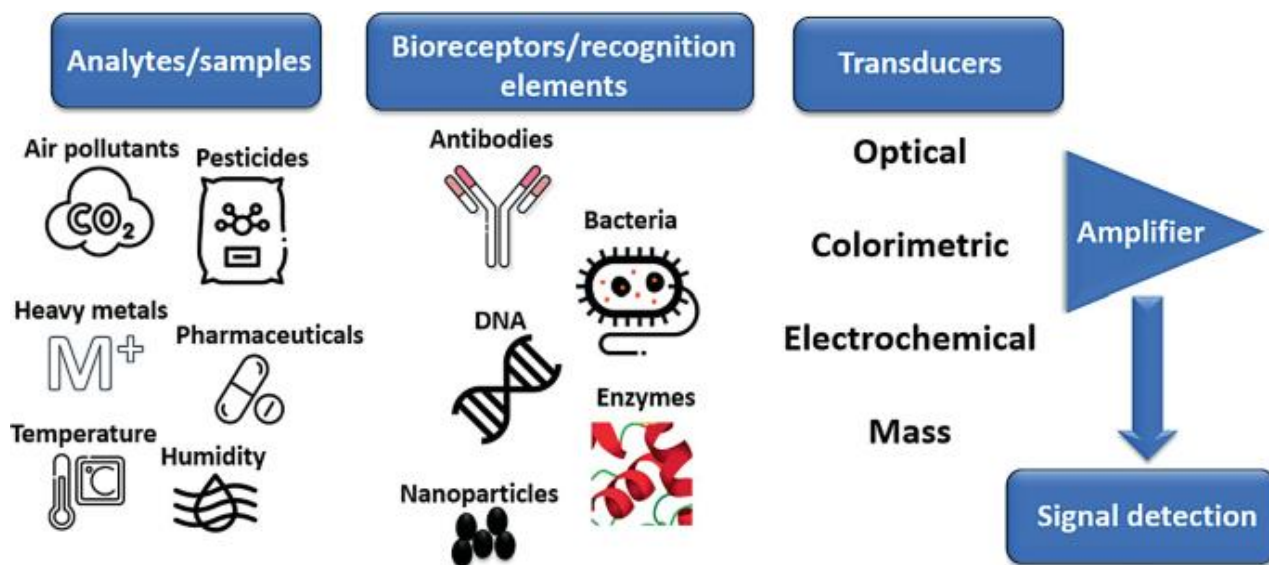


Figure 14 Components of nanosensors used in environmental monitoring

Nanosensors are sensors at the nanometer scale that have extremely high sensitivity and selective performance. They can detect very small changes caused by the presence of specific pollutants in water. These sensors are made from various nanostructured materials like metal nanoparticles (gold, silver), single-walled carbon nanotubes, quantum dots, magnetic nanoparticles, and especially advanced core-shell nanocomposites (for example, gold-silica or silver-silica). These materials are chosen because of their exceptional properties, such as very high surface area for interacting with pollutants, catalytic and photocatalytic activity, antimicrobial properties, and most

importantly, unique optical and electronic properties that form the basis of many detection mechanisms.

8-4-1 Key mechanisms for detecting pollutants using nanosensors

1. Detection based on optical properties and surface plasmon resonance: This is the most common and powerful mechanism in nanosensors. Metal nanoparticles, especially gold and silver, show a phenomenon called surface plasmon resonance (SPR). This happens when the electrons on the nanoparticle surface oscillate together in response to visible light, causing strong absorption of light at specific wavelengths and producing specific colors (for example, bright red for gold nanoparticles). The presence of heavy metal ions like Cd^{2+} , Hg^{2+} , Pb^{2+} , or certain pollutant molecules can affect this phenomenon in two main ways:
 - Nanoparticle aggregation: Metal ions can act like a bridge between nanoparticles and cause them to clump together. This aggregation strongly affects the surface plasmon properties and causes a clear change in the absorption wavelength, resulting in an observable color change in the solution. For example, gold-silica core-shell nanocomposites strongly aggregate even at 0.01 ppm of cadmium ions, and this change can be easily detected by optical spectroscopy or sometimes even by the naked eye. This color or spectrum change directly indicates the presence and amount of the pollutant.
 - Electron transfer and local environment change: Some ions like Hg^{2+} chemically bind directly to the surface of metal nanoparticles. This binding can cause electron transfer from the ion to the nanoparticle or vice versa. This electron transfer changes the density of free electrons in the nanoparticle and shifts the resonance frequency of the surface plasmon, which again appears as a change in the absorption spectrum or color. Nanosensors made from gold nanowires use this mechanism to detect very low amounts (ppb) of mercury.

Surface-enhanced Raman spectroscopy (SERS): Metal nanoparticles (especially silver and gold) can strongly amplify the electromagnetic field of light on their surface. When a pollutant molecule (like pesticides or dyes) is near this surface, its Raman spectroscopy signal can be enhanced by millions of times. This allows detection of single molecules and precise identification of the type of organic pollutant even at very low concentrations.

2. Electrochemical detection: Some nanosensors (like sensors based on carbon nanotubes or nanostructured platinum electrodes) work by measuring changes in electric current or potential caused by the interaction of pollutants with the sensor surface. The presence of pollutants may cause electron transfer, surface resistance changes, or a specific electrochemical reaction, which can be recorded as an electric signal. For example, sensors made from nanoporous platinum electrodes deposited on silicon accurately measure the dissolved oxygen (DO) in water through oxygen reduction electrochemical reactions.

Sensors based on single-walled carbon nanotubes also have very high sensitivity to gases like NO₂ and NH₃ even at room temperature due to their high electrical conductivity and large surface area.

3. **Biological detection:** This method uses biological components like enzymes, antibodies, or DNA attached to nanoparticles (usually magnetic or fluorescent). These components interact specifically with the target pollutant (for example, a microbial pathogen or a specific toxin). This interaction can cause changes in the optical properties (like fluorescence turning on or off) or magnetic properties of the nanoparticle, which is recorded as a detection signal. These sensors are very sensitive and selective for monitoring microbial contamination and organic toxins.

8-4-2 Wide applications of nanosensors in water quality monitoring

Real-time monitoring of heavy metals: Nanosensors, especially types based on surface plasmon resonance (like gold-silica nanocomposites) and electrochemical sensors, can detect toxic heavy metal ions like lead, cadmium, mercury, arsenic, and chromium with very high sensitivity even at ppb levels, often selectively in drinking water, industrial wastewater, and natural water sources. This allows continuous monitoring and fast warning of pollution.

Measurement of nitrate and phosphate: High concentrations of nitrate and phosphate (which cause eutrophication) in surface and groundwater are a big problem. Optical sensors based on nanocomposites (like silver nanoparticles in a polymethacrylate acid matrix) change their color or absorption spectrum in the presence of these ions, enabling fast and cost-effective measurement with high sensitivity.

Measurement of dissolved oxygen (DO) and pH: Advanced electrochemical sensors based on nanostructured platinum electrodes can precisely and continuously measure the vital amount of dissolved oxygen in water. Also, specific triboelectric or optical sensors have been developed for continuous monitoring of water pH, which is essential for controlling treatment processes and the health of aquatic ecosystems.

Detection of pesticides and insecticides: Sensors based on SERS and biological sensors are powerful tools to detect very low amounts and identify specific types of pesticides (like organophosphates) and insecticides in agricultural water and water sources.

Monitoring microbial contamination: Biological nanosensors using antibodies or specific DNA sequences attached to fluorescent or magnetic nanoparticles can rapidly and sensitively detect bacteria, viruses, and other microbial pathogens in water. This is vital for the safety of drinking water and aquaculture pools.

Control of water distribution networks ("smart pipes"): A new and important application is using arrays of nanosensors (combining sensors for flow, pressure, chemical quality, and microbes)

inside or on water pipes. These smart systems allow real-time and continuous monitoring of key parameters throughout the distribution network. These sensors can detect water flow rate, pressure inside pipes, stagnant or low flow points, exact leak locations, and most importantly, changes in chemical and microbial water quality (such as sudden pollutant entry or chlorine level drop) without disturbing the water flow. This enables optimal management of the network, reduces water loss, and ensures quality water delivery to consumers.

In conclusion, nanosensors, by using advanced detection mechanisms based on nanomaterial properties (especially surface plasmon resonance, SERS, electrochemistry, and biosensing), have created a great revolution in water quality monitoring. Their ability to detect a wide range of dangerous pollutants (heavy metals, nitrate/phosphate, pesticides, pathogens) and vital parameters like DO and pH quickly, sensitively, selectively, and often cheaply, makes them essential tools to ensure safe drinking water, manage industrial wastewater, protect aquatic environments, and optimize water distribution networks. Future research aims to make these sensors more stable, reusable, able to detect multiple pollutants at once, and integrated into automatic monitoring systems, which are important directions in this vital field.

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Chapter 9: Removing Soil Contaminants

Soil is one of the most important parts of our environment. Clean, non-polluted soil is very important for increasing agricultural productivity, making crops healthier, and protecting the health of people and nature. That's why cleaning up polluted soil is a very important issue. Nanotechnology has always been a useful tool to help solve environmental problems, and polluted soil is no exception. In this chapter, the most common soil pollutants are introduced, and then nanotechnology-based methods for cleaning them up are explained.

Because of industrial development in recent years, soil—one of the most valuable parts of nature—is getting polluted quickly. The dumping of chemicals from factories into the ground and the uncontrolled use of fertilizers, pesticides, and other farming chemicals are the main causes of soil pollution. The most important soil pollutants are heavy metals, metalloids, petroleum products, and pesticides.

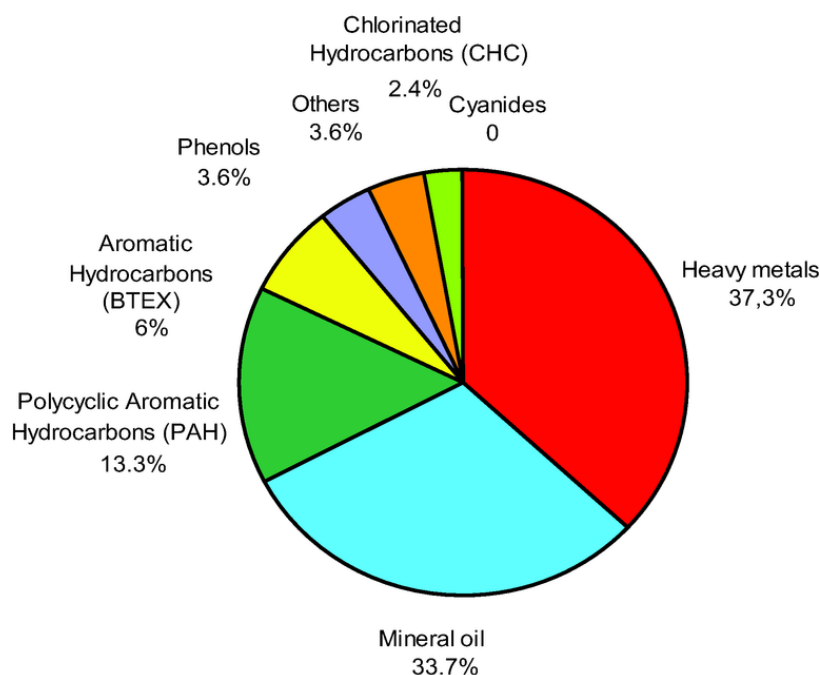


Figure 8 soil pollutants in Europe in 2024

Figure 15 shows soil pollutants in Europe in 2024, where 37.3% of the pollutants were heavy metals. Polluted soils can affect human health, ecosystems, farming, and the environment as a whole. So, cleaning polluted soil has become a major issue in environmental technologies. In this article, we will explore different ways of removing these pollutants and the role of nanotechnology in this field.

9-1 Types of Soil Cleanup Methods

This section briefly explains the main ways to clean up polluted soil.

- **Physical methods:** Physical cleanup includes methods like soil replacement and thermal desorption. In soil replacement, the polluted soil (or part of it) is replaced with clean soil. In thermal desorption, the soil is heated using direct heat or waves to make the pollutants unstable and able to evaporate. Then, gases carry these pollutants away. This method is mostly used for heavy metals like mercury and arsenic.
- **Chemical methods:** Chemical cleanup includes chemical washing, stabilization, and electrokinetic treatment. In chemical washing, the soil is washed with water, chemical solutions, or gas, which removes the pollutants. In stabilization, substances are added to the soil that react with the pollutants to turn them into non-toxic or immobile forms, stopping them from reaching plants or water. In electrokinetic treatment, two electrodes are placed in the soil, and voltage is applied. This creates an electric field that moves pollutants toward the electrodes and helps reduce their toxicity. (Figure19)

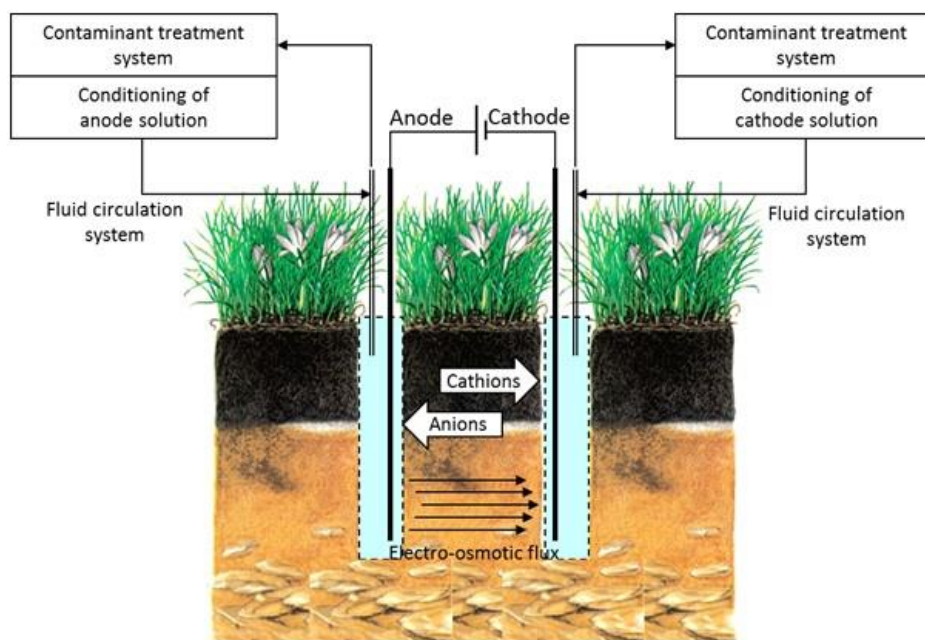


Figure 10 Principle of Electrokinetic Remediation (Cameselle et al., 2013).

- **Biological methods:** Biological cleanup uses living plants to absorb pollutants and clean the soil. Microorganisms are also used to stop the movement of pollutants within the soil.

9-2 Nanotechnology and Its Use in Soil Cleanup

Nanomaterials have a small size, many different shapes, and a very high surface area, which helps them react more, absorb better, and work as strong catalysts. In recent years, scientists have focused on using nanotechnology for cleaning water, air, soil, and more. Nanotechnology-based methods are effective, low-cost, and environmentally friendly. During the cleaning process, most reactions happen on the surface between the nanomaterial and the pollutant. Because nanomaterials have a large surface area and many active sites, the cleanup happens quickly and effectively. Their small size also helps them easily reach polluted areas. Nanoparticles can also be modified on their surface or structure to improve their performance. They can also be used for detecting pollutants, not just removing them.

In many parts of the world, both "on-site" and "off-site" cleanup methods are used. For on-site cleanup, a liquid form of nanoparticles (colloidal or water-based) can be sprayed or injected into the polluted soil using gravity or pressure. Once injected, these nanoparticles stay suspended and create a cleanup zone.

9-3 Types of Soil Pollutants and How Nanotechnology Helps

- **Heavy Metals:**

Heavy metals are the main pollutants in soil, so cleaning them is very important. These metals include arsenic, lead, cadmium, chromium, copper, mercury, nickel, and more. Nanoparticles can effectively remove them from soil. The main cleanup methods are **adsorption** (trapping pollutants on the surface) and **stabilization** (changing them into harmless forms). In general, materials used in cleanup are divided into two types: **adsorbents** and **reactive materials**.

Adsorbents trap pollutants on their surface or inside their structure. Adsorption is one of the best methods for removing heavy metals because it's simple, cheap, and effective. It is used for removing metal ions like copper, mercury, cadmium, and chromium. The effectiveness of this method depends on the material used. The small size and large surface area of nanomaterials make them excellent for this purpose.

Reactive materials remove pollutants using chemical reactions like acid-base, redox, precipitation, ion exchange, and photocatalysis. These reactions turn toxic substances into less harmful ones.

Nanoparticles made of zero-valent iron (nZVI) are especially useful. Iron is cheap, non-toxic, and widely available. These iron nanoparticles can easily reach polluted areas and stay stable in the soil. They can react with heavy metals like mercury, nickel, and cadmium to reduce their toxicity. Their surface area allows them to contact more pollutants and clean more efficiently. These particles are also corrosion-resistant and can adjust their chemical state in different environments.

Role of Nanotechnology in Sustainable Environmental Management

The main way nZVI works is by adsorbing the heavy metals and converting them into less toxic and more stable forms. The success of this depends on the standard electrode potential of each metal. Some metals like cadmium and zinc stick to the iron surface. Others like chromium, arsenic, and copper get reduced and form a solid in the soil. (Figure 17)

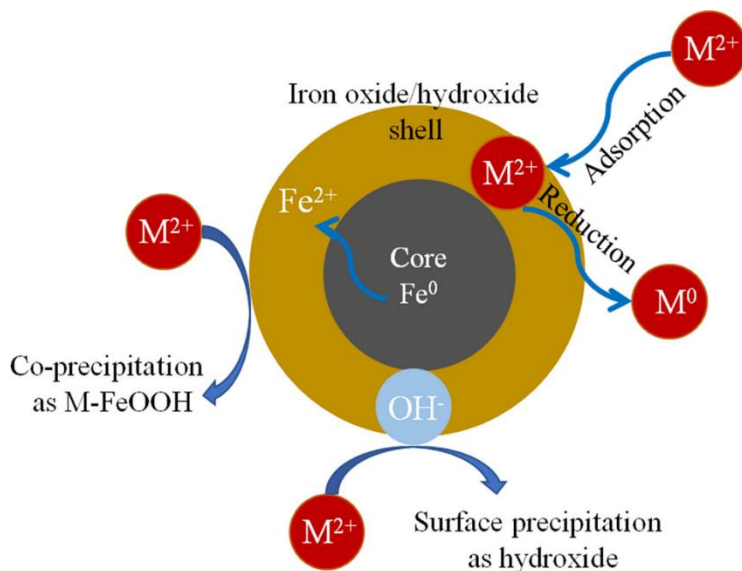


Figure 11 shows a core-shell structure of a zero-valent iron nanoparticle and how it breaks down pollutants.

One problem with nZVI is that it can oxidize quickly because of its high surface area. Over time, a layer of iron oxide forms, which reduces its cleaning ability. However, this layer still allows electrons to pass through and lets the nanoparticle keep its reducing power. It also helps the nanoparticle react with organic compounds. To slow down this oxidation, the particles are sometimes coated with other materials, like noble metals (e.g., palladium) or metal oxides. Adding small amounts of other metals boosts their activity, since the added metal acts as a catalyst.

nZVI particles tend to clump together, which limits their use. To prevent this, their surfaces are modified using polymers or surfactants, which help keep them stable and mobile. In one study, nZVI particles stabilized with carboxymethyl cellulose (CMC) were used to remove cadmium. These particles stayed stable in water and moved more easily in soil, improving cadmium cleanup.

In another study, a combination of electrokinetic method and a special barrier filled with nZVI converted 88% of hexavalent chromium into the less harmful trivalent chromium.

nZVI particles were first made in 1997 by reducing Fe²⁺ with NaBH₄. However, this method is expensive and only used in labs. Another method uses Fe(CO)₅, but it creates toxic byproducts. The industrial method involves reducing FeOOH or Fe₂O₃ with hydrogen gas at high temperatures.

Role of Nanotechnology in Sustainable Environmental Management

Although injecting nanoparticles into soil and groundwater raises concerns about their toxicity, studies show that nZVI is usually not toxic by itself. In fact, in real environmental conditions, the particles oxidize faster than in labs, reducing the risk. But they can still carry toxic substances long distances, which needs more research.

nZVI is useful not only for cleaning heavy metals but also for removing organic pollutants and certain anions (negatively charged ions). These include:

- Organic compounds: chlorinated solvents, phosphorus-organic compounds, chlorophenols, antibiotics, organic dyes, pesticides, and nitroamines
- Metals: alkaline earth metals, chromium, cobalt, copper, lead, molybdenum, nickel, silver, technetium, vanadium, zinc, cadmium, arsenic, and selenium
- Inorganic anions: nitrate, perchlorate, bromate

Iron sulfide nanoparticles are also effective, especially for removing mercury. When stabilized with CMC, these particles move well in soil and remove mercury through adsorption, precipitation, and stabilization.

Although phosphate-based materials like phosphoric acid and natural phosphate rocks have been used to clean heavy metals, they have some limits. Since iron is cheap, non-toxic, and abundant, it can solve these problems. Iron phosphate nanoparticles are safer for the environment and are often used to remove copper.

Iron oxide nanoparticles are strong adsorbents of heavy metals. Adding zirconium makes them more stable and increases their power to remove arsenic and lead. Other materials used include allophane (a type of clay) for copper and carbon black for copper and zinc.

- **Petroleum products and organic pollutants:**

Soil pollution from petroleum includes harmful substances like benzene, toluene, ethylbenzene, and xylene (BTEX). Other pollutants include polycyclic aromatic hydrocarbons (PAHs). The most common method for cleaning them is to use nZVI or photocatalytic materials to break chemical bonds, open benzene rings, and neutralize the pollutants. Nanoparticles like titanium dioxide (TiO_2), zinc oxide, and copper oxide, when exposed to light, can move electrons to their conduction bands and destroy or neutralize pollutants.

As industries grow and cities expand, soil pollution is increasing rapidly. The main pollutants are heavy metals and petroleum-related compounds. Traditional cleanup methods include physical, chemical, and biological techniques. But with the arrival of nanotechnology, more efficient methods are now available. Nanoparticles clean the soil by adsorbing or breaking down the

pollutants. The most common ones used are nZVI, iron sulfide, iron oxide, and photocatalyst nanoparticles.

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Chapter 10: Removing Air Pollutants

Air is one of the main elements for the survival of living beings, including humans. Therefore, protecting it from different types of pollution—whether caused by natural processes or by human activities—is very important.

The growing problems caused by air pollution have led scientists to use advanced, modern technologies to help nature and our environment. One of these modern solutions is nanotechnology and its applications in protecting the environment.

10-1 Types of pollution and their effects on health

In general, sources of pollution are created in two ways: natural and artificial. **Natural sources:** dust, smoke, and carbon monoxide from fires; radon gas from minerals in the Earth

Artificial sources: pollutants, disease-causing agents

Researchers say that right now, there are many uses of nanostructured materials to protect the environment. These include nanofilters to clean industrial wastewater, nanopowders to purify polluted gases from cars and factories, and nanotubes for storing very clean hydrogen fuel. The direct and indirect effects of nanotechnology on the environment and air pollution can be studied from different viewpoints, and the future use of this new technology is very broad.

10-2 Methods to fight pollutants and disease-causing agents

Scientists have always been inspired by nature in their research. Most natural living systems operate at the nanoscale. By using nature's nanoscale designs, researchers can, in the lab and using principles of physics, chemistry, engineering, and nanotechnology, create and study molecules and biological tools at the nanoscale. This has led to a new field called nano-biotechnology. Some of the most important scientific applications of nanotechnology in the environment and air pollution are nanosensors, nanofilters, and environmental catalysts.

One key use of nanotechnology in environmental protection is nanofilters that clean air and remove harmful pollutants. Silver nanofilters, for example, use the antibacterial properties of silver nanoparticles to sterilize public environments and remove bacteria-based pollutants. These filters interfere with bacterial cell metabolism, killing them and helping prevent diseases. The nanoparticles in these filters are about 550 nm in size and can remove more than 99% of bacteria. They also work well in hot or cold environments.

Carbon nanofilters, which use carbon particles sized 100–400 nm, are very good at capturing smells and volatile organic compounds (VOCs). Because they have a very large surface area, they are up to 15 times more effective than standard activated carbon. They can remove both polar and non-polar gases, and their active surface area is about 1,000 m² per gram.

Hybrid nanofilters are advanced filters made using a mix of organic and inorganic nanoparticles. They can both ventilate and purify air at once. Using particles like TiO₂ and copper, these filters can destroy resistant bacteria such as *Escherichia coli* and *Staphylococcus* with over 99% efficiency. They are also antifungal, stop mold growth, remove gases like ammonia and VOCs, are washable, stable, and electrostatic—so they capture dust and suspended particles too.

Another new method is nano-photocatalyst filters. These use stable titanium dioxide (TiO_2) nanoparticles to oxidize pollutants and kill microbes in the presence of ultraviolet light. Polluted air is pulled in by a fan, goes through several filter layers under UV light, and then is cleaned before it returns to the environment. These filters remove VOC gases and kill bacteria. The TiO_2 particles are about 5–10 nm in size, have a large surface area ($\sim 2,300 \text{ m}^2/\text{g}$), and use light energy to continuously oxidize and regenerate.

In addition, smart nano-coatings are applied to building exteriors, roads, and other city structures—especially in busy areas. These coatings use light-activated semiconductor metals like TiO_2 , zinc oxide (ZnO), and tungsten trioxide (WO_3). TiO_2 is most common because it is chemically stable, not toxic, and affordable. Activated by sunlight, these coatings turn pollutants like nitrogen oxides (NO_x) and VOCs into harmless carbon dioxide (CO_2) and water (H_2O).

There are also nano-air-purifying paints, made with acrylic or silicone resins and nanotechnology. Besides looking nice, they are water-repellent, dust-repellent, weather-resistant, and self-cleaning. When they absorb sunlight and contact moisture, they produce reactive radicals like hydroxyl and peroxide that quickly break down air pollutants into CO_2 and water. The nanoparticles act as catalysts and are not used up, so the cleanup continues as long as there is light and moisture. These paints are also antibacterial, deodorizing, and can decompose formaldehyde, benzene, VOCs, and fungi—making them effective for improving urban air quality.

In recent years, companies and research centers around the world have been developing and marketing nanotech-based air-cleaning products. One successful example is “Noxout”, the world’s first air-purifying paint. Developed through EU research projects and produced with help from the Philippines and UK-based Cristal Active, it uses photocatalytic chemistry to turn harmful nitrogen oxides (NO and NO_2) from vehicle exhaust, factories, and power plants into harmless substances. This makes Noxout effective at reducing air pollution.

Another innovative product is Air Scrubber Plus, which uses nanotube-coated TiO_2 honeycomb cells. In the presence of light, water, and oxygen, it activates photocatalysis to remove bacteria, viruses, and other pathogens from the air. Designed for homes, offices, and industrial spaces, it is a successful commercial nanotech solution for public health and improving indoor air quality. It removes 99.9% of surface bacteria.

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One recent innovation in personal protective equipment is a nano-mask developed in Hong Kong using nanofibers. These nanofibers are the main material of the filter and, because of their high surface area and density, can trap air pollutants very effectively. They especially catch aerosol nanoparticles—dust, vehicle exhaust particles, and industrial emissions that enter our lungs. While ordinary masks filter about 25% of these tiny particles, nanomasks can capture up to 80% without affecting breathing. This makes them safe and efficient for use in polluted cities and industrial places. They can also filter out bacteria and viruses smaller than 1 μm , and can be used as gas filters to clean air polluted with nitrogen oxides (NO_x) and VOCs—making them a powerful tool for personal health and environmental protection. (Figure 13)

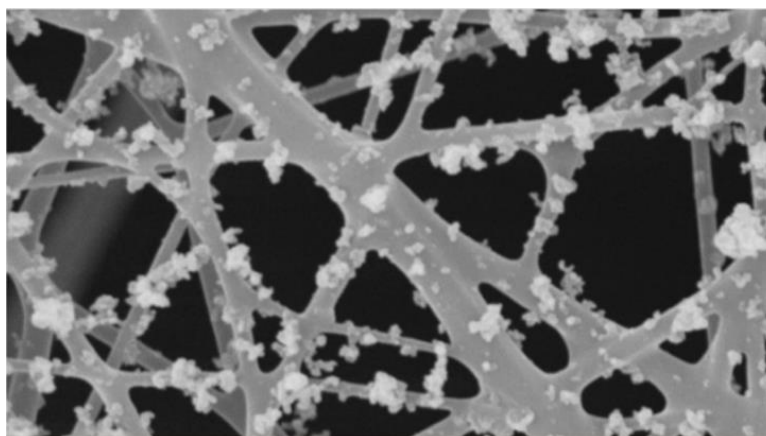


Figure 12 Microscopic image of nanofibers

In efforts to improve diesel fuel and reduce harmful emissions, researchers have created a nanocatalyst made of cerium dioxide and amide-functionalized multi-walled carbon nanotubes. This catalyst was tested in two fuel types—5% biodiesel + 95% diesel (B5), and 20% biodiesel + 80% diesel (B20)—at concentrations of 30, 60, and 90 ppm added to the fuel.

Using the cerium oxide–carbon nanotube nanocatalyst in B5 and B20 fuels at 30, 60, and 90 ppm reduced emissions of carbon monoxide (CO), unburned hydrocarbons (HC), nitrogen oxides (NO_x), and soot particles. [5] The experiments showed that the amide groups on the nanotubes match the biodiesel's molecules, helping the catalyst mix evenly in the fuel. This boosts the catalyst's stability and improves combustion. As a result, harmful emissions significantly decreased.

The studies show that nanotechnology, thanks to the unique properties of nanoscale structures, has a great potential to reduce air pollution and improve environmental quality. Using nanoparticles in air filters, anti-pollution coatings, and combustion catalysts has proven much more effective than traditional technologies. Silver, carbon, hybrid, and photocatalyst nanofilters use processes like surface adsorption, oxidation, antibacterial action, and photocatalysis to remove over 99% of pollutants, bacteria, and VOCs.

Nanofiber masks can filter 80% of aerosol particles without affecting breathing, enhancing respiratory safety in cities and factories. Cerium oxide–carbon nanotube catalysts distribute better

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in fuel and improve combustion while reducing harmful CO, HC, NO_x, and soot emissions from diesel engines.

Overall, the scientific evidence highlights that nanotechnology is not just a helpful tool but a fundamental solution for managing air pollution and advancing future environmental technologies. Continued interdisciplinary research—especially on safety, biodegradability, and industrial scalability—is essential to fully harness its potential.

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Section5: Safety and sustainable development

Chapter 11: Safety of Nanomaterials in the Environment and Health

11-1 Toxicity of Nanomaterials for Humans and the Environment

Nanotechnology, as one of the leading scientific fields of the 21st century, has created major advancements in various areas such as environment, medicine, energy, agriculture, and industry. However, with the rapid growth in the production and use of nanomaterials on an industrial and consumer scale, attention to their safety aspects has become a scientific and ethical necessity.

Nanomaterials, due to their extremely small size (less than 100 nanometers), have unique physical and chemical properties that make them different from regular materials. These properties, on one hand, increase their efficiency in removing pollutants, delivering drugs, improving mechanical properties of materials, and electronic applications; but on the other hand, they raise concerns about their unpredictable behavior in the environment and inside living organisms.

Unlike many common chemical substances, nanomaterials—because of their high surface area, high mobility, ability to penetrate cell membranes, and resistance to biological degradation—may accumulate in food chains or cause toxic and unwanted reactions in cells. Therefore, careful evaluation of the environmental and biomedical effects of nanomaterials is essential for the responsible and sustainable development of this technology.

International organizations like the OECD, WHO, and REACH have provided guidelines and legal frameworks to ensure that while keeping the benefits of nanotechnology, the long-term risks to human health and the environment are avoided. For this reason, learning about nanomaterial safety and understanding risk assessment principles are vital parts of any nanotechnology education program.

In this section, we focus on the physical, chemical, and surface properties of nanomaterials that determine their toxicity. Then we explore how different types of nanomaterials can affect humans, aquatic organisms, and plants. These factors include size, shape, surface charge, surface coating or modification, chemical composition, solubility, and particle aggregation behavior. Each of these properties can influence how much of the nanomaterial is absorbed by cells, what kind of cellular response it causes, and how it behaves in the environment.

11-1-1 Properties That Influence Nanomaterial Toxicity

1. Size

Size is one of the most important factors in the toxic behavior of nanomaterials. The smaller the particle size, the larger the surface area relative to its volume, which increases its reactivity. Nanoparticles smaller than 10 nanometers especially have a high ability to penetrate cell membranes and may reach organelles like mitochondria or even the nucleus. This penetration can lead to the production of reactive oxygen species (ROS) and cause oxidative stress.

Also, smaller nanoparticles can pass through biological barriers such as the blood-brain barrier or the placenta and may cause long-term effects on the nervous system or fetal development. For example, nano TiO₂ with a size of 5 nanometers has been shown to be more toxic than the same particle at 30 nanometers.

2. Shape

The shape of nanoparticles affects how they interact with cells and tissues. Nanorods, nanotubes, nanosheets, and nanospheres show different behaviors. For example, multi-walled carbon nanotubes (MWCNTs) with long lengths can act like asbestos fibers and cause severe inflammatory reactions in lung tissues.

Needle-shaped nanomaterials may physically pierce cell membranes and be more damaging than spherical particles. Additionally, sheet-like particles such as graphene or graphene oxide can form layers that block cell respiration or ion exchange. (Figure 19)

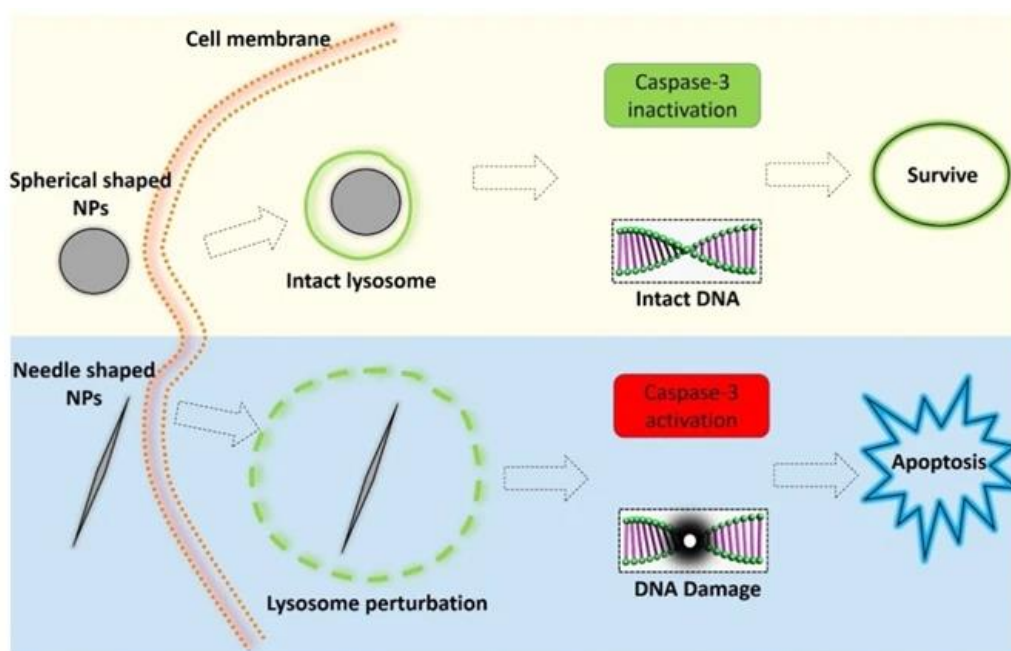


Figure 19 Schematic representation of the proposed cytotoxicity pathways for PLGA-PEG NPs with different morphological features

3. Surface Charge

The electric charge on the surface of nanoparticles determines how they interact with cells and biological molecules. Positively charged nanoparticles are more easily absorbed by cells due to the electrostatic attraction with the negatively charged cell membrane, and thus they usually show higher toxicity.

This charge can change depending on the environment (pH, ions, presence of proteins). For example, chitosan nanoparticles have a positive charge and can sometimes damage bacterial cell membranes. On the other hand, neutral or negatively charged nanoparticles usually have lower cell uptake but may aggregate more in natural environments and show unpredictable behavior.

4. Surface Coating and Chemical Modifications

Coating nanoparticles with organic materials or polymers can affect their stability, solubility, movement, and toxicity. Uncoated nanoparticles are usually more reactive and less stable, and may interact with proteins and enzymes in biological conditions.

In contrast, biocompatible coatings like polyethylene glycol (PEG) or citric acid can prevent direct contact between nanoparticles and cells, reducing toxicity. Surface modifications can also help direct the nanoparticle to specific pollutants or target cells, reducing contact with healthy cells. Therefore, surface engineering is an important method to reduce unwanted biological effects of nanoparticles.

5. Chemical Composition

The main chemical composition of nanoparticles is a key factor in their toxicity. In general, metal-based nanomaterials (like Ag, Cu, ZnO) tend to release metal ions, which can be toxic. For example, silver nanoparticles (AgNPs) release Ag^+ ions that have antibacterial properties but may also harm human cells at high concentrations.

On the other hand, relatively inert nanomaterials like nano SiO_2 or nano TiO_2 usually have low toxicity in the dark, unless activated by specific light (like UV). Some nanocomposites that are made of different types of materials may produce unexpected combined effects, and these must be evaluated separately.

6. Solubility and Environmental Stability

Nanomaterials that dissolve easily in biological or aquatic environments may break down faster and have lower toxic effects. In contrast, insoluble and stable particles like nano TiO_2 or carbon nanotubes tend to accumulate and stay in the environment.

Some of these particles may remain in sediments, wastewater sludge, or soil, and may enter the food chain over time. Also, the reactivity of nanomaterials in the environment can lead to the creation of new intermediate or toxic compounds. For example, nano ZnO dissolves in water and releases Zn^{2+} ions, which at high concentrations are toxic to algae or plankton.

7. Tendency to Aggregate and Form Agglomerates

The way nanoparticles clump together in the environment also affects their toxicity. When nanoparticles stick together (aggregate or agglomerate), their effective size increases, which may prevent them from entering cells. However, in some cases, aggregation reduces movement and causes them to settle in water ecosystems.

On the other hand, stable particles that stay in solution may have longer contact with cells and therefore cause more toxicity. Environmental conditions like pH, water hardness, ion content, and the presence of organic materials all influence how nanoparticles aggregate.

11-1-2 Effects of Nanomaterials on Aquatic and Soil Organisms

With the increasing release of nanomaterials into the environment through industrial wastewater, consumer products, and agricultural use, there are growing concerns about their effects on natural ecosystems—especially aquatic organisms and soil-dwelling creatures. These organisms are the first to come into contact with nanomaterials released into nature and can be used as biological

indicators to assess the risks of nanomaterials at different ecosystem levels. In this section, we explore the biological effects of nanomaterials on algae, plankton, fish, earthworms, fungi, and soil bacteria.

In aquatic environments, nanoparticles enter lakes, rivers, and groundwater through different sources such as wastewater discharge, surface runoff, or release from products like sunscreens and water filters. These nanoparticles can mix with organic or inorganic substances and remain in either a stable or suspended form. One of the most important groups of aquatic organisms directly exposed to nanomaterials are algae and phytoplankton. They are at the bottom of the food chain and play a key role in primary production in aquatic ecosystems. Studies have shown that metal-based nanoparticles such as silver (AgNPs), zinc oxide (ZnO), and copper oxide (CuO) can inhibit algae growth, disrupt photosynthesis, and cause cell death. The mechanisms behind these effects often include the production of reactive oxygen species (ROS) and damage to the cell membrane. Next in the food chain are zooplankton and small crustaceans like *Daphnia magna*, which are primary consumers exposed to suspended nanomaterials. Research has shown that exposure to silver or titanium dioxide nanoparticles can reduce feeding rates, lower mobility, delay growth, and decrease reproduction ability in these organisms. Nanoparticles may also deposit on their gills, interfere with gas exchange, and even cause death. The toxicity of nanoparticles to crustaceans greatly depends on their concentration, surface coating, and particle size. For instance, polymer-coated nano TiO₂ has shown lower toxicity compared to uncoated pure TiO₂ nanoparticles.

Fish, as aquatic vertebrates, are also affected by nanomaterials, especially during the stages of egg development and larval growth. Many studies on zebrafish (*Danio rerio*) have shown that silver nanoparticles can cause developmental defects, lower survival rates, gill damage, liver inflammation, and oxidative stress. Since nanomaterials can pass through cell membranes, they may reach internal organs of the fish and then be transferred through the food chain to other organisms, including birds and humans. A major concern in this area is the phenomenon of **bioaccumulation** and **biomagnification**, where nanoparticles build up in organisms' bodies and increase in concentration as they move up the food chain.

In soil ecosystems, small organisms like earthworms, fungi, microalgae, and beneficial bacteria play a vital role in biological cycles, soil aeration, decomposition of organic matter, and nitrogen fixation. The entry of nanomaterials into the soil—through the use of sewage sludge, nano-based pesticides and fertilizers, or direct dumping of industrial waste—can disrupt this natural balance. Earthworms, as indicators of soil health, are directly at risk from nanoparticles. Experiments have shown that long-term exposure to ZnO or silver nanoparticles can cause weight loss, skin epithelial damage, reproductive problems, and early death in earthworms.

Nanomaterials may also affect the composition and activity of soil microbial communities. Nitrogen-fixing bacteria such as *Rhizobium* or *Azotobacter*, which are essential for soil fertility, may reduce their activity or die in the presence of silver or copper nanoparticles. Nanoparticles can interact with cell membranes or disrupt respiratory pathways, thereby inhibiting microbial growth. This not only damages the nitrogen cycle but also slows the natural breakdown of pollutants, leading to the accumulation of toxic compounds in the soil.

The type, size, and surface behavior of nanoparticles play a key role in how strongly they affect soil organisms. For example, nanoparticles modified with organic or biological coatings are less likely to attach to bacterial cells and usually show lower toxicity. In addition, natural organic compounds in soil such as humic acids or dissolved organic matter can react with nanoparticles and reduce their movement or neutralize them.

It is important to note that the toxicity of nanomaterials in natural environments is not always predictable and may vary depending on environmental factors like pH, temperature, organic substances, and the presence of other pollutants. Also, the chronic and long-term effects of nanoparticles on soil and water ecosystems still require further research. The results of such studies can help in creating safety policies, designing low-risk nanomaterials, and promoting safe and controlled use of nanotechnology in the environment.

11-1-3 Effects of Nanomaterials on Plants

Due to the increasing use of nanomaterials in agriculture, the environment, and industry, plants are among the first organisms in soil and aquatic ecosystems to come into direct or indirect contact with nanoparticles. Nanomaterials may enter plants through soil, irrigation water, or even air and can affect various aspects of growth, metabolism, physiology, cell structure, and gene expression. Understanding these effects is essential to ensure the safe use of nanotechnology in agriculture and environmental management.

Nanomaterials can enter plants either by being absorbed through the roots or by penetrating leaves and surface pores. Once inside the plant tissues, depending on their size, composition, surface charge, and coating, nanoparticles may move to different parts of the plant and cause either positive or negative effects. At low concentrations, some nanomaterials may promote growth, increase photosynthesis, improve resistance to environmental stress, and enhance the uptake of nutrients. However, at high concentrations or when toxic nanomaterials are present, they may inhibit growth, change root morphology, cause oxidative stress, and damage cell structures.

One major negative effect of nanomaterials is the disruption of water and nutrient uptake in plants. Some nanomaterials like TiO₂, ZnO, and Ag at high concentrations can alter the structure of root cell walls and reduce their permeability. As a result, the plant may lose its ability to absorb water and minerals, leading to physiological drought. Additionally, certain nanomaterials—especially silver nanoparticles—can produce reactive oxygen species (ROS), which damage cell membranes, DNA, proteins, and organelles, interfering with vital processes like respiration and photosynthesis. Studies have shown that nanoparticles can enter plant cells either actively or passively. Active entry usually involves endocytosis and transport through protein channels, while passive entry may occur through simple diffusion or interactions with organic soil compounds. Once inside the cell, nanomaterials may accumulate in vacuoles, chloroplasts, or even the nucleus. For example, gold nanoparticles (AuNPs) have been found to accumulate in the nuclei of *Arabidopsis thaliana* plant cells, changing the expression of genes related to growth and defense. These interactions may affect the cell cycle, cell division, or the expression of genes involved in photosynthesis and respiration.

Another known effect of nanomaterials on plants is changes in root morphology. Some metallic nanomaterials such as CuO and ZnO can cause root shortening, fewer lateral roots, and even tissue necrosis. These changes result from a combination of factors including direct toxicity of released metal ions, oxidative stress, and interactions with growth hormones such as auxins and gibberellins. Some studies also suggest that nanoparticles can disturb hormonal signaling or enzyme secretion, leading to abnormal plant growth.

Photosynthesis, a vital process in plants, is also affected by nanomaterials. Nanoparticles like TiO₂, which have photocatalytic properties, can enhance light absorption and speed up photosynthesis at low and controlled doses. However, at high doses—especially under UV light—these same particles can cause light stress, reduce chlorophyll production, damage chloroplast membranes, and disrupt the photosynthetic system. A reduction in chlorophyll and interference with the Calvin cycle has been observed in plants like tomatoes and rice when exposed to nano ZnO or nano Ag. Nanomaterials also influence plant oxidation and defense systems. In response to nanomaterial-induced stress, plants activate their antioxidant systems. Enzymes such as catalase, superoxide dismutase, and peroxidase increase their activity to fight ROS. However, if ROS production exceeds the plant's defense capacity, toxicity appears, resulting in early aging, cell death, or growth problems. These physiological responses have been clearly reported in crops like corn, beans, soybeans, and peas when exposed to silver and copper oxide nanoparticles. (Figure 20)

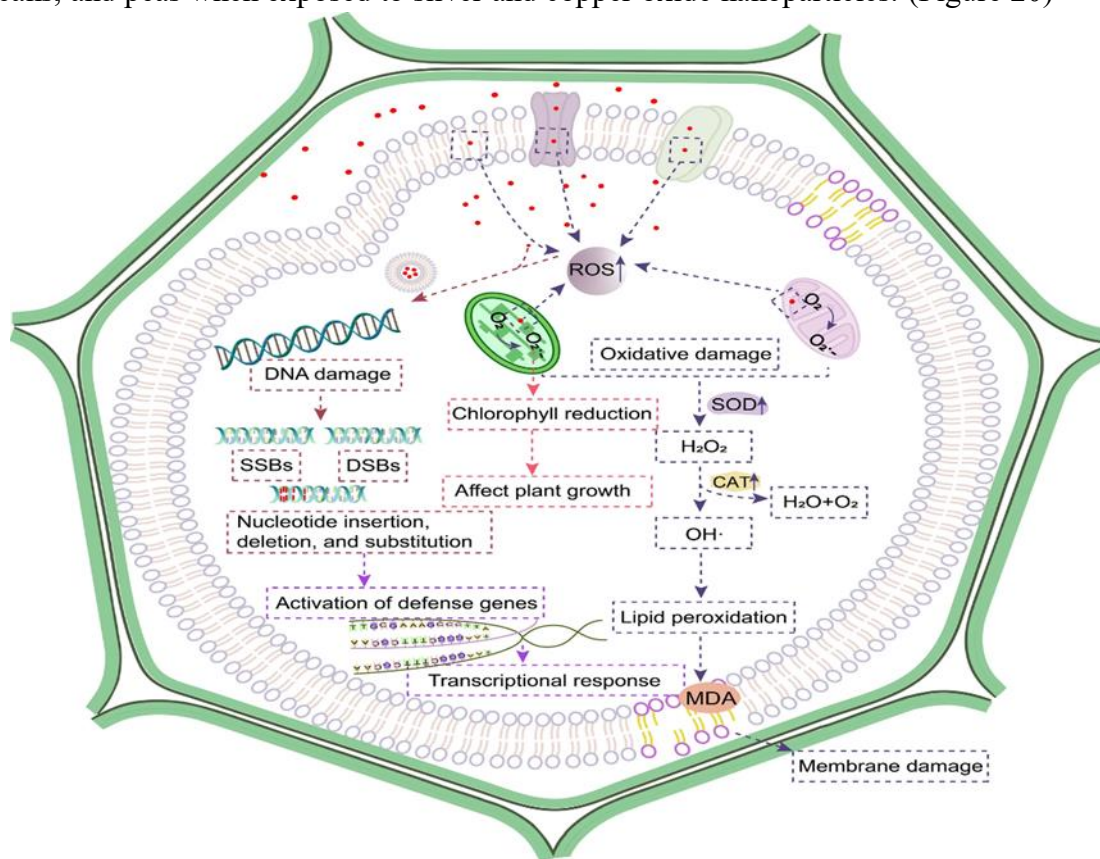


Figure 20. Mechanism of phytotoxicity induced by nanoparticles (NPs)

Figure 1: Mechanism of phytotoxicity induced by NPs. CAT = catalase, DSBs = double-strand breaks, MDA = malondialdehyde, ROS = reactive oxygen species, SOD = superoxide dismutase, SSBs = single-strand breaks. The red dots represent nanoparticles (NPs).

From a molecular point of view, nanoparticles can increase or decrease the expression of certain genes. For example, exposure to ZnO nanoparticles has been shown to increase the expression of defense-related genes like **PR1** and **NPR1**. Changes in the transcription of genes related to photosynthesis or hormone pathways have also been observed in some species when exposed to nano Fe₃O₄ or graphene oxide, showing that nanoparticles can affect plant genetic regulation. These changes may be temporary and reversible or, in some cases, long-lasting and permanent.

A key point in interpreting the effects of nanomaterials on plants is the interaction between the properties of the nanomaterial and the environmental conditions. For example, nanomaterials that are stable in neutral or alkaline environments may break down in acidic soil and release toxic metal ions. Also, in the presence of organic compounds, nanoparticles may form complexes (chelates), reducing their direct contact with plant cells. Therefore, by controlling environmental conditions, the toxicity of nanoparticles can be reduced and their harmful effects prevented.

Nevertheless, some studies have reported positive effects of nanomaterials on plants. For instance, nano iron (Fe₃O₄) and nano zinc (ZnO) at low concentrations have improved nutrient absorption, enhanced photosynthesis rates, and increased plant tolerance to drought and salinity. The use of nanomaterials in slow-release fertilizers and nano-carriers for pesticides, if used responsibly, may increase agricultural productivity and reduce input consumption. Such uses, when risk-assessed carefully, can support sustainable farming in the future.

In conclusion, the effects of nanomaterials on plants depend heavily on the material type, concentration, exposure time, plant species, environmental conditions, and soil characteristics. Many negative effects have been seen only in high concentrations or laboratory conditions, and with careful, well-engineered use in natural environments, nanomaterials can be applied safely.

11-1-4 Health Risks of Nanomaterials for Humans

While nanotechnology has brought amazing opportunities to improve human life—especially in fields such as medicine, energy, water purification, and environmental protection—there are growing concerns about the possible risks of nanomaterials to human health. Unlike larger materials, nanoparticles have a very small size, large surface area, and high reactivity, allowing them to enter the body, pass through biological barriers, and accumulate in organs. These properties make them useful for medical treatments and diagnostics, but without proper safety checks, they may also pose serious health risks.

Nanomaterials can enter the human body through various routes: inhalation, skin contact, ingestion, or direct injection. In non-medical uses like clothing, cosmetics, air filters, and food packaging, the main entry points are the skin and respiratory system. In medicine and pharmaceuticals, ingestion and injection are more common. Once inside the body, depending on features like size, surface charge, coating, and solubility, nanoparticles can spread through different tissues and may even cross barriers like the blood-brain barrier (BBB) or the placenta.

A major concern is the production of reactive oxygen species (ROS) and the creation of oxidative stress in human cells. Metal-based nanoparticles such as Ag, CuO, and ZnO can generate radicals when in contact with oxygen or water, damaging cell membranes, DNA, proteins, and mitochondria. These effects can lead to cell death (apoptosis), genetic mutations, or even cancer over time. For example, lab studies have shown that high doses of silver nanoparticles can cause double-strand breaks in DNA of human liver cells and interfere with cell cycle regulation.

The respiratory system is one of the most vulnerable systems to airborne nanoparticles. These tiny particles can be inhaled and reach the lung alveoli, causing inflammation, tissue damage, and abnormal immune responses. Nanoparticles with fiber-like structures—such as carbon nanotubes—can act like asbestos fibers, leading to lung fibrosis or immune cell activation in the lungs. Animal studies also show that nano TiO₂ may increase the risk of lung tumors in mice over time. However, this depends strongly on surface coatings and particle size; particles with inert or biocompatible coatings usually show lower toxicity.

Skin contact with nanomaterials is also a concern in cosmetics, sunscreens, and antibacterial fabrics. Although human skin normally acts as a strong barrier, if there are wounds, dryness, or inflammation, nanoparticles may enter more easily. Nano ZnO and TiO₂ are commonly used in sunscreens and are generally safe, but long-term exposure or UV light may damage skin cells like keratinocytes and cause structural changes.

The digestive system can also be exposed to nanomaterials through accidental or intentional ingestion—for example, through nano-treated water, packaged foods, or nano-based medicines. In the intestines, nanoparticles may react with digestive enzymes or food compounds, disturbing nutrient absorption. Some nanomaterials may also change the balance of gut microbiota, which can lead to chronic inflammation, metabolic issues, or weakened immunity.

Another concern is the crossing of the placenta and entry into the fetus. Nanoparticles smaller than 50 nm may pass through the placenta and build up in fetal tissues. Animal studies have shown that high doses of Ag or TiO₂ nanoparticles may reduce fetal weight, delay organ development, and harm the nervous system. Although human data is still limited, these findings suggest caution when using nanomaterials in products for pregnant women.

From an immune system perspective, nanomaterials can either over-stimulate or suppress immune responses. In some cases, nanoparticles activate immune cells like macrophages, leading to inflammation or increasing the risk of autoimmune diseases. Nanoparticles can also attract certain proteins to their surface, forming a layer called the “protein corona”. This can change how the immune system recognizes them—sometimes helping them hide from detection, and other times causing unwanted immune reactions.

11-2 Challenges of Nanomaterial Release and Accumulation in the Environment

Understanding the pathways through which nanomaterials enter the environment is a fundamental step in risk analysis and the development of control strategies. Nanomaterials may be released into the environment at various stages of their life cycle, including during production, transportation,

application, use, and disposal. In the industrial production phase, the leakage of nanoparticles from equipment or industrial waste into air, soil, or wastewater can be a major source of pollution. Additionally, during the use of nanomaterial-containing products—such as paints, coatings, antibacterial agents, sunscreens, water filters, and even smart textiles—nanoparticles may be released into natural environments.

The transport routes of nanomaterials also vary depending on their type and the physical and chemical conditions of the environment. In aquatic environments, nanomaterials can bind with colloidal particles or organic matter and settle into sediments or be absorbed by aquatic organisms. In the air, nanoparticles can remain suspended and be easily inhaled by humans or animals. In soil, these particles may penetrate into micropores or transfer to plant roots. Also, when industrial sludge or wastewater is used in agriculture, there is a higher likelihood of nanomaterials entering the food chain indirectly. In all these cases, awareness of nanoparticle release and distribution pathways can assist in better nanomaterial design, waste management, and risk reduction.

Once nanomaterials enter the environment, processes such as transport, chemical transformation, bioaccumulation, and fate determine their ultimate environmental behavior. The distribution of nanomaterials in soil, water, or air is influenced by factors such as particle size, surface charge, chemical coatings, water solubility, and interaction with organic and inorganic environmental components. For instance, metal-based nanoparticles (like Ag, ZnO, and TiO₂) may form stable complexes in the presence of organic compounds like humic acids, or they may undergo aggregation, which reduces their mobility. Conversely, in the presence of dispersants or at certain pH levels, nanomaterials may become more mobile and penetrate deeper layers of soil or underground water sources.

One of the main challenges is the unpredictability of nanomaterial behavior in natural environments. Laboratory studies typically use simple, controlled environments to examine nanoparticle behavior. However, real-world conditions are highly variable and complex. For example, acidic soils have different capacities for absorbing and stabilizing nanomaterials compared to alkaline soils. Moreover, clay type, organic matter content, environmental temperature, soil microbiota, and the presence of other pollutants like heavy metals or pesticides can all influence the stability and movement of nanomaterials. This complexity makes it difficult to accurately model the environmental behavior of nanomaterials and to fully assess their potential ecological risks.

Another major issue is the bioaccumulation of nanomaterials in the food chain. (Figure 21) Nanomaterials can be absorbed by primitive organisms such as algae, bacteria, plankton, and earthworms. These particles are then passed along the food chain to higher organisms, eventually accumulating in fish, birds, and even humans. Bioaccumulation is harmful not only to individual species but also poses the risk of transmitting toxic levels of nanomaterials to humans. Some nanomaterials, such as silver nanoparticles or carbon nanotubes, tend to accumulate in fatty tissues or organs like the liver and brain and are difficult to eliminate from the body. Long-term accumulation may lead to genetic damage, hormonal disruptions, or even carcinogenic effects.

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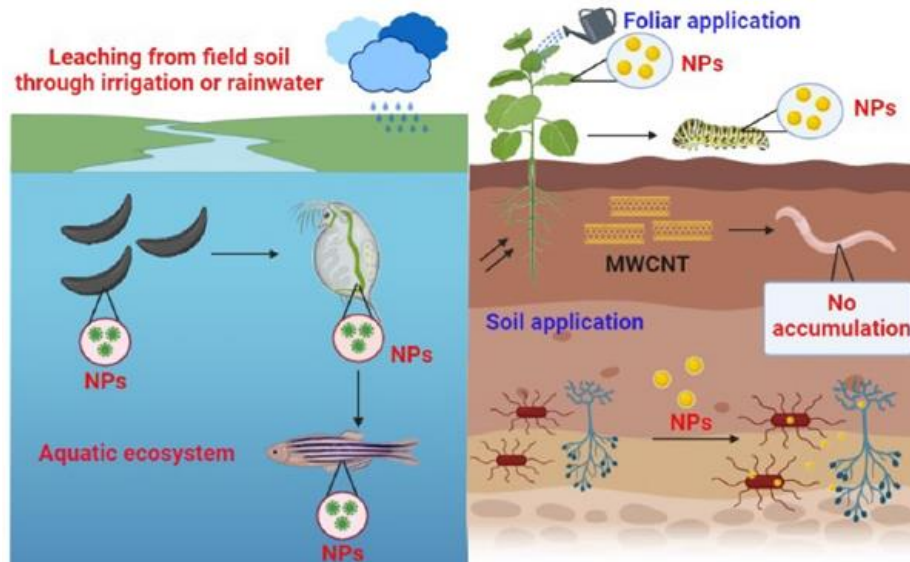


Figure 21 A representative diagram showing the bioaccumulation of nanoparticles in aquatic and soil ecosystems.

The phenomenon of chemical transformation of nanomaterials in the environment is another complex challenge. Nanomaterials may undergo alterations due to environmental conditions. For example, metal nanoparticles may release free metal ions, which are themselves toxic. Exposure to UV light or chemical agents in the environment may degrade nanoparticle coatings, thereby altering their behavior. As an example, nano TiO_2 exposed to UV light in aquatic environments may become photocatalytically active and generate reactive oxygen species (ROS). These transformations change the original properties of the nanomaterial and make it difficult to predict its effects.

Moreover, the disruption of natural microbiota by nanomaterials is another environmental concern. Nanomaterials may lead to the death of beneficial microorganisms or disturb their population balance. Such effects can influence key processes like organic matter decomposition, nitrogen fixation, or phosphorus cycling. Changes in microbiota can indirectly reduce soil fertility, increase plant diseases, or alter aquatic ecosystems. Some nanoparticles, like AgNPs, have strong antibacterial properties that, although advantageous under controlled conditions, can be harmful to beneficial environmental microbes in nature.

The persistence of nanomaterials in the environment is an additional challenge. Some nanomaterials—especially those synthetically produced with resistant coatings—may remain in the environment for years without undergoing biodegradation. This persistence means that even at low concentrations, long-term and widespread use can lead to gradual accumulation in the environment, causing chronic and long-term effects. Long-term environmental studies are still in their early stages, and the consequences of such widespread persistence are not yet fully understood.

In conclusion, the safe and sustainable development of nanotechnology depends on properly understanding, monitoring, and managing the release and accumulation of nanomaterials. Without

this, nanotechnology, despite its scientific and economic benefits, may bring unforeseen long-term risks to both the environment and human health.

11-3 risk assessment of nanomaterials methods

The risk assessment of nanomaterials is a critical process aimed at understanding and managing their potential impacts on human health and the environment. This process involves estimating or measuring the magnitude, frequency, and duration of exposure to nanoparticles. A typical risk assessment framework includes several key steps: problem formulation, where specific nanomaterials of concern and potential exposure routes are identified; exposure characterization, which involves measuring or estimating nanoparticle concentrations; exposure modeling, utilizing mathematical models to predict exposure levels; and finally, data analysis and interpretation to understand exposure patterns and associated risks. The ultimate objective is to develop a comprehensive understanding of a nanomaterial's safety for its intended application, while explicitly acknowledging and addressing any inherent uncertainties.

11-3-1 Toxicity Testing Methods

In Vitro Methods: These controlled experiments on cell cultures are widely used to evaluate nanotoxicity. Cytotoxicity assays, such as MTT (tetrazolium-based assays) and LDH (lactate dehydrogenase) release assays and ROS detection assay (for measuring active oxygen forms) are employed to assess cell viability and membrane integrity. Genotoxicity tests, including comet and micronucleus assays, examine DNA damage and chromosomal alterations following nanoparticle exposure. By exposing specific cell lines that model various biological barriers (e.g., epithelial cells for skin, lungs, or gastrointestinal tract), these methods provide crucial insights into how nanomaterials interact with different physiological systems.

In Vivo Methods: These studies evaluate how nanomaterials behave within living organisms, typically using rodent models. Researchers track bioaccumulation and long-term effects on vital organs such as the liver, kidneys, and brain.[10] Exposure routes in these studies are designed to mimic real-world scenarios, including inhalation, ingestion, and injection.[10] While invaluable for understanding systemic effects, traditional in vivo testing is resource-intensive and often requires a large number of laboratory animal studies, raising both practical and ethical concerns.

Computational Toxicology (In Silico Models): This approach applies computational models to predict nanotoxicity by analyzing the physicochemical properties of nanoparticles.[10] Techniques like Quantitative Nanostructure-Toxicity Relationship (QNTR) and Quantitative Structure-Activity Relationship (QSAR) modeling utilize descriptors such as particle size, surface charge, aggregation state, and solubility to estimate biological interactions and toxic potential.[10] These models offer an efficient alternative to traditional toxicity assessments, significantly reducing reliance on animal studies, minimizing costs, and enabling high-throughput screening of a vast array of nanomaterials.

Integrated Approaches to Testing and Assessment (IATA): Recognizing the limitations of conventional testing, IATA integrate existing knowledge bases on chemical classes with results

from biochemical and cellular assays, computational predictive methods, and exposure studies. The aim is to identify targeted testing requirements or develop comprehensive assessment conclusions. IATA can refine, reduce, or even replace selected conventional tests, particularly animal toxicity studies, thereby saving significant resources in terms of time and evaluation. This strategic shift towards computational and integrated approaches is driven by the sheer volume and complexity of new nanomaterials entering the market. It implies that future regulatory approval and safe development will increasingly depend on advanced predictive modeling and data integration, necessitating substantial investment in computational infrastructure and expertise, as well as rigorous validation of these new methods specifically for nanomaterials.

Despite their advantages, these methods face limitations. The accuracy of computational models, for example, is highly dependent on the quality of the experimental data they are trained on. Furthermore, defining the domain of applicability for QSAR models, especially given the diverse and complex nature of nanomaterials (which often exist as aggregates or mixtures), presents a significant challenge. Developing mechanistically-based models for the novel mechanisms of action exhibited by some nanomaterials also remains difficult.

11-3-2 Nanoparticle Characterization and Exposure Assessment

Effective risk assessment fundamentally relies on comprehensive characterization of nanoparticles and a thorough understanding of potential exposure.

Nanoparticle Characterization: This crucial step involves measuring the physicochemical properties of nanoparticles, such as their size, shape, and surface chemistry.⁸ These properties are paramount as they directly influence how nanoparticles behave in various environments, their ultimate fate, and their potential toxicity.^[8] Several advanced techniques are SEM, AFM & some others you see on season 7.

Exposure Assessment and Dosimetry: This process quantifies the magnitude, frequency, and duration of human or environmental exposure to nanoparticles.^[8] Nanoparticles can enter the human body through multiple pathways: inhalation into the lungs, ingestion via contaminated food and water, and dermal penetration through the skin. Notably, some nanomaterials possess the ability to cross critical biological barriers, such as the human blood-brain barrier or the placental barrier, in ways that larger particles cannot.

Dosimetry, the determination of the actual dose received, is essential for understanding exposure-response relationships and informing risk assessment.^[8] It aids in developing exposure limits and guidelines, with the dose typically calculated as $(\text{Concentration} \times \text{Exposure duration}) / \text{Body weight}$.

11-4 National and International Legal and Regulatory Frameworks

11-4-1 European Union: REACH Regulation and the Role of ECHA

The European Union has taken a proactive stance in regulating nanomaterials, recognizing nanotechnology as a Key Enabling Technology (KET) crucial for innovation and industrial progress. Under the EU's REACH (Registration, Evaluation, Authorization and Restriction of Chemicals) regulation, nanomaterials are specifically referred to as "nanoforms".

The definition of nanomaterials under REACH was updated on June 10, 2022, to include natural, incidental, or manufactured substances composed of particles or aggregates smaller than 100 nanometers, which must constitute more than 50% of the material (though this threshold can be adjusted for certain safety assessments).

Since January 1, 2020, new legal requirements mandate that any company manufacturing or importing nanomaterials into the EU must register these substances and provide detailed safety information. This includes a clear definition of the nanoform, its toxicological profile, and a full risk assessment. The European Chemicals Agency (ECHA) plays a central role in overseeing the regulation of nanomaterials under both REACH and CLP (Classification, Labelling and Packaging) frameworks. ECHA provides updated guidance documents, supports training programs, and established the Nanomaterials Expert Group (NMEG) in 2012 to ensure the safe development of nanomaterials and resolve technical regulatory issues. This proactive adaptation of a comprehensive chemical regulation framework, contrasting with the acknowledged limitations of the U.S. EPA, positions the EU as a model for balancing innovation with safety by providing a clearer, albeit stringent, pathway for market entry.

11-4-2 OECD Initiatives and International Harmonization Efforts

The Organisation for Economic Co-operation and Development (OECD) is actively involved in assisting governments with policies that ensure the responsible development of nanotechnologies, with a strong focus on the safety evaluation and assessment of manufactured nanomaterials. The OECD promotes international cooperation on the human health and environmental safety aspects of manufactured nanomaterials for regulatory purposes.

A key initiative was the 'Sponsorship Programme for the Testing of Manufactured Nanomaterials,' a seven-year experimental testing program conducted from 2007 to 2014.¹⁵ This program investigated 11 commercially viable nanomaterials using over 110 different chemical tests covering physical and chemical properties, environmental fate, ecotoxicology, and toxicology. The program concluded that, for the most part, standard test guidelines used for regular chemical substances are suitable for nanomaterials.

However, the conclusion that standard chemical test guidelines are "for the most part suitable" for nanomaterials carries a significant nuance. While a baseline of existing tests may apply, other OECD documents and national frameworks, such as Canada's, emphasize that nanomaterials can exhibit highly divergent properties compared to their bulk forms. These unique nanoscale phenomena, including increased reactivity due to high surface area or the ability to cross biological barriers, necessitate specific adaptations and the development of new guidelines. This means that international harmonization efforts, while crucial, must extend beyond simply applying existing chemical tests. It highlights the ongoing challenge of developing universally accepted, nano-specific test methods and regulatory approaches that truly account for the novel properties and potential hazards, rather than merely attempting to fit them into existing chemical paradigms. This continuous evolution requires sustained scientific research to inform regulatory policy.

The OECD actively develops and updates Test Guidelines (TGs) and Guidance Documents (GDs) for nanomaterial safety testing. These cover a range of aspects, including physical-chemical

properties (e.g., Test No. 125 for Particle Size and Size Distribution, Test No. 126 for Hydrophobicity Index), effects on biotic systems, environmental fate and behavior (e.g., Test No. 318 for Dispersion Stability), and human health effects.[14] International organizations like the International Organization for Standardization (ISO) and the OECD actively collaborate with national governments to develop shared testing protocols and data submission templates, aiming to overcome challenges in global alignment of nanomaterial regulation.

11-4-3 National Regulatory Approaches (e.g., US EPA, Canada CEPA)

National regulatory approaches to nanomaterials vary significantly, reflecting different levels of regulatory maturity and strategic priorities.

In the United States, the Environmental Protection Agency (EPA) possesses the statutory authority to regulate nanomaterials. However, the agency acknowledges that it currently lacks sufficient information and established processes to effectively manage the human health and environmental risks associated with these materials. The EPA's approach often involves regulating nanomaterials as chemicals, which means its success is inherently linked to the existing limitations of applicable chemical statutes and a reliance on industry-submitted data. This situation points to a reactive rather than proactive regulatory stance, where the rapid pace of nanomaterial development outstrips the agency's capacity for comprehensive risk assessment and management.

In contrast, Canada has developed a specific framework for the risk assessment of manufactured nanomaterials under the Canadian Environmental Protection Act, 1999 (CEPA). This framework provides guidance on adapting existing chemical risk assessment practices to account for the novel properties exhibited by substances at the nanoscale. The Canadian framework is structured to provide an overview of general CEPA risk assessment practices, followed by detailed nanomaterial-specific considerations, including modifications for unique properties.

Key adaptations in Canada include:

- A specific working definition of a nanomaterial, encompassing substances at or within the 1 to 100 nm nanoscale or exhibiting nanoscale properties.
- Grouping strategies that go beyond mere chemical composition, considering properties like agglomeration and particle size distribution.
- The use of detailed physical-chemical properties for identification, as standard Chemical Abstracts Service Registry Numbers (CAS RNs) do not differentiate between bulk and nanoscale forms.
- An acknowledgment that traditional environmental fate models for organic chemicals are often not applicable to nanomaterials, necessitating the development of adapted models.

The divergence in regulatory maturity and approach across major economies like the EU, Canada, and the U.S. creates significant challenges for global trade and innovation. Manufacturers face inconsistent data requirements and regulatory hurdles across different jurisdictions, which can slow commercialization and increase costs. This situation underscores the critical need for continued international cooperation and harmonization efforts, such as those promoted by the OECD, to establish a more coherent global regulatory landscape that ensures both safety and efficient market access for nanotechnology products.

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Chapter12: Socio-economic considerations and sustainable development

12-1 Cost-Benefit Analysis and Market Growth Potential

Nanotechnology is a powerful catalyst for innovation and entrepreneurship, particularly evident in sectors such as energy, healthcare, and electronics. It facilitates the creation of novel materials and products endowed with unique properties, driving advancements across these industries. The technology can substantially enhance manufacturing processes, leading to improved efficiency and reduced operational costs, thereby boosting overall productivity. The emergence of nanotech-based products and services is actively generating new markets and employment opportunities, contributing significantly to economic expansion. Nations and enterprises that strategically invest in nanotechnology are well-positioned to secure a competitive advantage in the global economic arena.

Projections for the global nanotech industry indicate robust growth, with estimates suggesting a market value of \$125 billion by 2025. Other forecasts suggest the market could exceed \$100 billion by 2030.[1] In the United States, the economic impact of nanotechnology was estimated to be between \$67 billion and \$83 billion in 2022, accumulating to nearly a trillion dollars over the preceding two decades. This substantial impact significantly surpasses the U.S. Government's investment of approximately \$40 billion over the same period, highlighting a considerable return on public investment. When closely related industries, such as microelectronics and semiconductors, are included in the analysis, the estimated economic contribution in 2022 alone surged to between \$268 billion and \$297 billion.

This substantial economic contribution, particularly the observation that government investment is dwarfed by the broader economic impact, points to a powerful multiplier effect. Nanotechnology's economic value is often diffused and challenging to quantify directly because it functions as an enabling technology embedded within numerous industries that do not exclusively identify as "nanotech R&D". This suggests that strategic, sustained public investment in foundational research and development acts as a significant catalyst for broader economic growth, yielding returns far exceeding the initial outlay. Policymakers should therefore recognize nanotechnology as a critical, high-leverage investment area. Economic assessments need to adopt more sophisticated methodologies that account for its indirect and embedded contributions across the economy to fully capture its pervasive benefits and justify continued public and private investment.(table 3)

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Table 3 Economic Benefits of Nanotechnology Across Key Industries

Industry	Potential Economic Benefits
Healthcare	Improved treatment outcomes (e.g., targeted drug delivery, personalized medicine), reduced healthcare costs, enhanced early diagnosis, real-time health monitoring.
Energy	Improved energy efficiency (e.g., solar panels doubling energy conversion, thermal insulation), reduced energy costs, stronger and lighter wind turbines, improved fuel efficiency, enhanced energy storage (e.g., lithium-ion batteries).
Electronics	Improved device performance (e.g., smaller, faster, more efficient microchips, quantum nanowires), reduced production costs, development of flexible touchscreens.
Food	Enhanced food safety (e.g., pathogen detection via nanobiosensors), improved food production (increased mechanical/thermal resistance, decreased oxygen transfer in packaging), extended shelf life, reduced waste.
Textile	Development of smart fabrics (stain/wrinkle resistant), stronger, lighter, and more durable materials for various equipment (e.g., motorcycle helmets, sports gear).
Manufacturing	Improved efficiency of production processes, reduced waste, improved product quality, development of new nanomaterials and nanoelectronics.
Environment	Air purification (ions), wastewater purification (nanobubbles, nanofiltration for heavy metals), more efficient and less polluting chemical reactions (nanocatalysts), environmental remediation (breaking down/immobilizing pollutants).

Although nanotechnology has lots of economic benefits across industries, it still faces significant cost-related challenges in research, development, and implementation. In the pharmaceutical industry, the development of a new nanodrug may cost up to 500 million, much of which is allocated to safety and efficacy testing, laboratory-scale production, and standardization processes. In the energy sector, incorporating nanomaterials into fuel cells or lithium-ion batteries can increase raw material costs by 20 to 30 percent, although, in the long term, these technologies help reduce energy loss and extend the operational lifespan of systems.

In agriculture, nano-fertilizers and nano-pesticides may be 2 to 5 times more expensive than conventional products. However, they offer advantages such as reduced usage, improved precision targeting, and less environmental contamination, making them economically viable on a larger scale. Still, the biggest economic barrier to industrial adoption of nanotechnology lies in scalability costs and manufacturing infrastructure. According to OECD reports, over 70% of nanotechnology projects in developing countries remain at the laboratory stage due to lack of financial support and appropriate infrastructure. To achieve sustainable development and global integration of nanotechnology, targeted investments and supportive policies for scaling up are essential.

12-2 Societal Impacts and Acceptance

Public perception is a critical determinant of nanotechnology's adoption and development. The level of public awareness and understanding significantly influences its acceptance. Misinformation or a lack of understanding can lead to negative perceptions and resistance to its widespread adoption. Public perception is shaped by a complex interplay of factors, including media coverage, the level of trust in regulatory bodies, and personal experiences. Studies on public attitudes consistently show that trust in regulatory agencies and the perceived benefits of nanotechnology are key determinants of public acceptance. This highlights that trust serves as the fundamental basis for societal legitimacy. Without transparent communication and robust regulatory frameworks, public skepticism can become a significant barrier to commercialization and societal benefit, even for demonstrably safe and beneficial applications. This underscores the need for nanotechnology to earn a "social license to operate."

Ethical considerations are another crucial aspect where societal values directly influence nanotechnology's trajectory. Concerns include the potential for misuse, such as nano-enhanced surveillance technologies that challenge traditional notions of privacy and raise questions about the balance between security and individual rights. Furthermore, the impact on human health and the environment, as well as issues related to equity and access, are central to ethical debates, as nanotechnology could potentially exacerbate existing social inequalities. Addressing these ethical concerns through responsible innovation practices is essential for maintaining public trust and promoting the overall legitimacy of nanotechnology.

To ensure that nanotechnology develops in a manner that aligns with societal values, responsible development and robust stakeholder engagement are paramount. Responsible development involves proactively considering potential societal impacts from the outset. This includes conducting thorough risk assessments, engaging in transparent communication with the public, and implementing robust regulatory frameworks.

Stakeholder engagement is critical for ensuring that nanotechnology development aligns with societal values. This involves not only policymakers and industry leaders but also the broader public and civil society organizations. Effective stakeholder engagement demands a commitment to transparency, inclusivity, and responsiveness to societal concerns. Strategies for enhancing the legitimacy of nanotechnology include improving public engagement and education through outreach programs and educational materials, implementing transparent and inclusive regulatory processes that encourage public participation, and actively addressing ethical concerns through responsible innovation practices. These practices encompass conducting thorough risk assessments, ensuring transparency and accountability in development and deployment, and fostering a culture of innovation that prioritizes human well-being and environmental sustainability.

12-3 Impact on Employment, Industries, and Agriculture

Nanotechnology's influence on employment is characterized by a dual impact: it is expected to create new job opportunities while simultaneously leading to job displacement in certain sectors due to increased automation.⁴ This pattern is consistent with historical technological advancements.

However, the increasing use of automation and nanotechnology-enabled processes may lead to job displacement, particularly in industries where tasks are repetitive or can be easily automated, such as manufacturing. This also implies a significant shift in traditional employment structures and industries, as new nanotechnology-enabled products and services may lead to the decline of existing industries and associated job losses. This represents nanotechnology as a driver of industrial restructuring and workforce evolution, a classic pattern of technological disruption. To effectively navigate this transition, societies must proactively invest in education, vocational training, and lifelong learning programs. Without a skilled workforce capable of adapting to nanotech-driven changes, the economic benefits might be unevenly distributed, potentially exacerbating social inequalities and leading to unemployment in sectors that fail to adapt. This highlights the need for foresight and strategic workforce planning.

To mitigate the negative impacts of job displacement, education and retraining programs are crucial. These initiatives aim to equip workers with the necessary skills to adapt to the changing job market. Furthermore, encouraging innovation and entrepreneurship in nanotechnology can also contribute to the creation of new job opportunities and drive economic growth.(table 4)

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Table 4: Nanotechnology's Dual Impact on Employment by Industry

Industry	Potential Impact of Nanotechnology	Potential Job Displacement	New Job Opportunities
Manufacturing	Increased automation and efficiency through nanotechnology-enabled processes.	High.	Nanotech manufacturing, Nanotech R&D.
Healthcare	Development of new medical devices and treatments.	Medium.	Nanotech R&D, Medical devices.
Electronics	Creation of smaller, faster, and more efficient devices.	Medium.	Nanotech R&D, Electronics.
Energy	Development of more efficient solar cells and energy storage systems.	Low.	Nanotech R&D, Energy systems.

12-4 Nanotechnology's Contribution to Sustainable Development Goals (SDGs)

Nanotechnology is a critical enabler for achieving the United Nations Sustainable Development Goals (SDGs), offering innovative solutions across a wide range of sectors.²⁰ Its applications directly and indirectly contribute to numerous global sustainability targets. This broad alignment demonstrates nanotechnology's profound potential to address pressing global challenges, positioning it not merely as a technological advancement but as a strategic tool for achieving a sustainable future. Investments in nanotechnology R&D and commercialization can yield multifaceted benefits across environmental, social, and economic dimensions of sustainable development.

Specific contributions include:

- **SDG 2 (Zero Hunger/Food Security):** Nanotechnology improves agricultural practices by enhancing the precise delivery of nutrients and pesticides, which leads to more efficient resource use and reduced environmental impact. It is also being explored for its potential to improve food storage and preservation, significantly contributing to the reduction of global food waste.
- **SDG 3 (Good Health and Well-being):** Nanotechnology directly impacts health through improved diagnosis and treatment of diseases, enabling targeted therapies and personalized

medicine. It facilitates highly sensitive biosensing and advancements in tissue engineering and regenerative medicine.

- **SDG 6 (Clean Water and Sanitation):** Advanced filtration systems utilizing nanomaterials are capable of removing even the smallest contaminants from water, providing safe drinking water, particularly in regions facing severe water scarcity or pollution. Nanomaterials also play a role in environmental remediation by breaking down or immobilizing pollutants in water, soil, and air.
- **SDG 7 (Affordable and Clean Energy):** Nanotechnology drives major improvements in renewable energy and energy storage technologies. Nanomaterials enhance the efficiency and reduce the costs of solar cells, fuel cells, and energy storage devices like batteries and supercapacitors.
- **SDG 9 (Industry, Innovation, and Infrastructure):** In the industrial sector, nanotechnology enables the development of stronger, lighter, and more durable materials used in construction (e.g., nano-cement, nano-concrete) and transportation, fostering more sustainable and efficient infrastructures.
- **SDG 12 (Responsible Consumption and Production/Waste Management and Circular Economy):** Nanotechnology plays a crucial role in waste management by enabling the recovery of valuable materials from complex waste streams, such as electronic waste. This reduces landfill waste and conserves resources by reintroducing recovered materials into the production cycle.
- **SDG 13 (Climate Action/Emissions Reduction):** Nanotechnology contributes to environmental sustainability by developing nanomaterials that improve energy efficiency and reduce greenhouse gas emissions. For instance, integrating nanomaterials into building materials can significantly enhance insulation, reducing energy required for heating and cooling.

In addition to its wide-ranging applications in water purification, air remediation, energy efficiency, and precision agriculture, nanotechnology also contributes significantly to sustainable development goals by promoting cleaner production methods and reducing environmental footprints. A growing area within this field is green nanotechnology, which focuses on the design, synthesis, and application of nanomaterials using environmentally friendly approaches. Green nanotechnology emphasizes the use of non-toxic solvents, renewable raw materials, low-energy synthesis methods, and waste minimization strategies. This branch of nanotechnology not only aims to reduce the negative environmental impact of nanomaterial production but also enhances the overall sustainability of nanotechnology-enabled products across industries.

12-5 Case Studies: Successful Environmental Projects and Sustainable Development Approaches

Different countries exhibit distinct drivers and strategic emphases in their approach to nanotechnology, often shaped by unique geopolitical, economic, and environmental contexts. This diversity implies that international collaboration must account for these varied priorities and regulatory philosophies, and that the success of nanotechnology for sustainable development is context-dependent.

- **China:** China has emerged as a global leader in nanomaterial manufacturing, holding close to 70% of the global graphene production capacity. Environmental Nanotechnology is a key focus

area within China's National Key R&D Program, driven by ambitious government campaigns to address severe water, soil, and air pollution. Chinese researchers are actively developing and scaling up technologies such as novel composites for wastewater decontamination and graphene-TiO₂ photocatalytic nets for river cleanup. China is the second-largest producer of Nano Environmental, Health, & Safety (EHS) research, rapidly gaining ground on the U.S. This problem-driven approach highlights how urgent societal needs can significantly accelerate technological development and deployment.

- Singapore: As a small city-state with limited natural resources, Singapore is strategically striving to increase its food self-production through technology. Nanotechnology applications in food, such as smart packaging and nanosensors for spoilage detection, are highly relevant to its food security goals. Public perception of nanotechnology in Singapore is influenced by exposure to risk-related information, with food applications causing the most concern among consumers.
- South Korea: South Korea's nano industry has experienced rapid development since 2001, bolstered by strong government support through initiatives like the National Nanotechnology Initiative. Significant investment in R&D and infrastructure has led to a substantial rise in its technological capabilities, reaching 66% of developed countries' levels by 2005. Environmental nanotechnology companies accounted for 5% of the nano industry products. Government plans emphasize facilitating R&D, promoting cooperation among industrial, academic, and research circles, expanding research infrastructure, and fostering international collaboration.
- Germany: The Federal Ministry for the Environment (BMUV) led the "NanoDialogue" from 2006 to 2023. This long-term, continuous exchange aimed to provide socio-political support for scientific and technological developments in nanotechnologies, focusing on opportunities, risks, assessment tools, risk management, traceability, and sustainability (termed "green nano"). This approach underscores a commitment to responsible and sustainable use through multi-stakeholder dialogue.
- United States: While a major investor, with federal funding for nanotechnology exceeding \$30 billion by 2024 since the National Nanotechnology Initiative (NNI) in 2001, only about 7% of this spending was allocated to study environmental, health, and safety implications. This contrasts with China's more explicit focus on EHS research driven by pollution challenges.

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Chapter 13: Challenges and the Future of Nanotechnology in the Environment

13-1 Limitations of Nanotechnology in Environmental Projects

Despite its remarkable potential to improve the quality of water, soil, and air resources, nanotechnology still faces a range of scientific, technical, economic, and social limitations that hinder its full-scale deployment in environmental projects. Many of these constraints arise from the inherent properties of nanomaterials or inefficient structures in commercialization pathways. However, through innovative approaches in material design, the development of sustainable synthesis methods, and interdisciplinary collaborations, these challenges can gradually be overcome.

One of the most significant challenges is the instability of nanomaterials under natural environmental conditions. Many nanoparticles undergo structural transformations when exposed to variable pH levels, high salt concentrations, sunlight, or oxygen, which diminishes their environmental performance. For example, photocatalytic nanomaterials such as TiO_2 exhibit lower efficiency in degrading pollutants in turbid or organic matter rich water. An innovative solution in this area is the development of doped, coated, or hybrid nanomaterials that exhibit higher chemical resistance and can perform effectively under real-world environmental conditions.

The second challenge is the uneven distribution and difficult recovery of nanoparticles after application. In projects such as water purification using nano adsorbents or nano catalysts, the particles often remain in the environment post-reaction, making their complete retrieval difficult or costly. This issue not only increases environmental risks but also raises operational costs. Utilizing magnetic nanoparticles (such as Fe_3O_4) or filtration-friendly platforms like nanomembranes and stabilized nanocomposites can be an effective method to overcome this limitation and enable easier separation of nanoparticles from the system.

Another major limitation is the incompatibility of some nanomaterial synthesis methods with environmental requirements. While the aim of many nano-projects is to protect the environment, certain synthesis processes—such as those involving toxic solvents, energy-intensive steps, or heavy metal precursors—may themselves cause pollution or pose hazards. Developing and localizing green synthesis methods, such as using plant extracts, aqueous media, and low temperatures, provides an ethical and viable solution to this contradiction.

A fourth challenge is the gradual decline in the performance of nanomaterials with prolonged or repeated use. Many nanoparticles, particularly adsorbents and photocatalysts, suffer from surface poisoning, clogging, or structural changes after several usage cycles, resulting in reduced efficiency. Rather than constant replacement, the appropriate scientific solution is to develop regenerable nanomaterials that can be restored through processes such as washing, heating, or light exposure, thereby extending their lifecycle.

On the other hand, the lack of international standards for assessing the environmental performance of nanomaterials makes it difficult to compare and select the most suitable nanomaterial for a specific application. Currently, various parameters such as adsorption capacity, chemical stability, toxicity, biodegradability, and cost are reported in different studies. However, there is no comprehensive framework for unified evaluation. Developing composite performance indices or creating open-access databases for environmental testing results could enhance transparency and improve decision-making in nano-based projects.

From a social perspective, the low public acceptance of nanotechnology and resistance to products containing nanoparticles—especially in areas like drinking water or agriculture—pose serious commercialization barriers. This challenge stems from a lack of public education, the absence of transparent product labeling, and safety concerns. The solution lies in implementing public education programs, transparent regulatory policies, and stakeholder participation in decision-making processes to build public trust in nanotechnology.

Furthermore, many environmental nano projects in developing countries remain at the laboratory stage due to a lack of industrial infrastructure and sufficient technical knowledge to produce and localize nanomaterials. While developed nations can afford sophisticated equipment for nanoparticle synthesis and characterization, resource-limited countries require low-cost platforms, local technology adaptation, and knowledge transfer through international networks. Innovations such as tabletop synthesis micro-platforms, the use of natural waste in synthesis, or academic collaborations with local companies can significantly help bridge this gap.

Finally, the disconnect between research and the market—known as the "lab-to-market gap"—is one of the most persistent issues in environmental nanotechnology projects. Many researchers, despite having excellent scientific achievements, struggle to find a suitable pathway for turning their innovations into commercial products. Establishing specialized environmental nanotechnology incubators, green-tech accelerators, and targeted venture capital investments can provide the foundation for overcoming this barrier.

13-2 The Future and Research Pathways of Nanotechnology in the Environment

In recent decades, nanotechnology has emerged as one of the key tools in the management and restoration of the environment. However, what will ultimately shape the future of this technology is not only its current capabilities, but also future innovations, interdisciplinary research approaches, and its ability to respond to complex environmental crises. Considering the dynamic nature of challenges such as climate change, emerging pollutants, depletion of natural resources, and the demands of sustainable development, the future trajectory of nanotechnology in environmental applications requires strategic and in-depth reconsideration in research and development.

One of the key future directions will be the focus on multifunctional nanomaterials. The next generation of environmental nanomaterials must be capable of simultaneously identifying and removing multiple types of pollutants. For instance, an advanced nano adsorbent should be able to capture or degrade heavy metals, volatile organic compounds, and microorganisms all in a single

step. Designing such materials requires a deep understanding of surface interactions at the nanoscale and the use of hybrid compounds such as metal–carbon nanocomposites or multiphase photocatalysts. Researchers are now working on developing nanomaterials that demonstrate adaptive behavior—materials that can alter their structure or surface activity in response to different pollutants.

Another promising research pathway involves the advancement of smart nano sensors and real-time environmental alert systems. These tools play a critical role in non-destructive and real-time monitoring of environmental contaminants. Nano sensors that can be embedded in soil or water and transmit precise data on pollutant type and concentration via the Internet of Things (IoT) are expected to constitute a significant part of the future of environmental surveillance. The development of nano-biosensors for detecting toxins, heavy metals, and organic compounds is a rapidly growing field, and their integration with digital technologies offers high potential for environmental crisis management.

A current challenge receiving increasing attention is the degradability and recyclability of used nanomaterials. In the future, there will be greater emphasis on developing nanomaterials that, after fulfilling their environmental function, degrade into harmless substances without leaving ecological footprints. Research in biodegradable nanomaterials, including those based on natural polymers (e.g., chitosan, alginate) or protein-based nanostructures, is expanding. Additionally, efforts are being made to design nanomaterials with time-programmed degradation, allowing them to break down or become recyclable after a specified operational period.

In line with energy efficiency and cost reduction goals, nature-inspired and biomimetic synthesis methods are also gaining significant attention. The future of environmental nanotechnology is moving toward the adoption of low-energy, solvent-free, and zero-waste processes. Examples of this approach include synthesizing nanomaterials under physiological conditions, utilizing microorganisms to produce metallic nanoparticles, or harnessing natural sunlight as an energy source. These methods are not only aligned with sustainable development principles, but they also contribute to lowering production costs.

Furthermore, future research will focus heavily on the management of emerging contaminants, such as microplastics, pharmaceutical residues in water, nano-pollutants, and PFAS compounds. These pollutants are difficult to eliminate using traditional methods and require nanomaterials with targeted surface activity and optimized structures. Designing nanostructures capable of selectively adsorbing, degrading, or immobilizing these compounds—particularly through the use of novel photocatalysts and modified nano adsorbents—has become a priority research area in many scientific centers around the world.

From a scientific policy perspective, the development of Life Cycle Assessment (LCA) models for environmental nanomaterials will play a key role in the future. Research in this domain involves the comprehensive analysis of the environmental impacts of nanomaterials throughout their lifespan—from production to usage and recycling. Creating global databases, conducting environmental modeling, and applying artificial intelligence to predict effects will help researchers design nanomaterials from the outset based on ecological principles.

Role of Nanotechnology in Sustainable Environmental Management

Looking further ahead, innovations such as environmental nanobots—tiny robotic systems designed to detect and remove pollutants at the nanoscale—or the use of nanomaterials activated by electromagnetic waves for remote purification, represent revolutionary ideas for intelligent and autonomous environmental management. Although these concepts are still in theoretical or laboratory stages, the rapid technological progress in materials science, electronics, and machine learning is paving the way for their commercialization in the near future.

Ultimately, the research pathway of nanotechnology in environmental applications requires the strengthening of interdisciplinary collaboration among chemists, environmental engineers, biologists, and data scientists. The integration of these fields, along with smart policymaking, education of the next generation, and private sector participation, paints a promising future where nanotechnology not only serves to eliminate pollution but also acts as a solution for prevention, restoration, and sustainable coexistence with nature.

13-3 The Role of Students and the Next Generation

In the twenty-first century, complex environmental crises such as climate change, pollution of vital resources, ecosystem degradation, and loss of biodiversity demand solutions that arise from science, technology, and creativity. Nanotechnology, as one of the most innovative scientific fields, has the potential to lie at the heart of these solutions. However, this technology cannot realize its true potential without a future generation that is informed, creative, and committed. Today's students are the researchers, innovators, entrepreneurs, and policymakers of tomorrow—and their role in advancing nanotechnology begins now.

The first and most important role of students is to develop foundational knowledge and interdisciplinary understanding of nanotechnology and the environment. Since nanotechnology is an inherently interdisciplinary field, students need to build strong proficiency in core sciences such as chemistry, physics, biology, and mathematics—especially in areas like material structure, surface properties, nanoscale interactions, and environmental reactions. Furthermore, strengthening skills such as experimental design, data analysis, modeling, and working with digital tools will prepare them for entry into the research arena.

The younger generation can also engage in the development of nanotechnology through social innovation and technology-driven entrepreneurship. Today, many successful startups in the field of clean technologies have been founded by creative students and youth. Particularly in developing countries, this generation can identify local needs—such as access to clean drinking water, polluted agricultural soils, or poor urban air quality—and design small but impactful nano-based projects to address them. With the support of educational institutions or young investors, they can take these projects from concept to production.

Ultimately, students are the future-makers who, by combining science, ethics, creativity, and environmental commitment, can guide the development of nanotechnology toward a path that is not only profitable and efficient but also aligned with human values, social justice, and the preservation of the planet for future generations. Investing in nanotechnology education at the

secondary and pre-university levels is not only a step toward scientific advancement for nations, but also an investment in the protection of Earth itself.

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